

# Acknowledgement

First of all, we are sincerely grateful to everyone who supported and guided us throughout our final year project for the diploma computer engineering (semester 5-6)

To begin with, we are deeply thankful to our internal guide, Prof. Foram N. Chovatiya, Assistant Professor, for her constant support, insightful feedback, and valuable guidance. Her expertise and encouragement played a key role in shaping the success of StyleSurfer.

We also want to express our sincere thanks to our project coordinator for their constant support and timely help. Their encouragement kept us organized and focused, allowing us to successfully achieve our project goals.

We would also like to give credit to all the team members who brought unique ideas and contributed their hard work in building StyleSurfer. Their dedication played a crucial role in making this project a reality, and we are proud of what we've accomplished together.

Lastly, we are deeply grateful to our families for their constant support and encouragement throughout the project. Their belief in us kept us motivated and inspired.

Harshad Soyliya – 92200938212

Purav Chauhan – 92200938220

Mansi Kadvani – 92200938221

Bhumika Kanzariya - 92200938234

Jay Kaneriya – 92200938239

## Abstract

*StyleSurfer is an affordable and eco-friendly fashion rental platform that allows people to wear stylish, high-quality clothes without the need to buy them. Based in Rajkot, the platform aims to reduce fashion waste by promoting reuse, making trendy and premium outfits more accessible to everyone.*

*Built with Python and Django for the backend, and HTML, CSS, Bootstrap, and JavaScript for the frontend, StyleSurfer ensures a seamless and user-friendly experience.*

*Customers can easily sign up, manage their profiles, browse outfits with advanced filters, and rent clothes with flexible options, including extended rentals and damage protection. Unique features like the Buy Back option (where users can customize outfits and return them within a fixed duration), Reselling (where pre-rented outfits are sold at a discount), and a fun Spin-to-Win discount game enhance the experience.*

*Sellers can list and manage products, track sales, and receive secure payments, while delivery partners benefit from automated order assignments, an easy-to-use dashboard, and navigation support for efficient deliveries.*

*By combining technology, sustainability, and convenience, StyleSurfer makes fashion more affordable and eco-friendlier, benefiting customers, sellers, and delivery partners alike.*

## List of Table

|    |                                               |    |
|----|-----------------------------------------------|----|
| 1  | Table 1.4.1 Definitions & Abbreviations ..... | 36 |
| 2  | Table 1 Customer.....                         | 56 |
| 3  | Table 2 Seller .....                          | 56 |
| 4  | Table 3 DP .....                              | 56 |
| 5  | Table 4 FAQ .....                             | 57 |
| 6  | Table 5 Category.....                         | 57 |
| 7  | Table 6 Product.....                          | 57 |
| 8  | Table 7 RCategory .....                       | 58 |
| 9  | Table 8 Resell_Product .....                  | 59 |
| 10 | Table 9 Sell With Us .....                    | 60 |
| 11 | Table 10 Cart.....                            | 61 |
| 12 | Table 11 Cart_Item .....                      | 62 |
| 13 | Table 12 Order .....                          | 62 |
| 14 | Table 13 OrderItem.....                       | 63 |
| 15 | Table 14 ReviewRating.....                    | 64 |
| 16 | Table 15 SpinToWin.....                       | 64 |
| 17 | Table 16 Voucher.....                         | 65 |
| 18 | Table 17 RedeemVoucher.....                   | 65 |
| 19 | Table 18 DPOrder .....                        | 66 |
| 20 | Table 19 HelpRequest.....                     | 67 |
| 21 | Table 20 BuyBack.....                         | 67 |
| 22 | Table 21 CustomDesign.....                    | 68 |

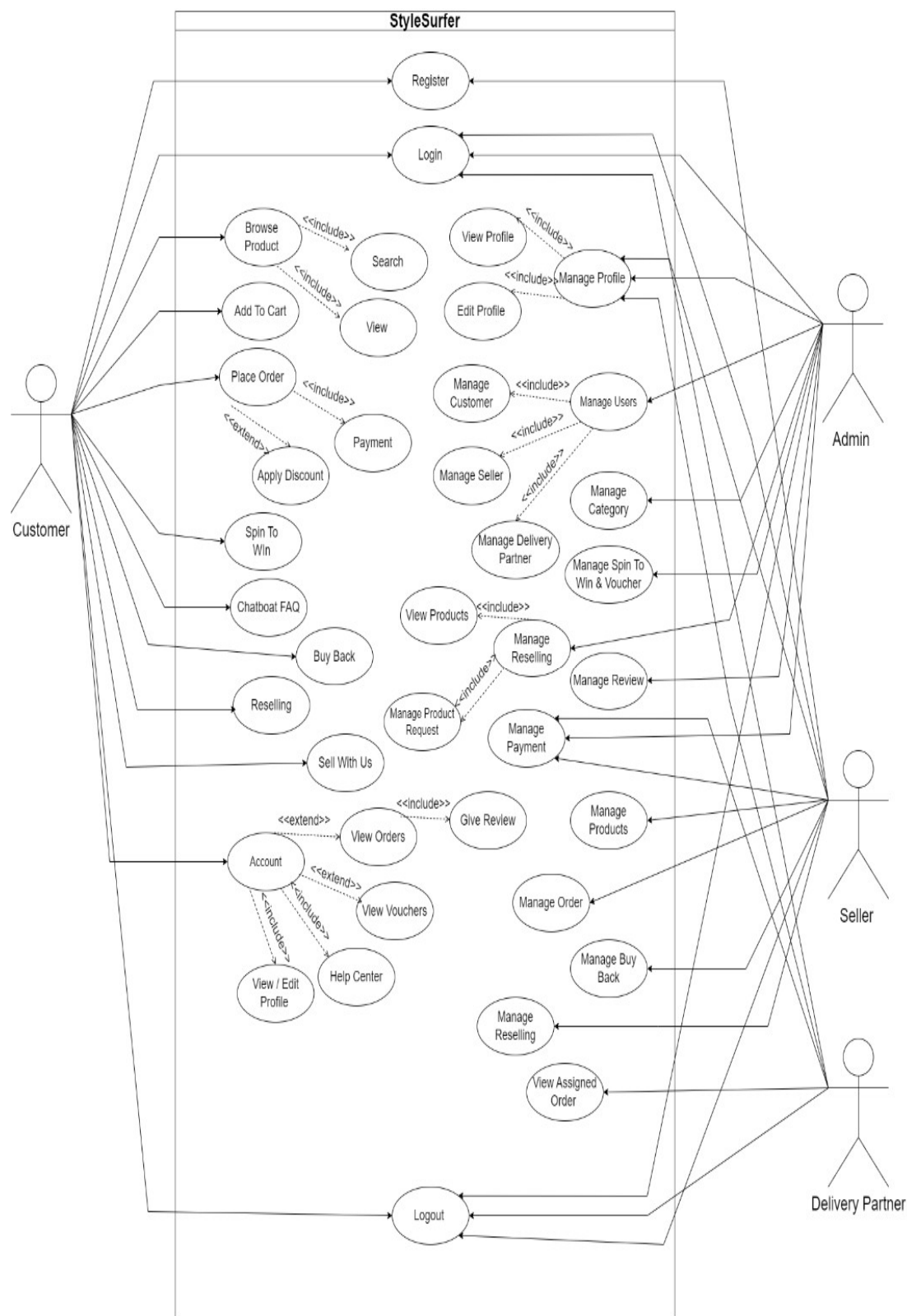
## List of Figures

|     |                                                    |    |
|-----|----------------------------------------------------|----|
| 1   | Figure 1 Use Case Diagram.....                     | 1  |
| 2   | Figure 2 E.R Diagram.....                          | 2  |
| 3   | Figure 3 Activity Diagram.....                     | 4  |
| 3.1 | Figure 3.1 Customer Activity Diagram.....          | 4  |
| 3.2 | Figure 3.2 Seller Activity Diagram.....            | 6  |
| 3.3 | Figure 3.3 Delivery Partner Activity Diagram.....  | 7  |
| 3.4 | Figure 3.4 Admin Activity Diagram .....            | 8  |
| 4   | Figure 4 Sequence Diagram .....                    | 10 |
| 4.1 | Figure 4.1 Customer Sequence Diagram .....         | 10 |
| 4.2 | Figure 4.2 Seller Sequence Diagram.....            | 12 |
| 4.3 | Figure 4.3 Delivery Partner Sequence Diagram ..... | 14 |
| 4.4 | Figure 4.4 Admin Sequence Diagram .....            | 15 |
| 5   | Figure 5 DFD (Level 0).....                        | 16 |
| 6   | Figure 6 DFD (Level 1).....                        | 17 |
| 6.1 | Figure 6.1 Customer DFD (Level 1).....             | 17 |
| 6.2 | Figure 6.2 Seller DFD (Level 1).....               | 19 |
| 6.3 | Figure 6.3 Delivery Partner DFD (Level 1).....     | 20 |
| 6.4 | Figure 6.4 Admin DFD (Level 1) .....               | 21 |
| 7   | Figure 7 DFD (Level 2).....                        | 22 |
| 7.1 | Figure 7.1 Customer DFD (Level 2).....             | 22 |
| 7.2 | Figure 7.2 Seller DFD (Level 2) .....              | 23 |
| 7.3 | Figure 7.3 Delivery Partner DFD (Level 2).....     | 24 |
| 7.4 | Figure 7.4 Admin DFD (Level 2) .....               | 25 |
| 8   | Figure 8 Flowchart Diagram.....                    | 26 |
| 8.1 | Figure 8.1 Customer Flowchart Diagram.....         | 26 |



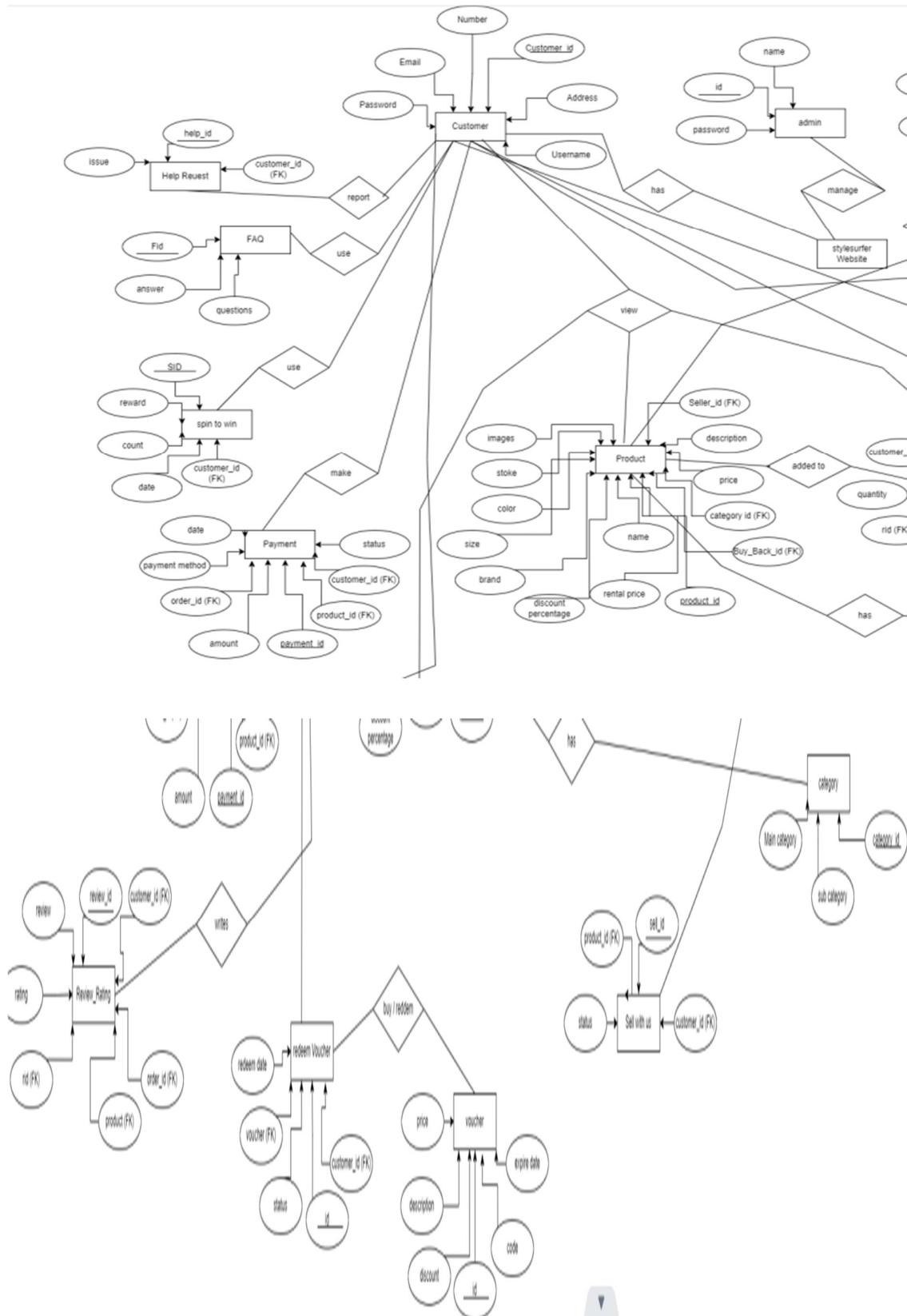
|     |                                                     |    |
|-----|-----------------------------------------------------|----|
| 8.2 | Figure 8.2 Seller Flowchart Diagram.....            | 29 |
| 8.3 | Figure 8.3 Delivery Partner Flowchart Diagram ..... | 30 |
| 8.4 | Figure 8.4 Admin Flowchart Diagram.....             | 31 |
| 9   | Figure 9 Class Diagram .....                        | 33 |

**Use Case Diagram :-**



**Figure 1 Use Case Diagram**

## E.R. Diagram :-



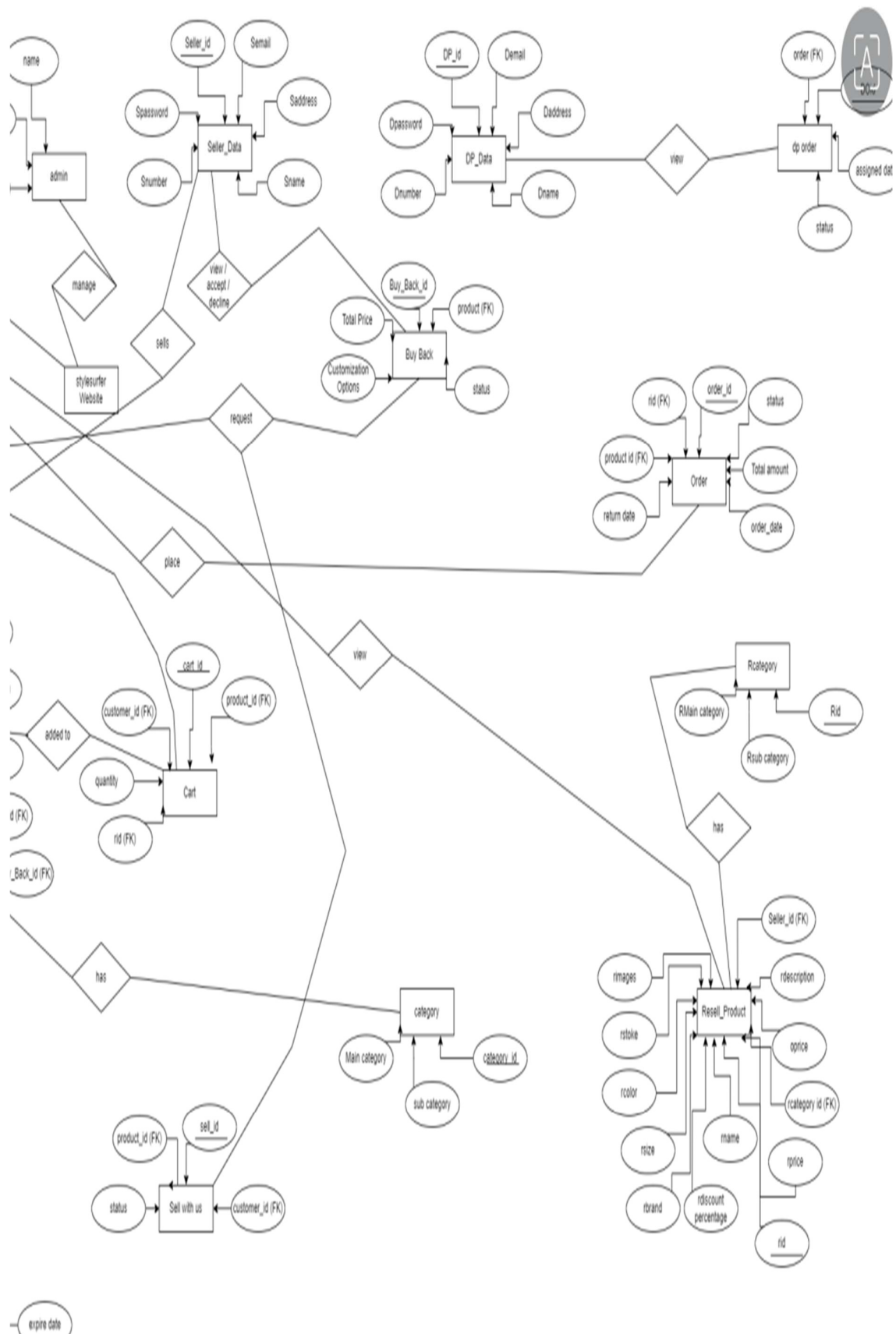
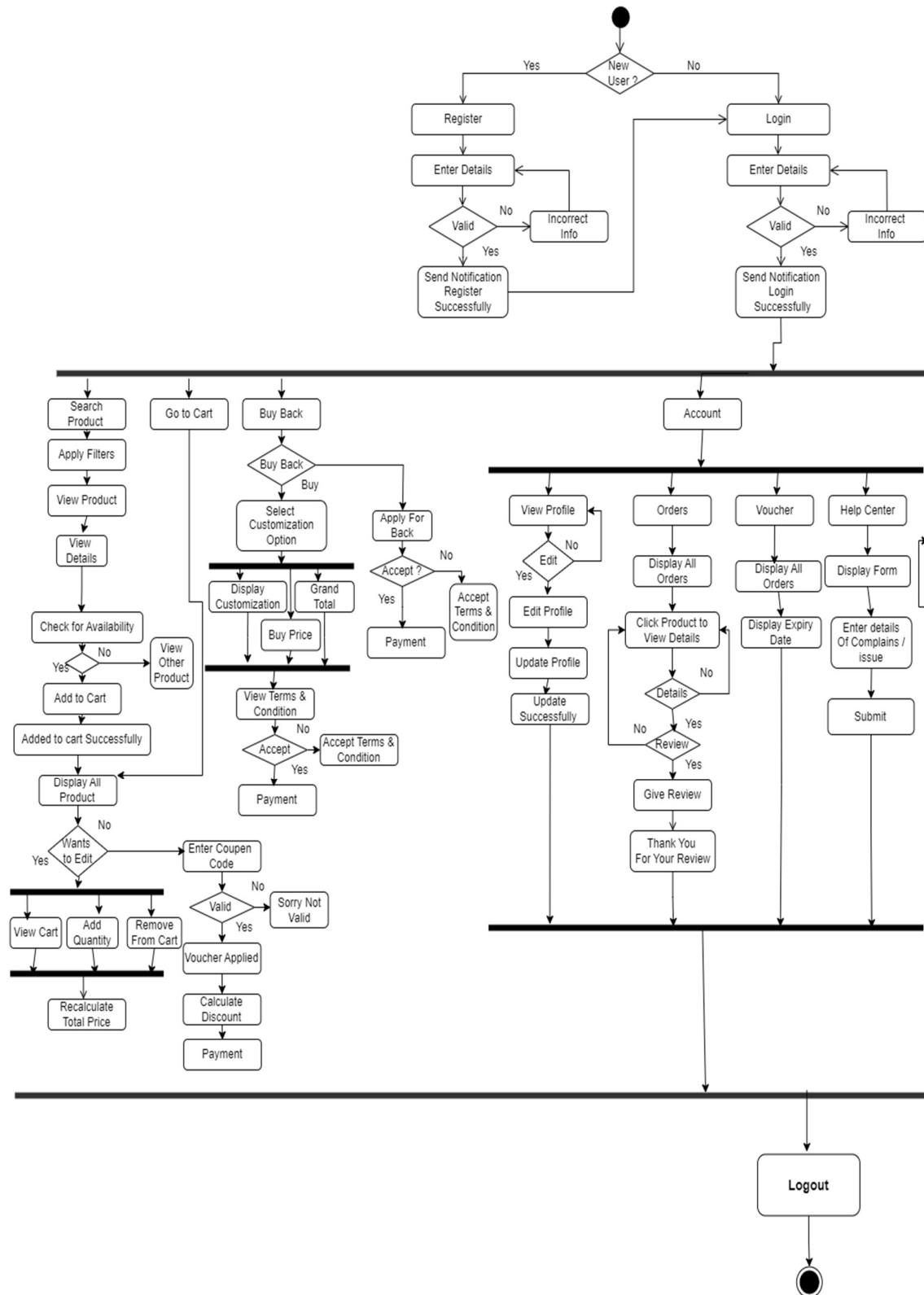
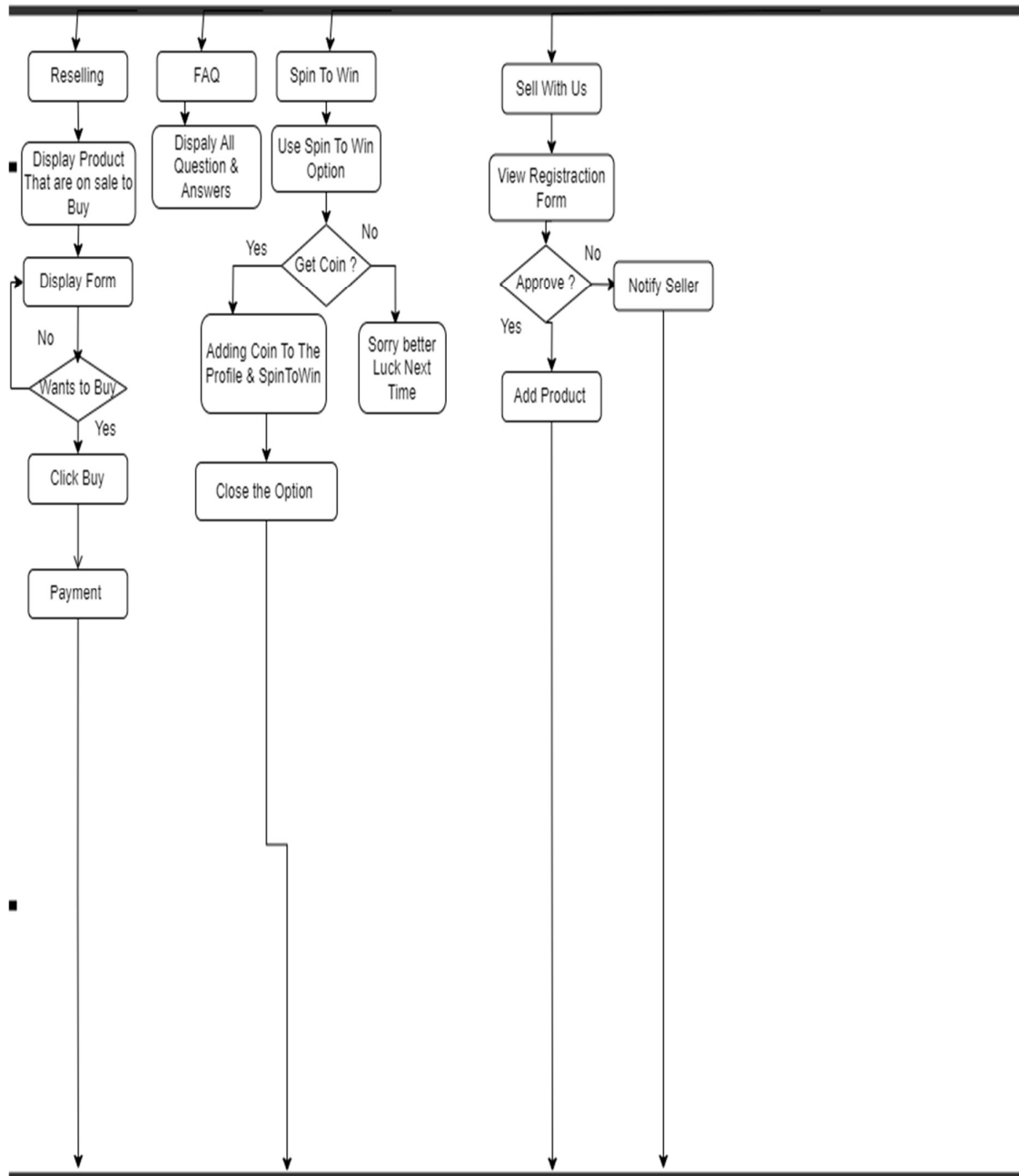


Figure 2 E.R. Diagram

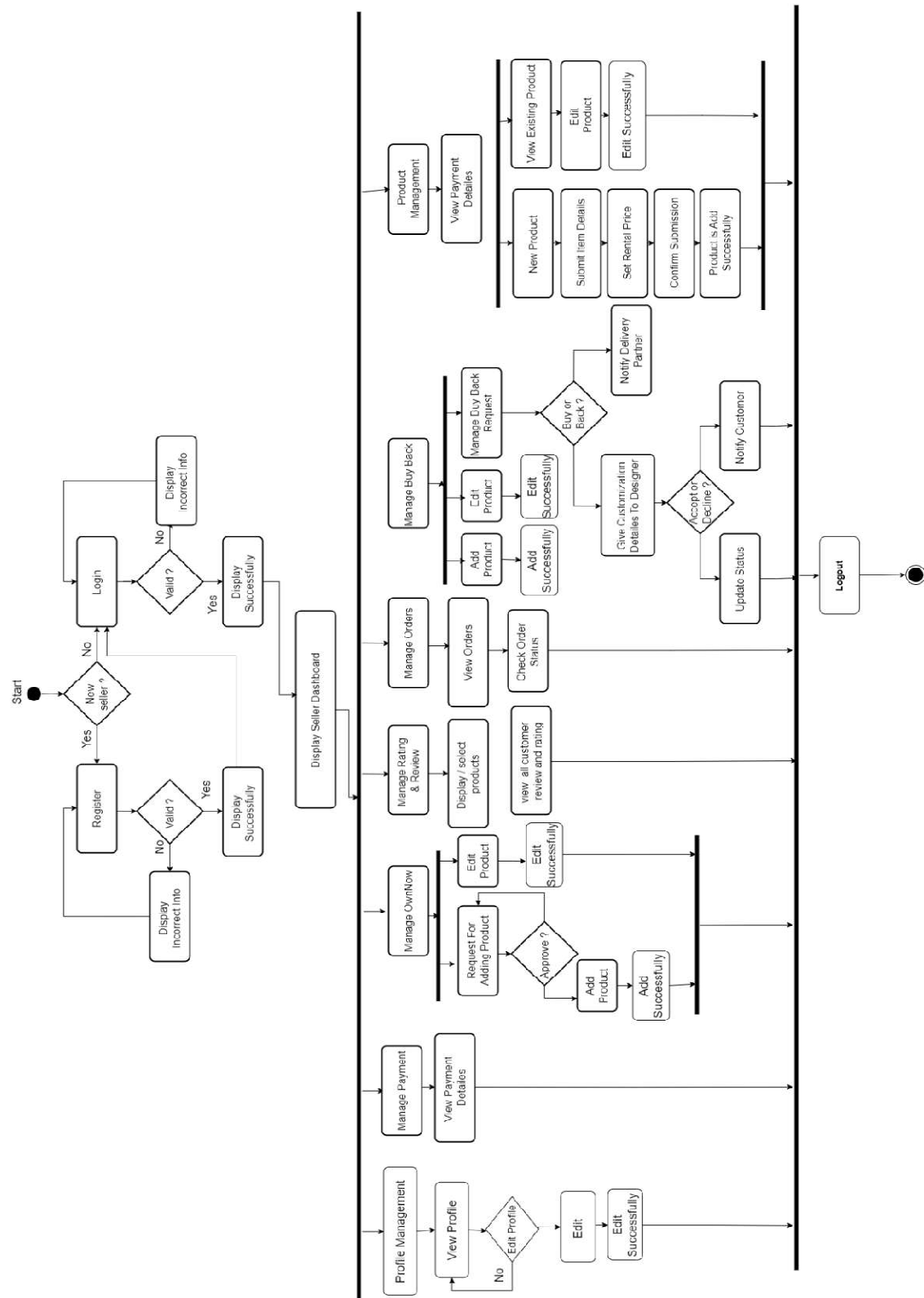
## Customer Activity Diagram :-





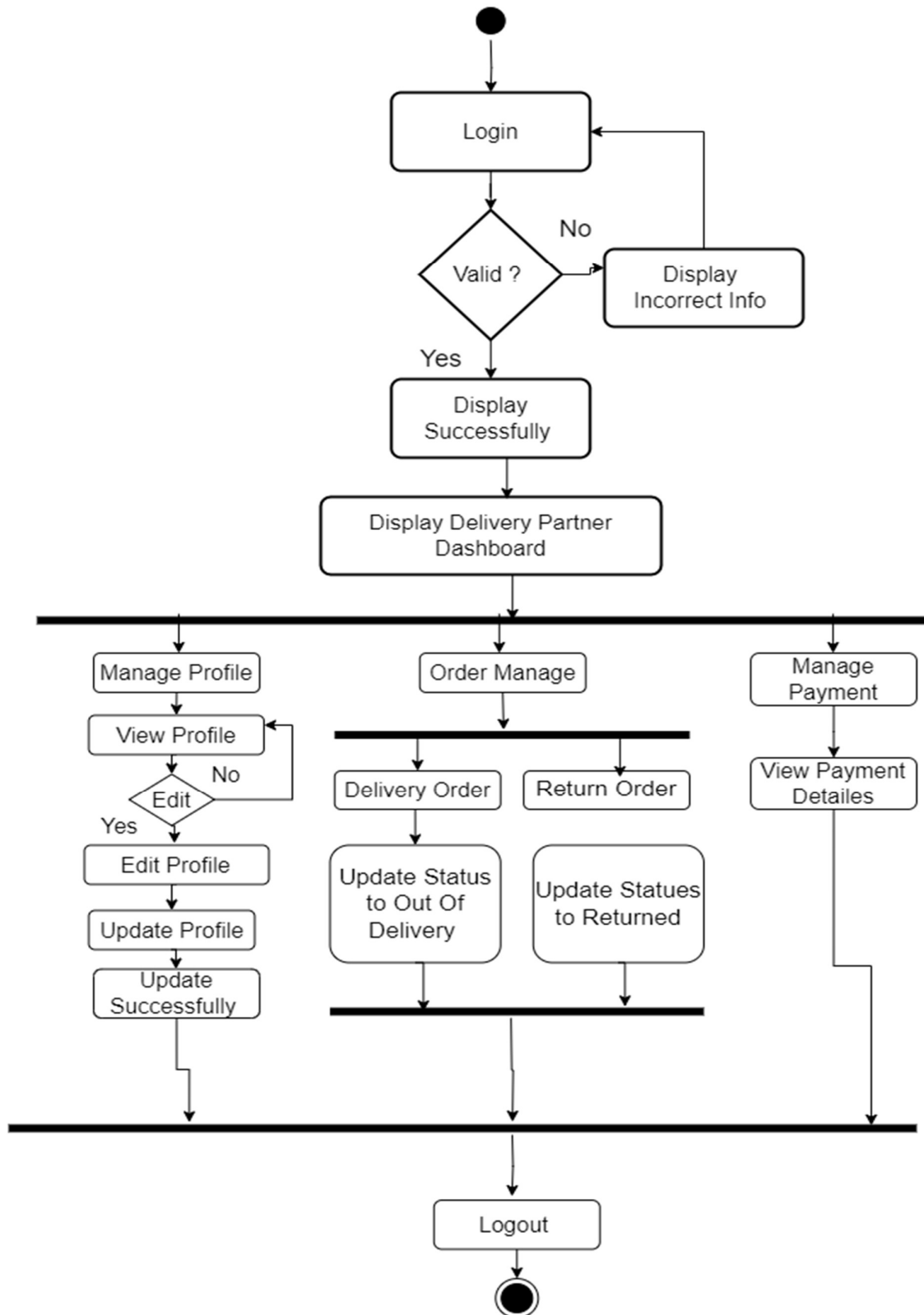
**Figure 3.1 Customer Activity Diagram**

# **Seller Activity Diagram :-**



**Figure 3.2 Seller Activity Diagram**

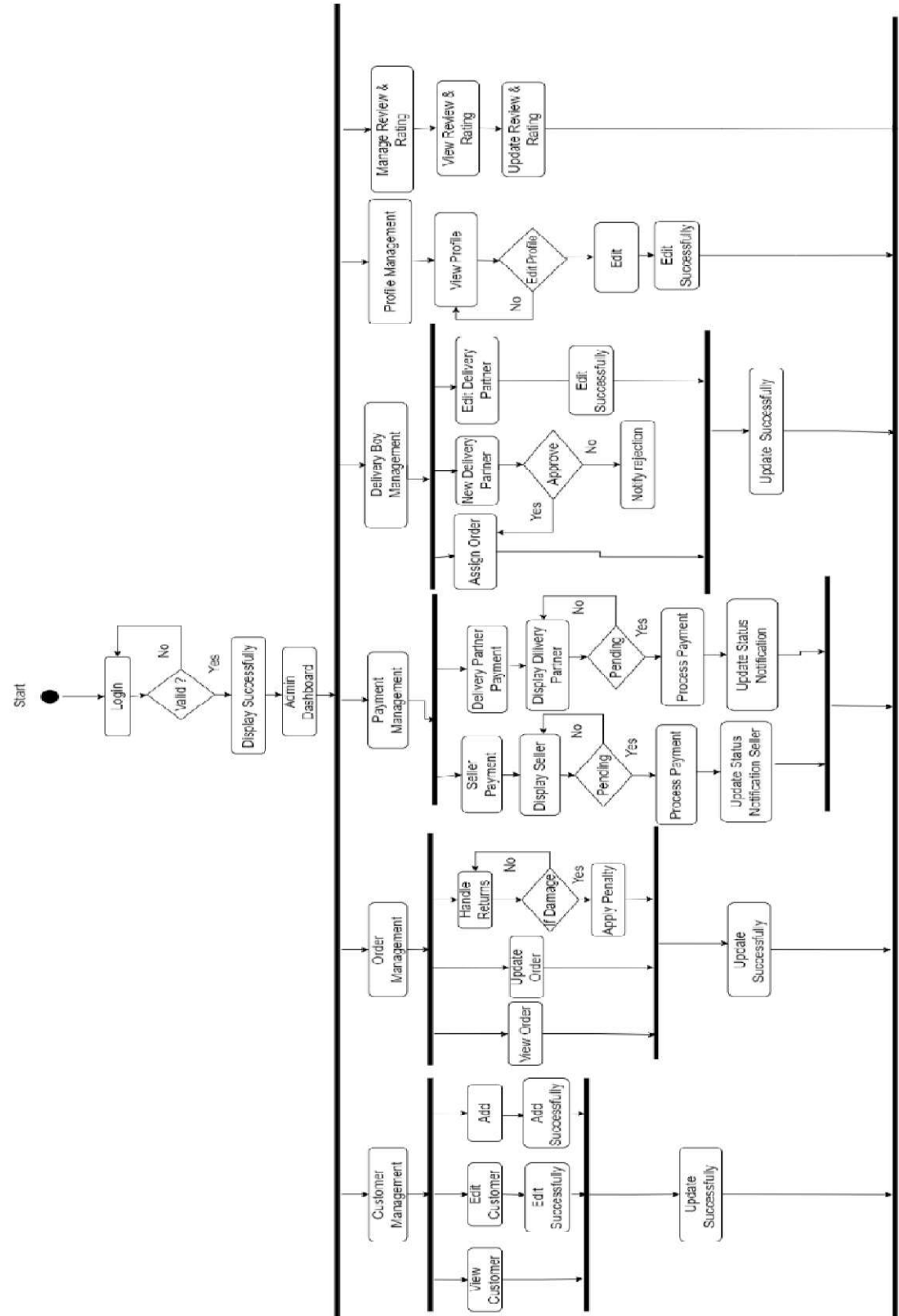
**Delivery Partner Activity Diagram :-**



**Figure 3.3 Delivery Partner Activity Diagram**



# Admin Activity Diagram :-



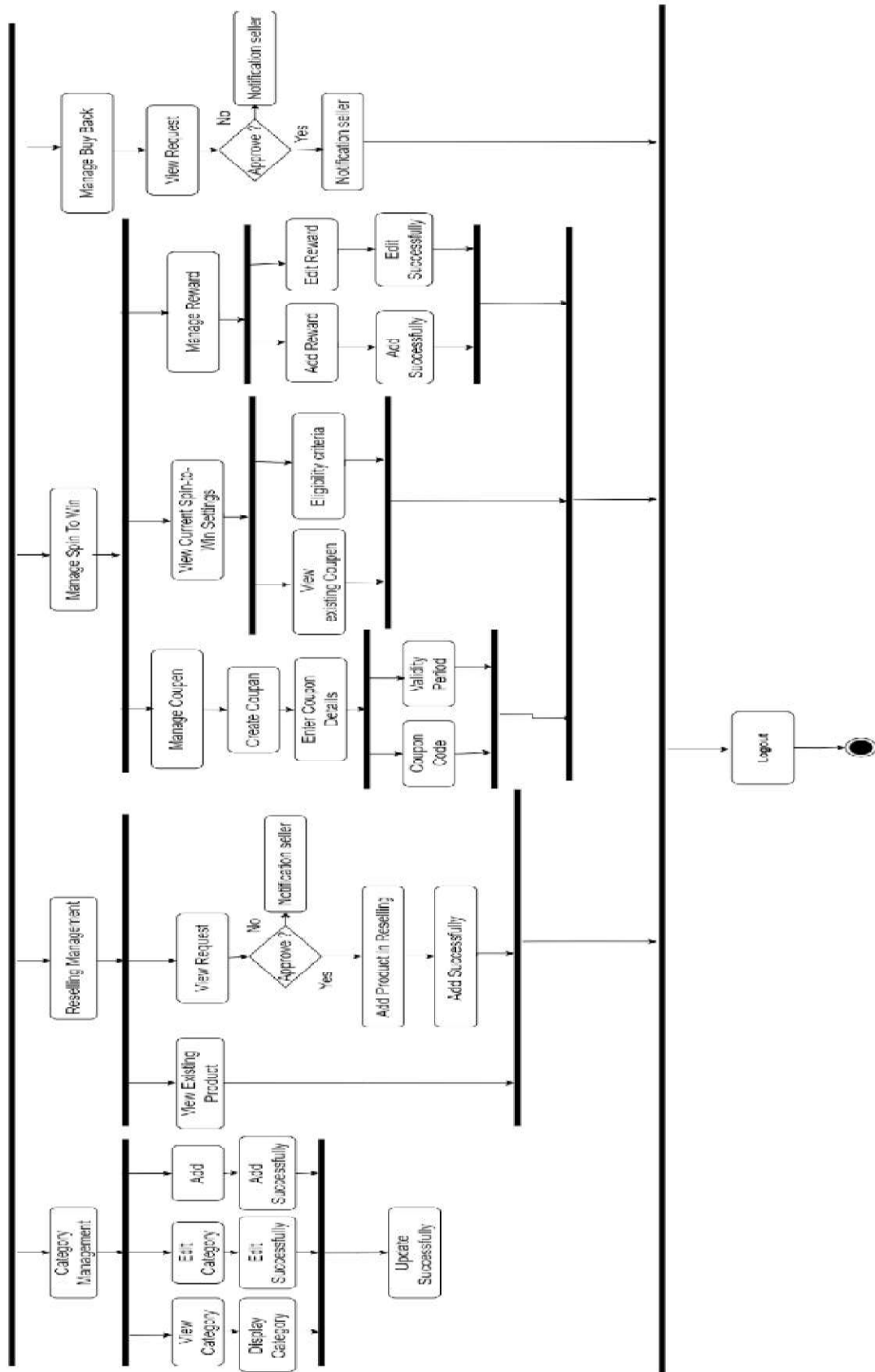
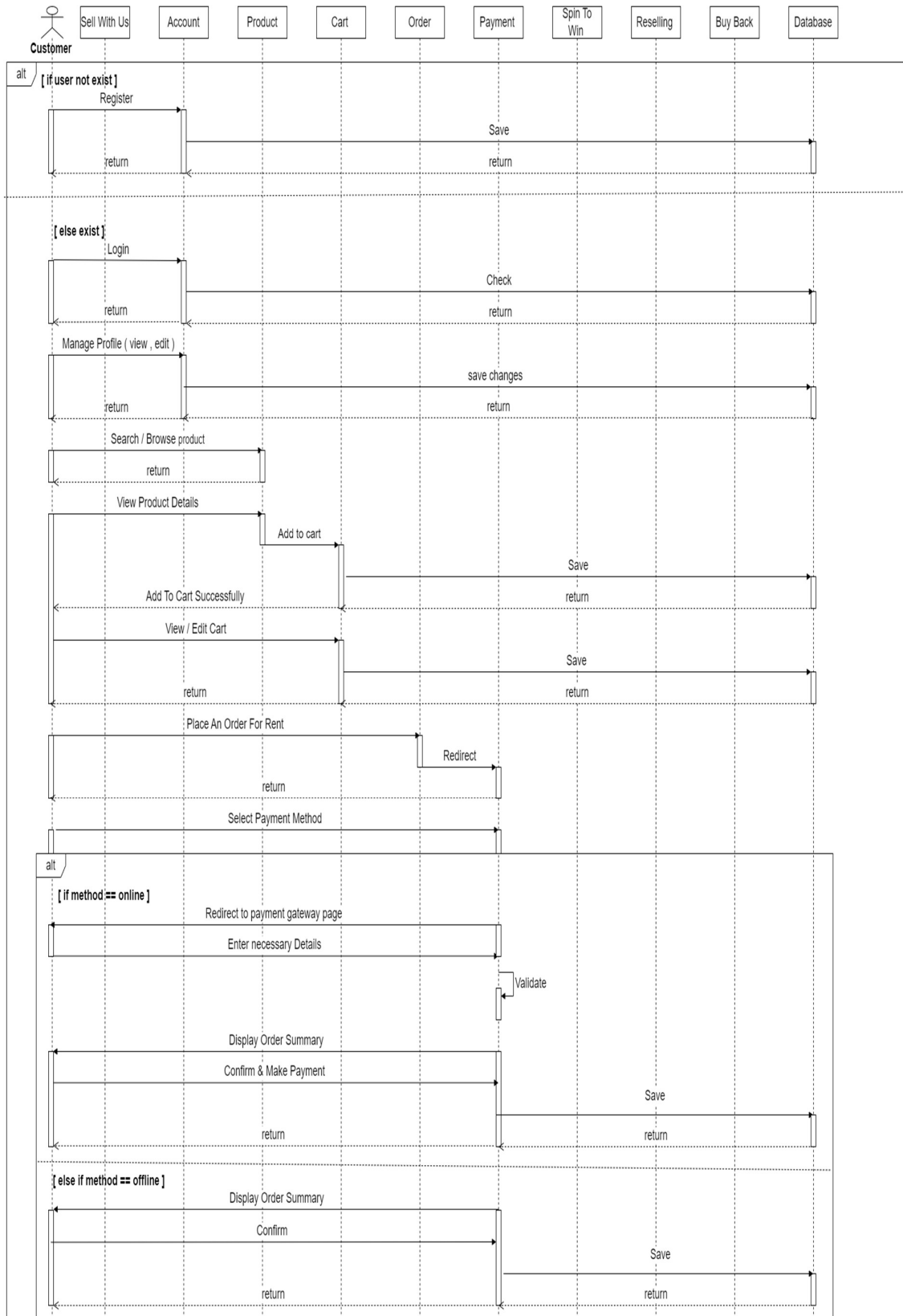
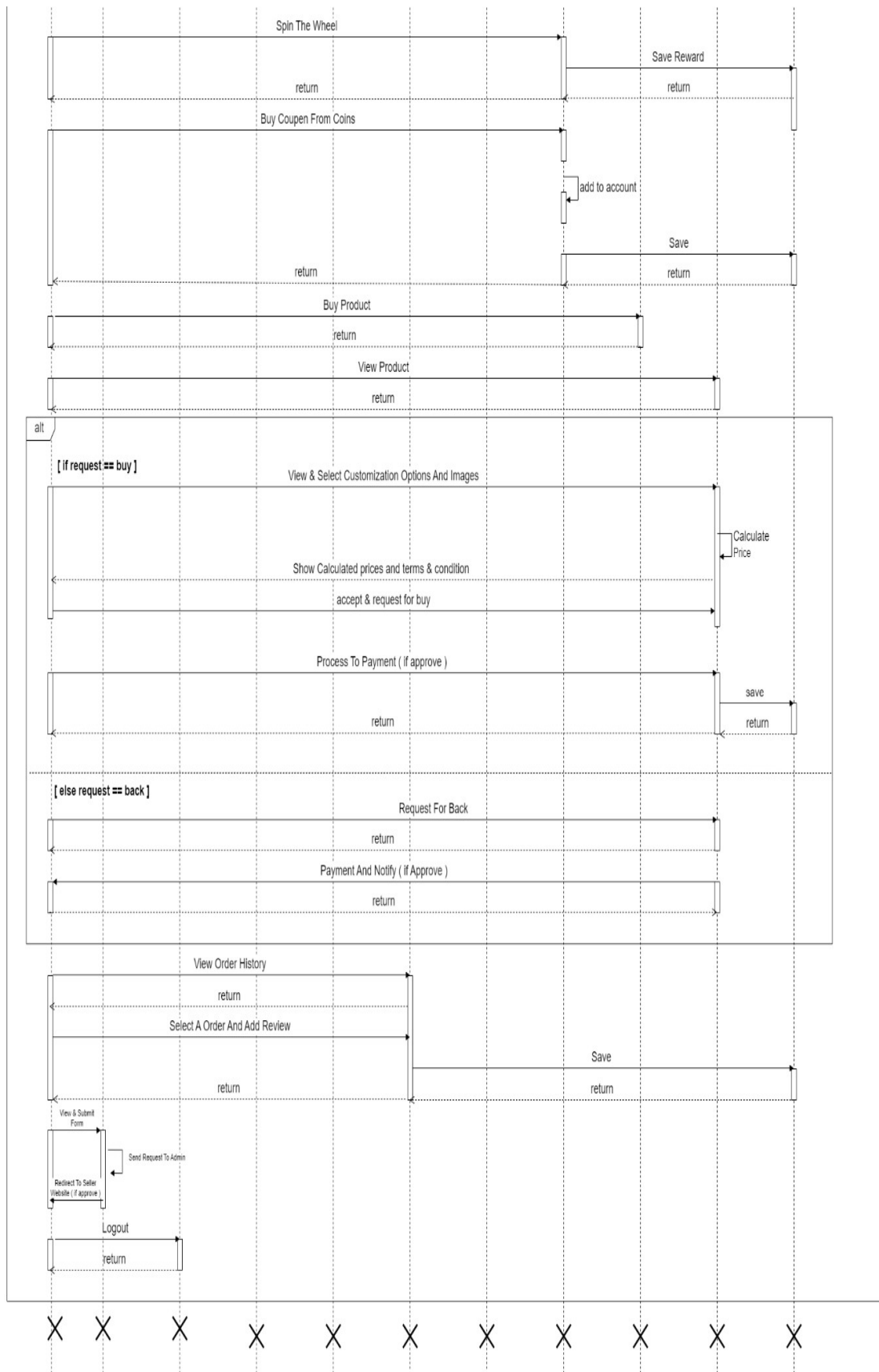


Figure 3.4 Admin Activity Diagram

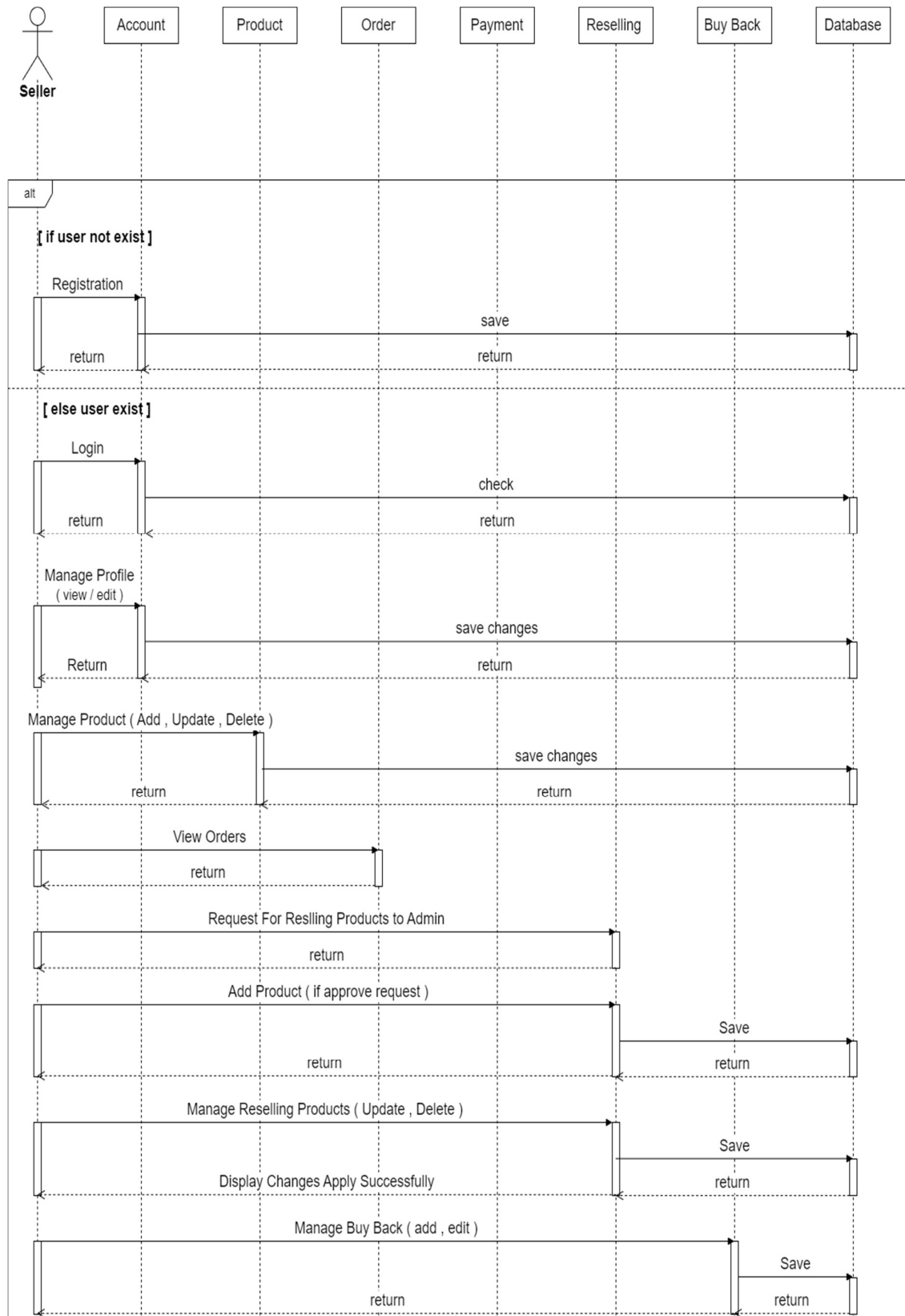
## Customer Sequence Diagram :-

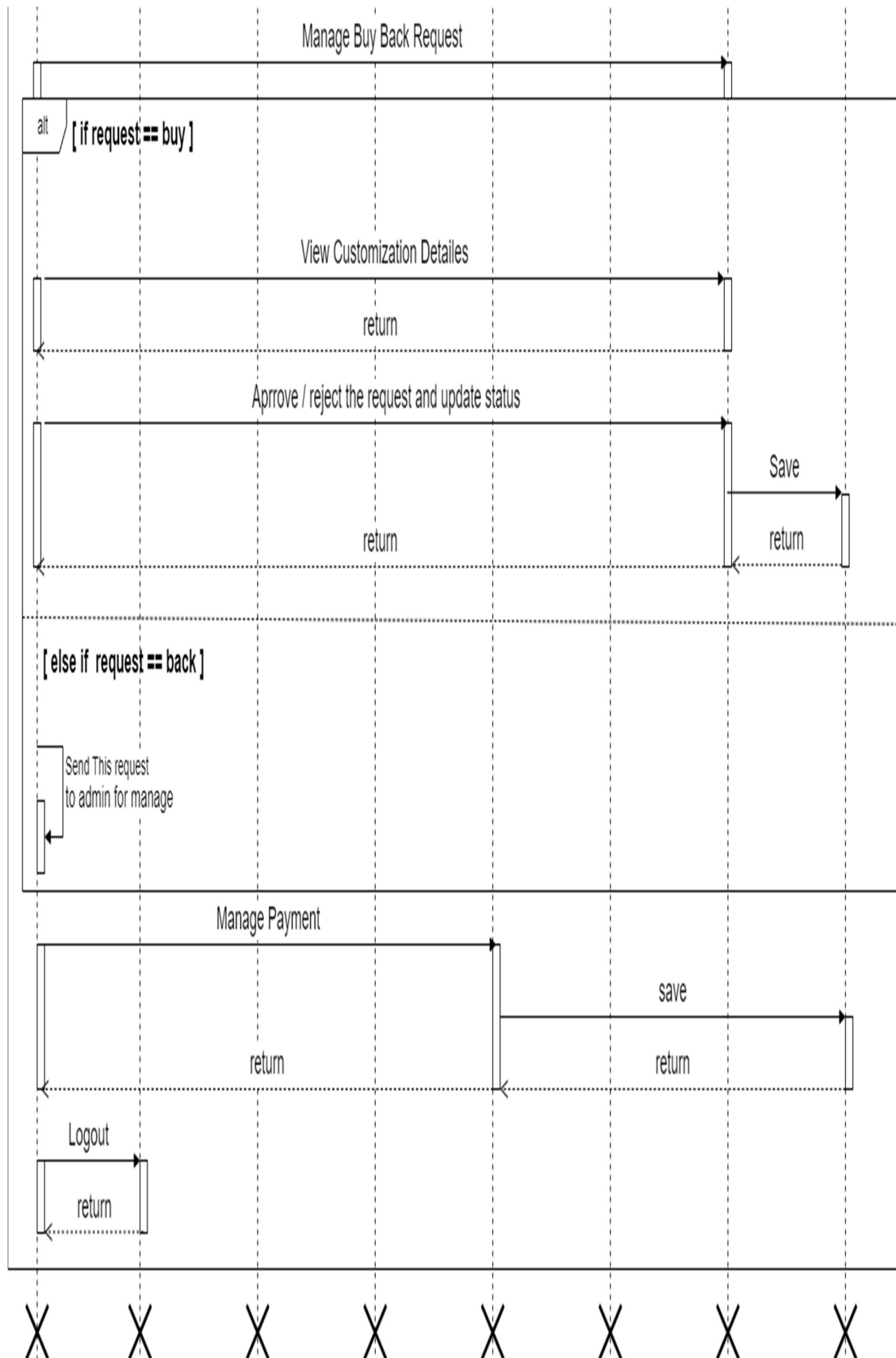




**Figure 4.1 Customer Sequence Diagram**

## Seller Sequence Diagram :-





**Figure 4.2 Seller Sequence Diagram**

## Delivery Partner Sequence Diagram :-

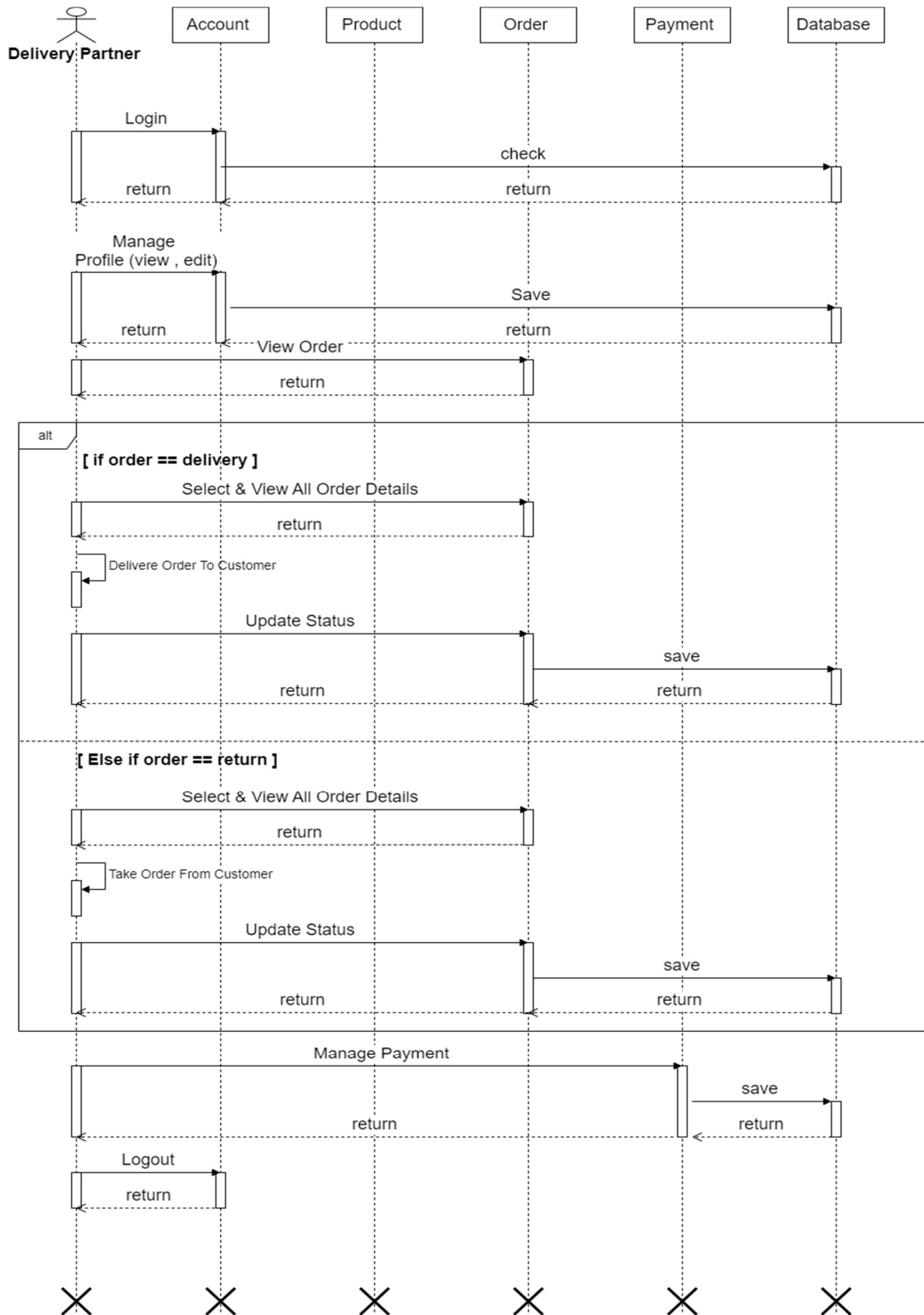
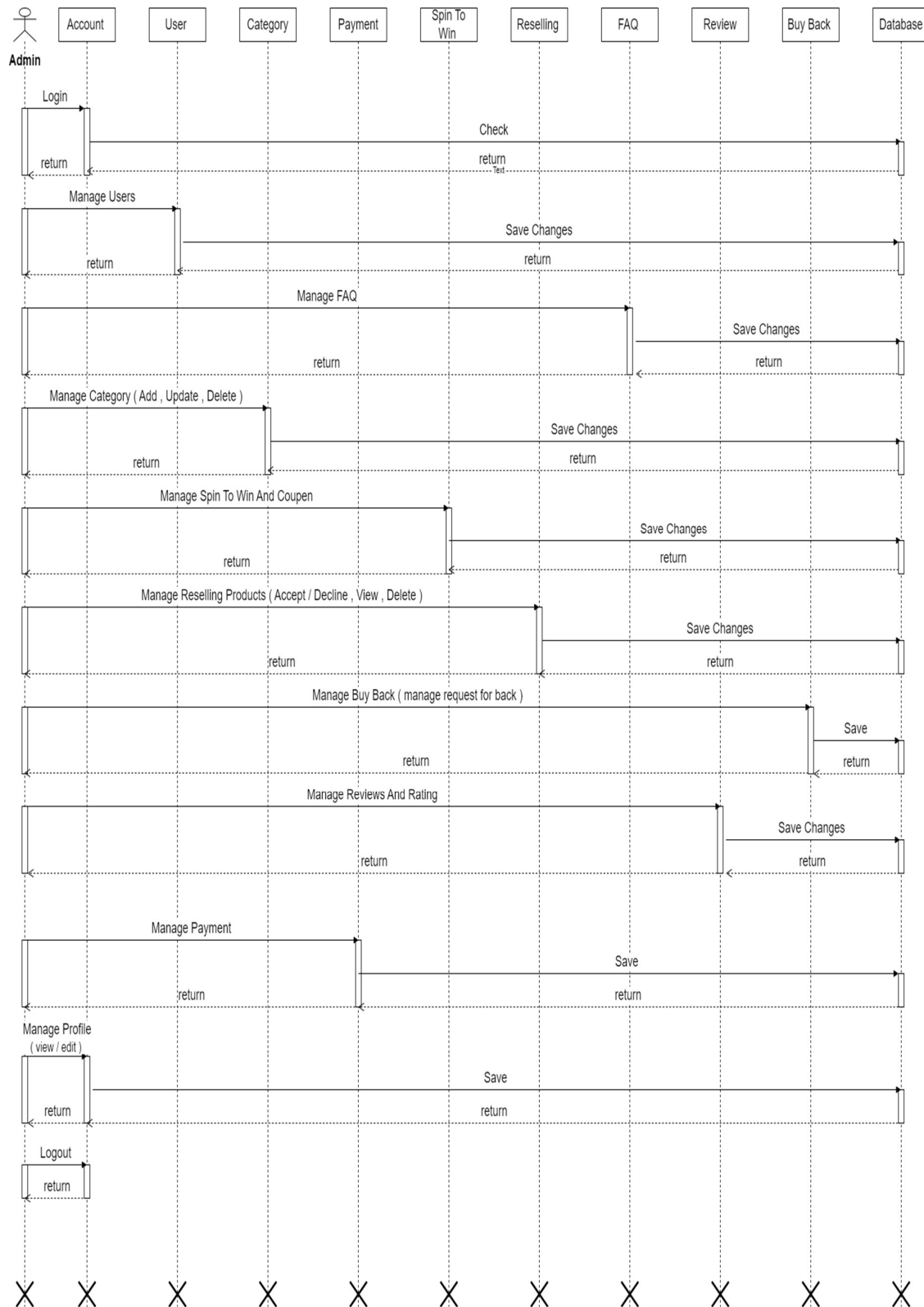


Figure 4.3 Delivery Partner Sequence Diagram

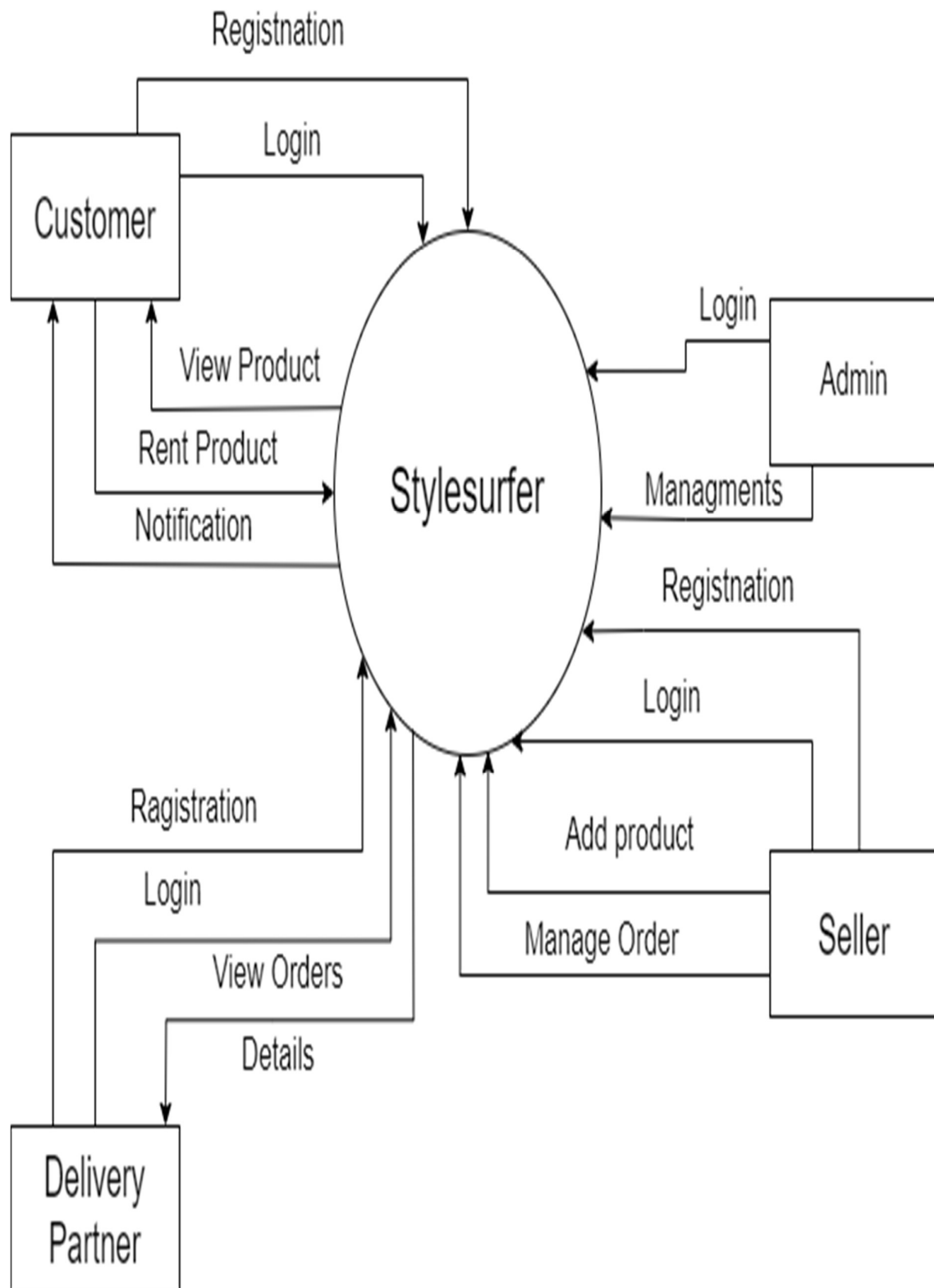
## Admin Sequence Diagram :-



**Figure 4.4 Admin Sequence Diagram**

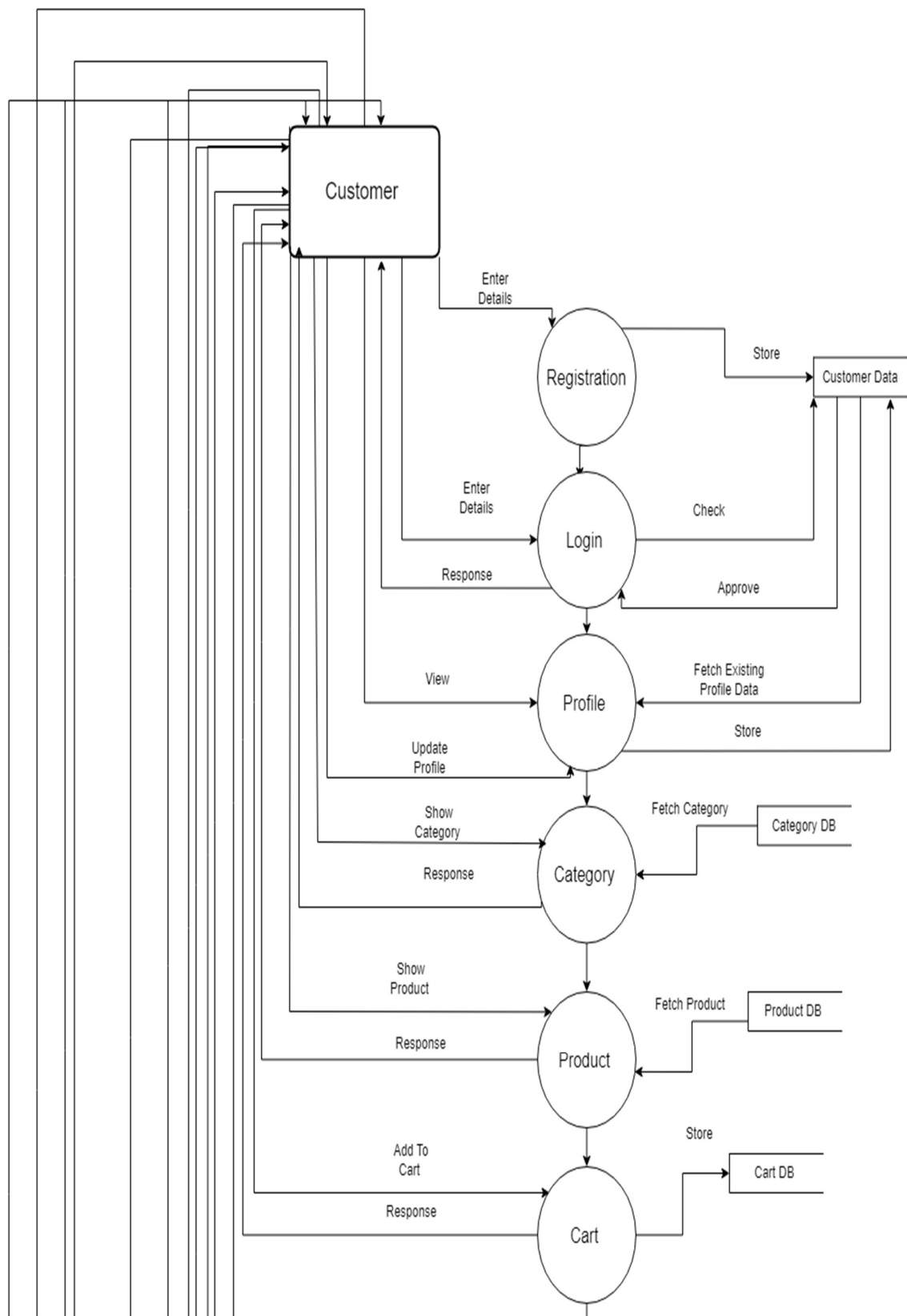


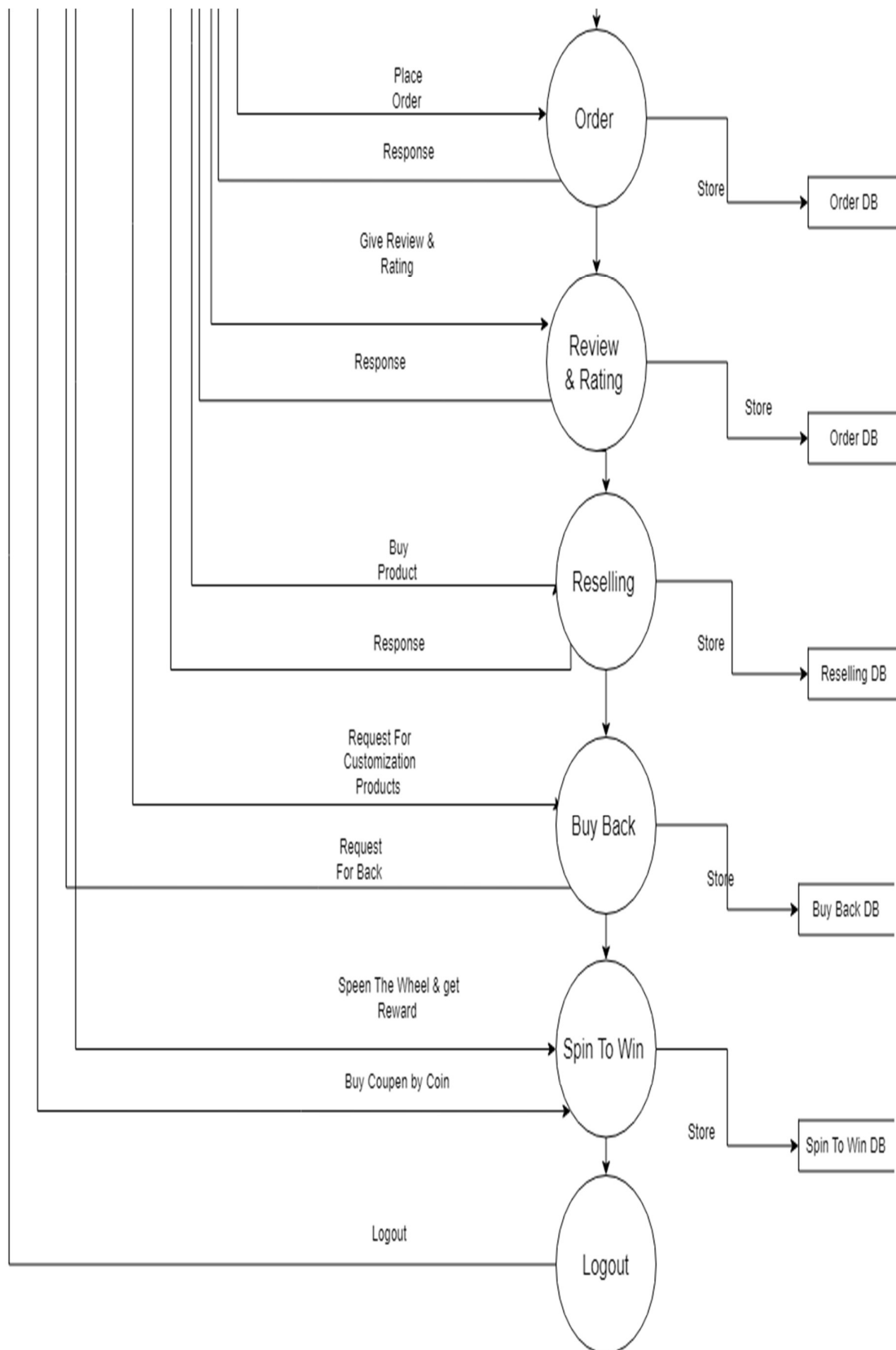
**DFD (Level 0) :-**



**Figure 5 DFD (Level 0)**

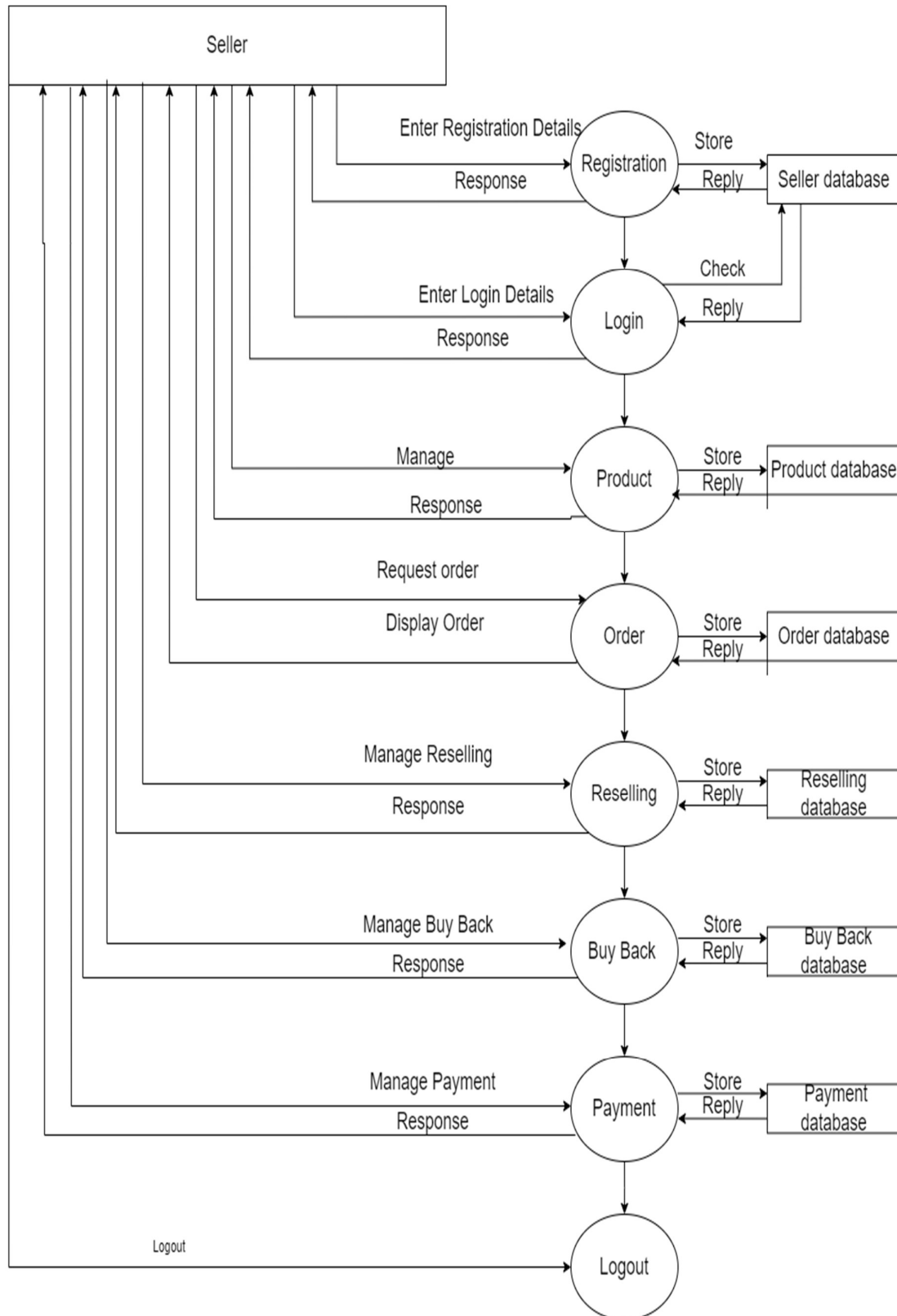
**Customer DFD (Level 1) :-**





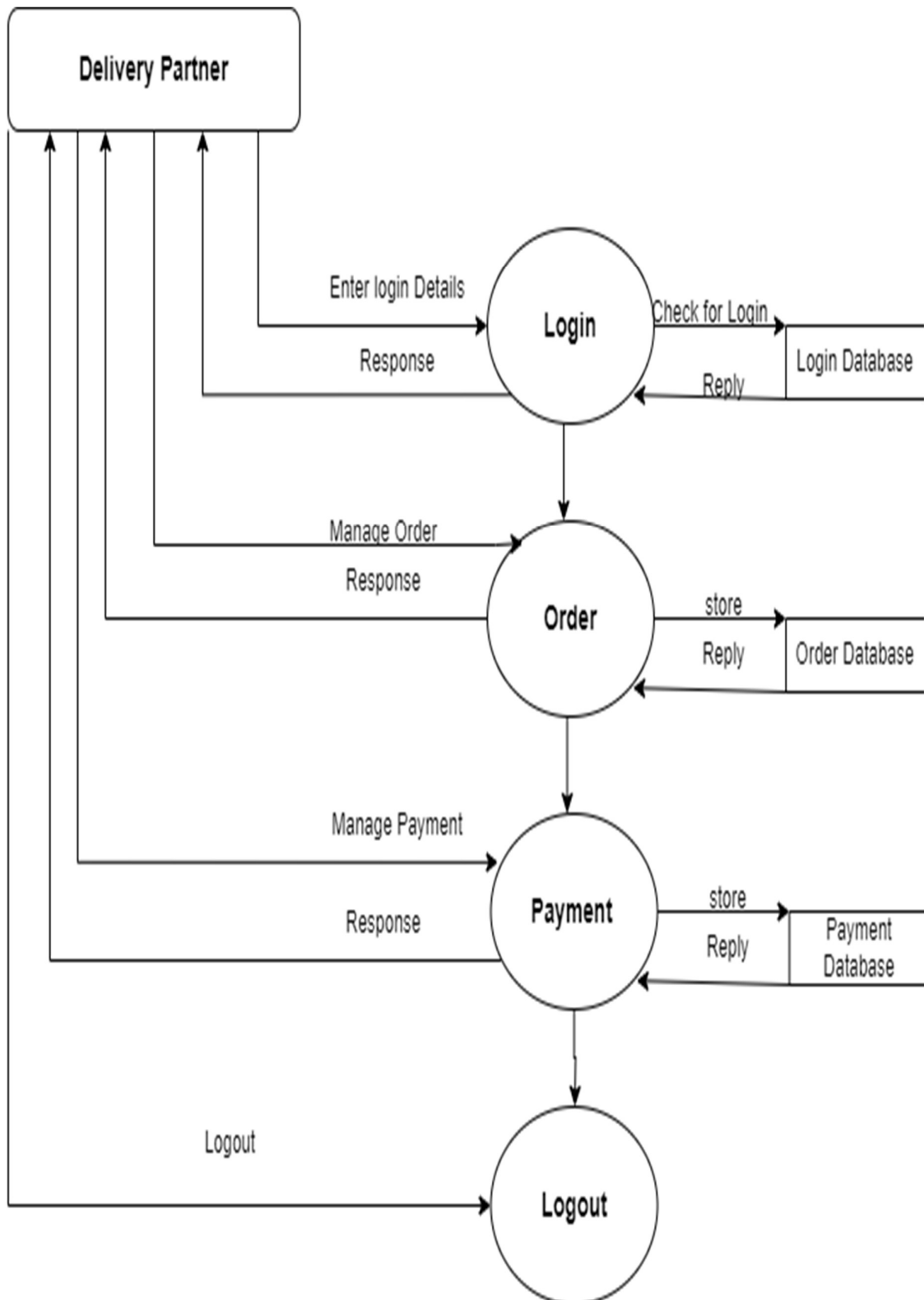
**Figure 6.1 Customer DFD (Level 1)**

### Seller DFD (Level 1) :-



**Figure 6.2 Seller DFD (Level 1)**

**Delivery Partner DFD (Level 1) :-**



**Figure 6.3 Delivery Partner DFD (Level 1)**

### Admin DFD (Level 1) :-

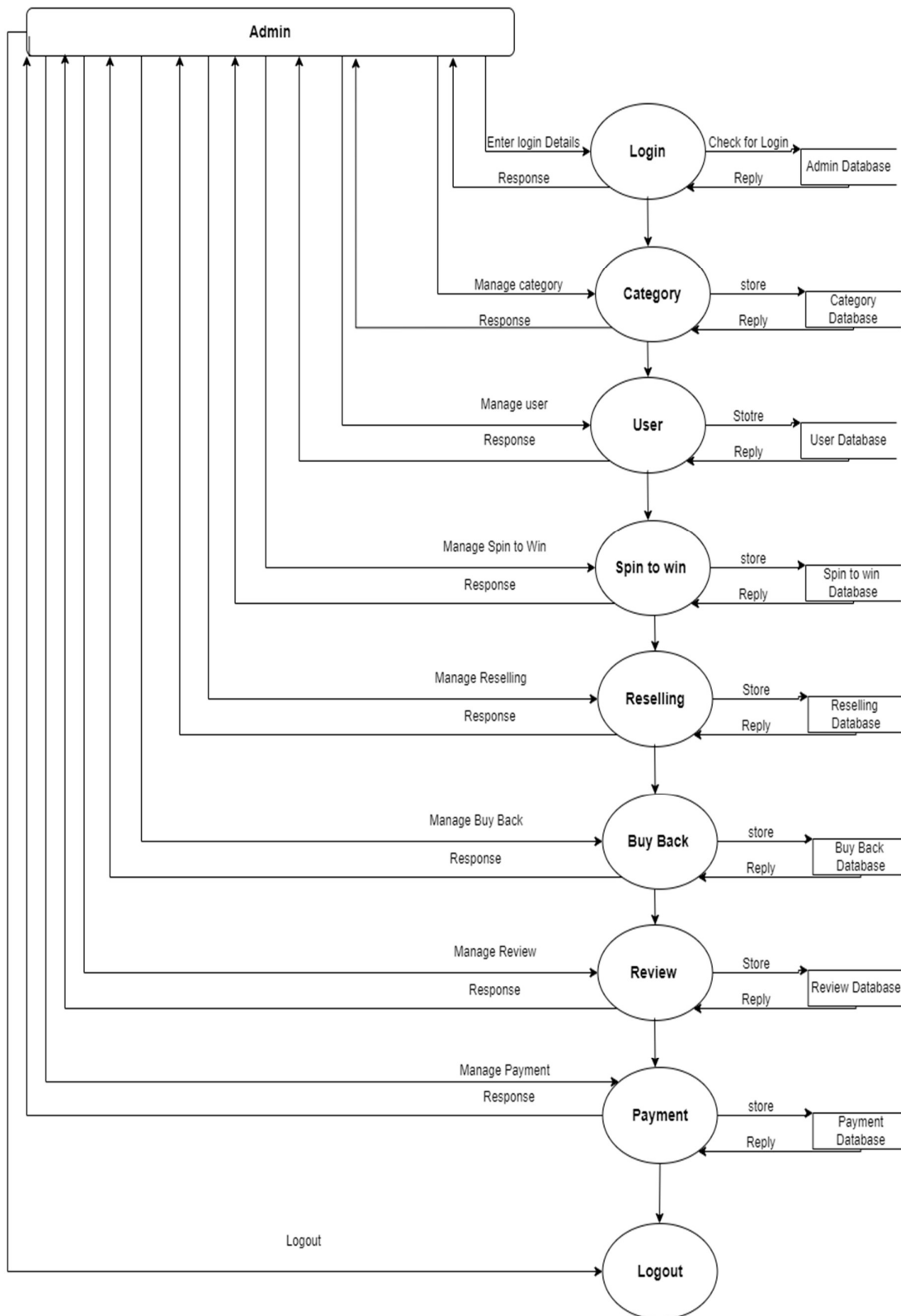
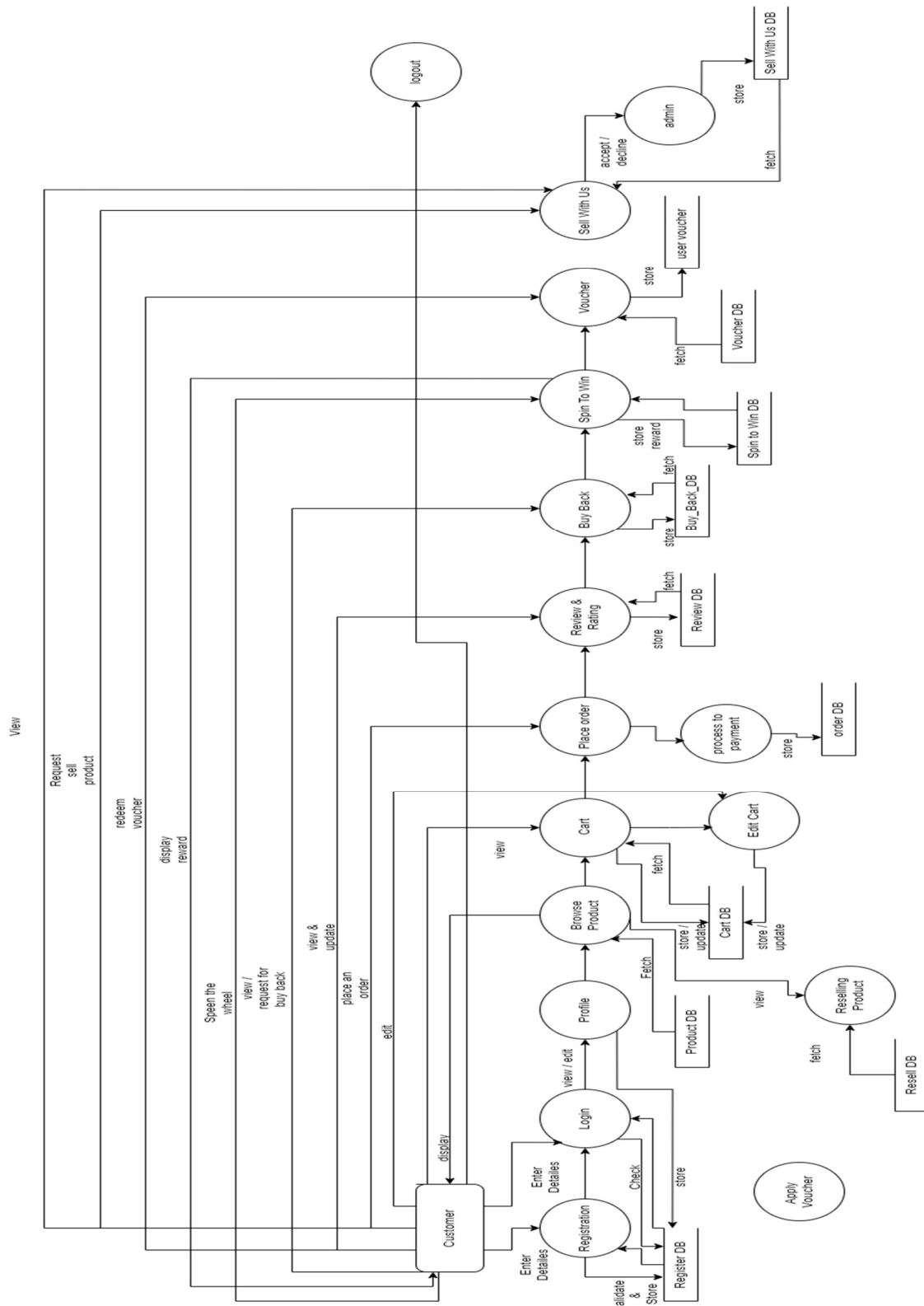


Figure 6.4 Admin DFD (Level 1)

### Customer DFD (Level 2) :-



**Figure 7.1 Customer DFD (Level 2)**

## Seller DFD (Level 2) :-

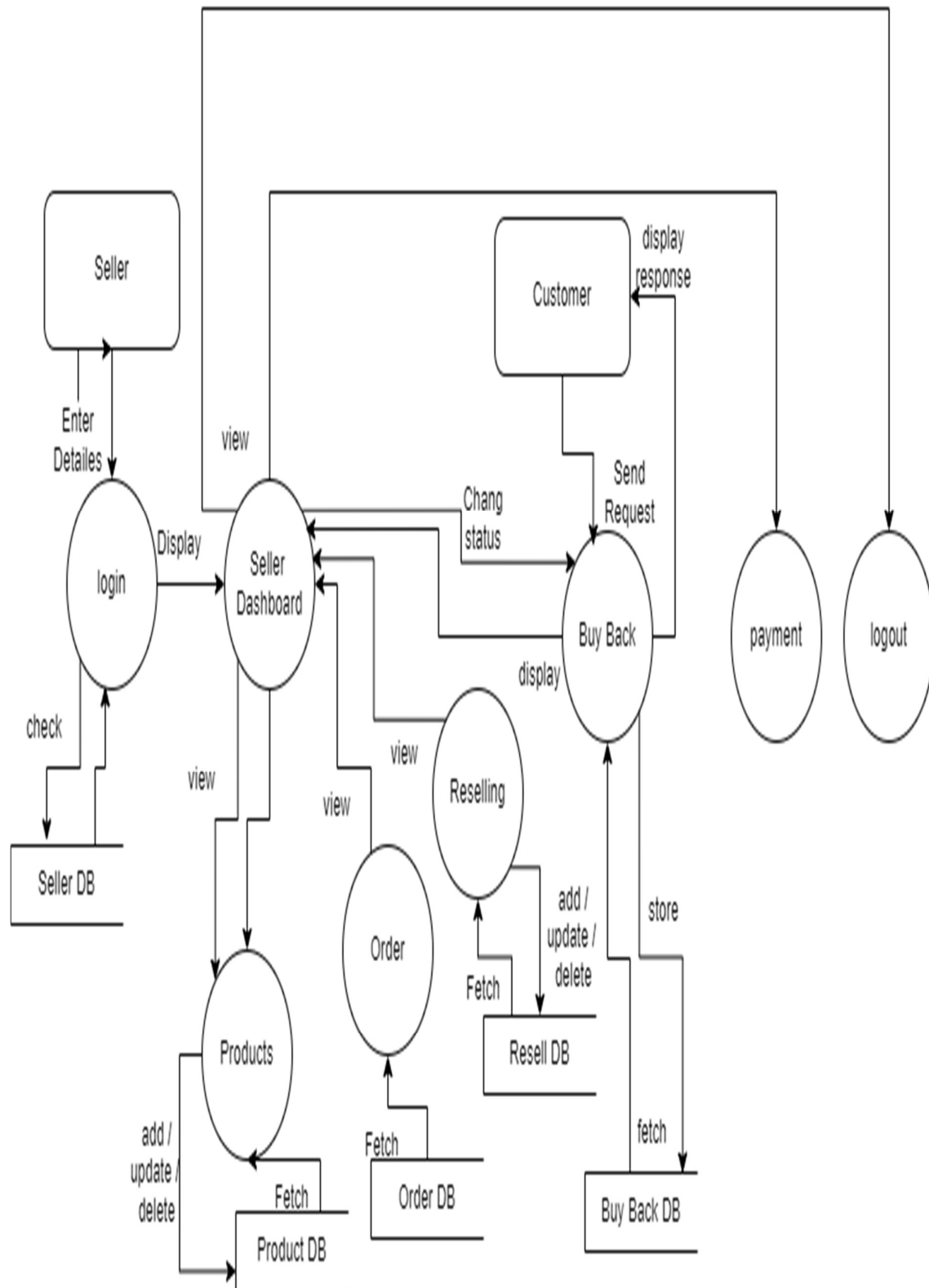
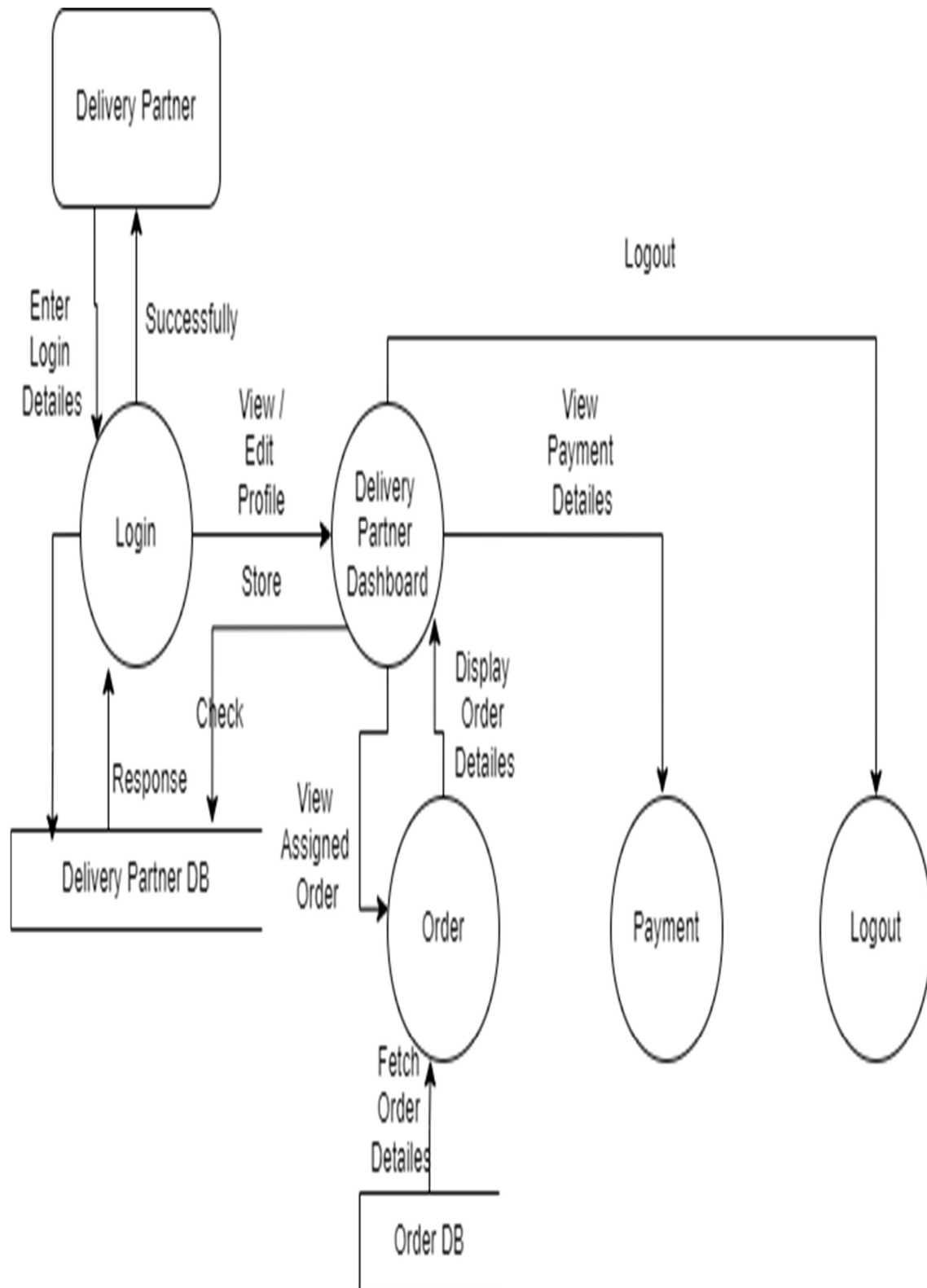


Figure 7.2 Seller DFD (Level 2)



**Delivery Partner DFD (Level 2) :-**



**Figure 7.3 Delivery Partner DFD (Level 2)**

## Admin DFD (Level 2) :-

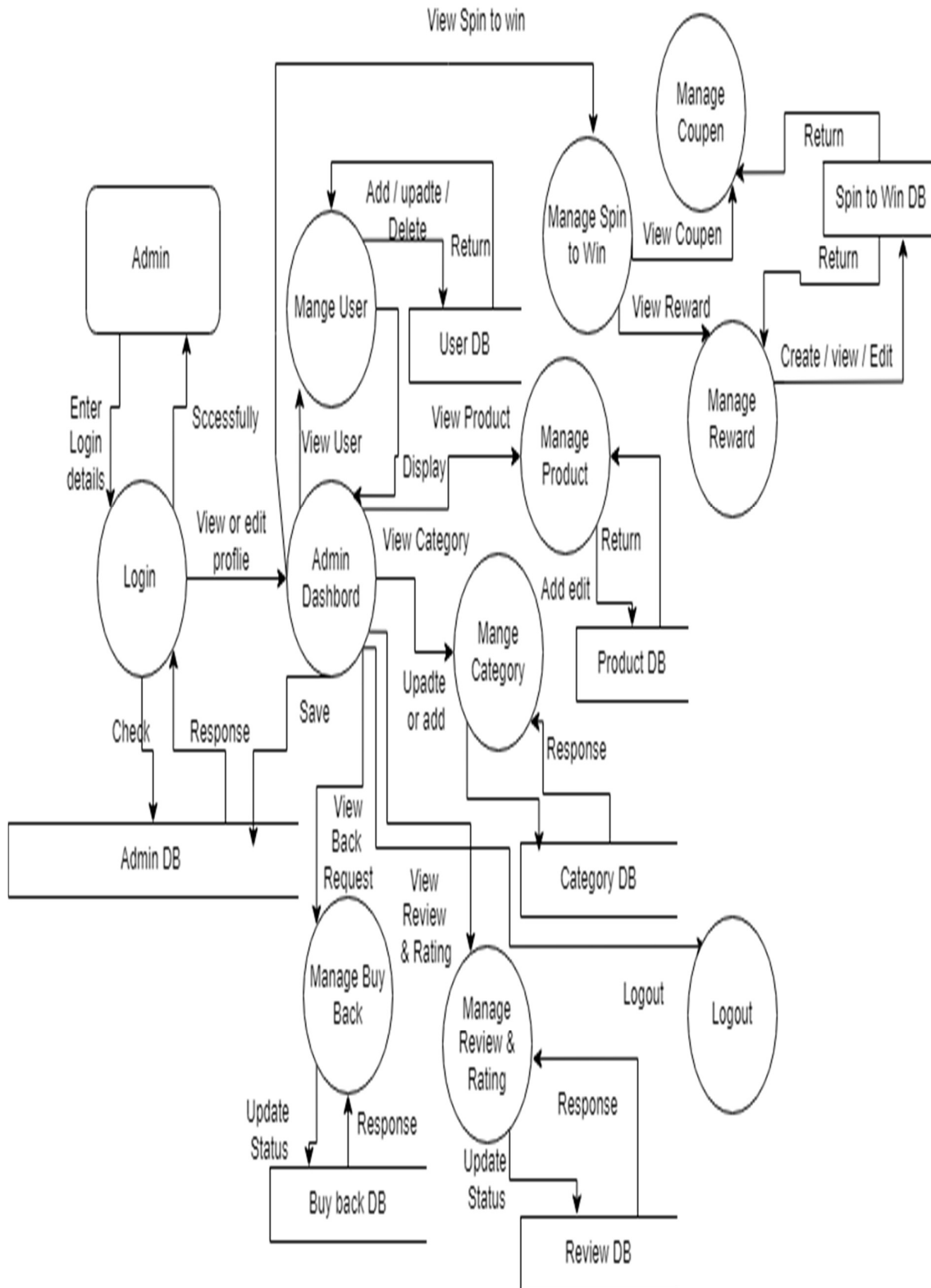
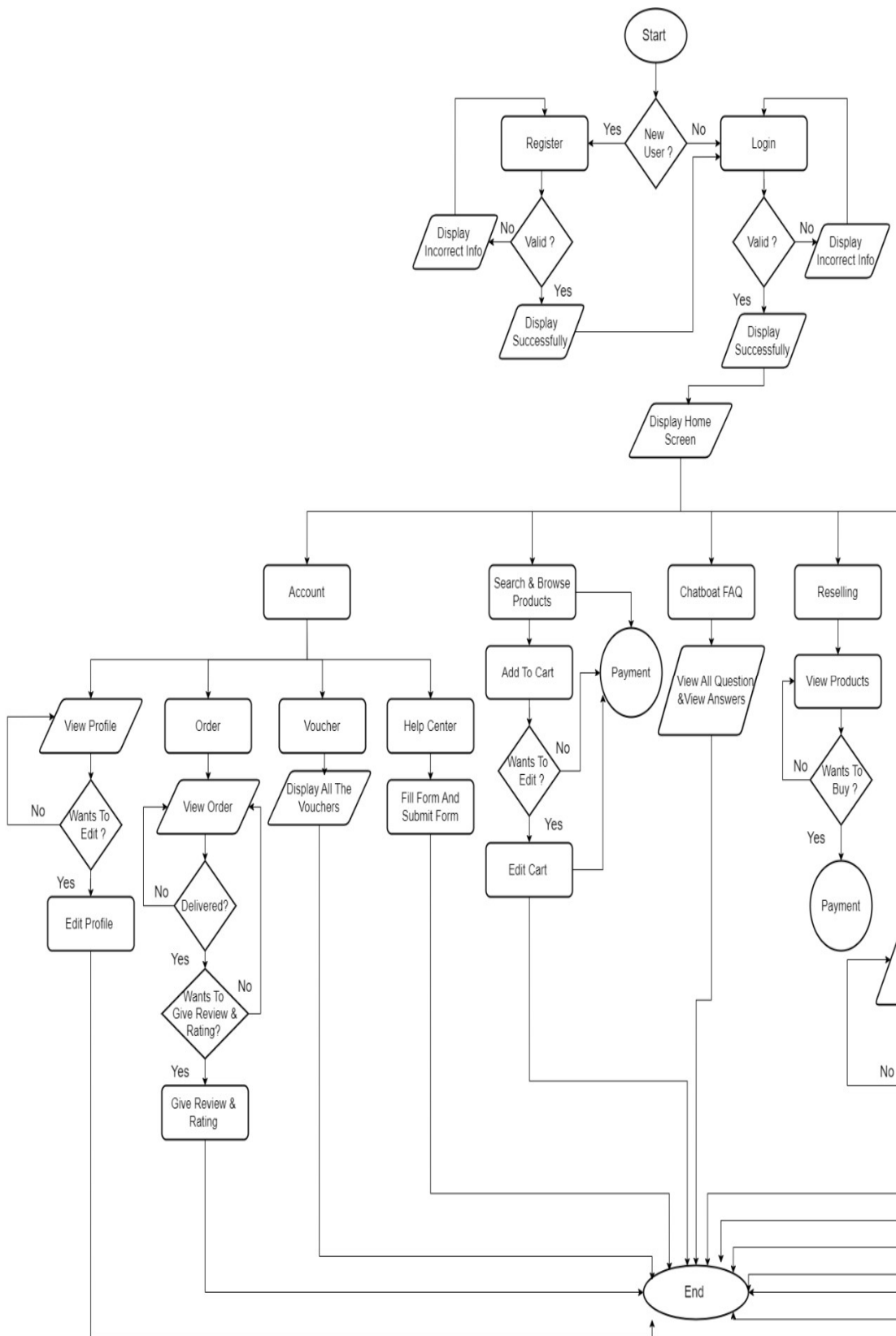
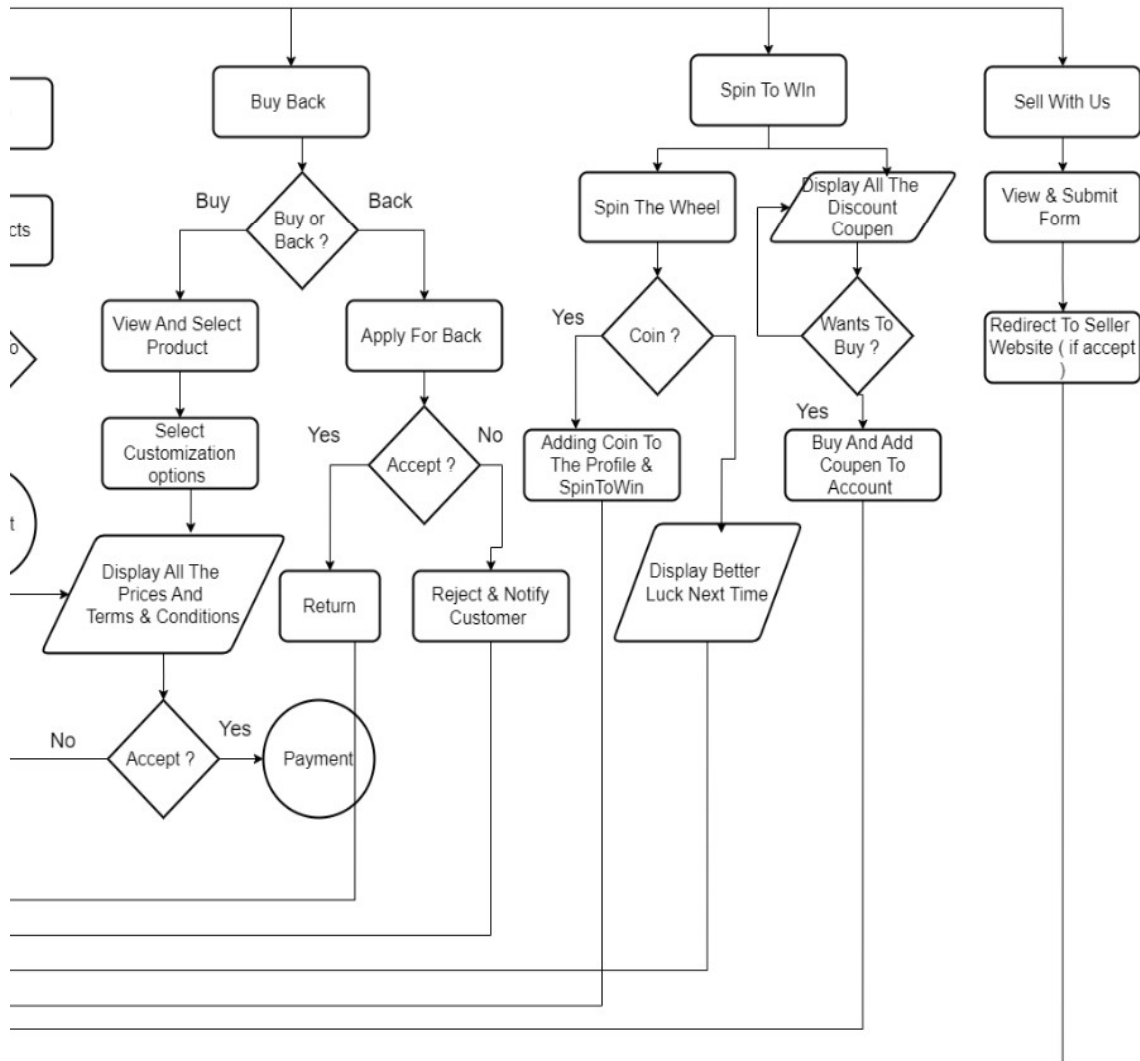
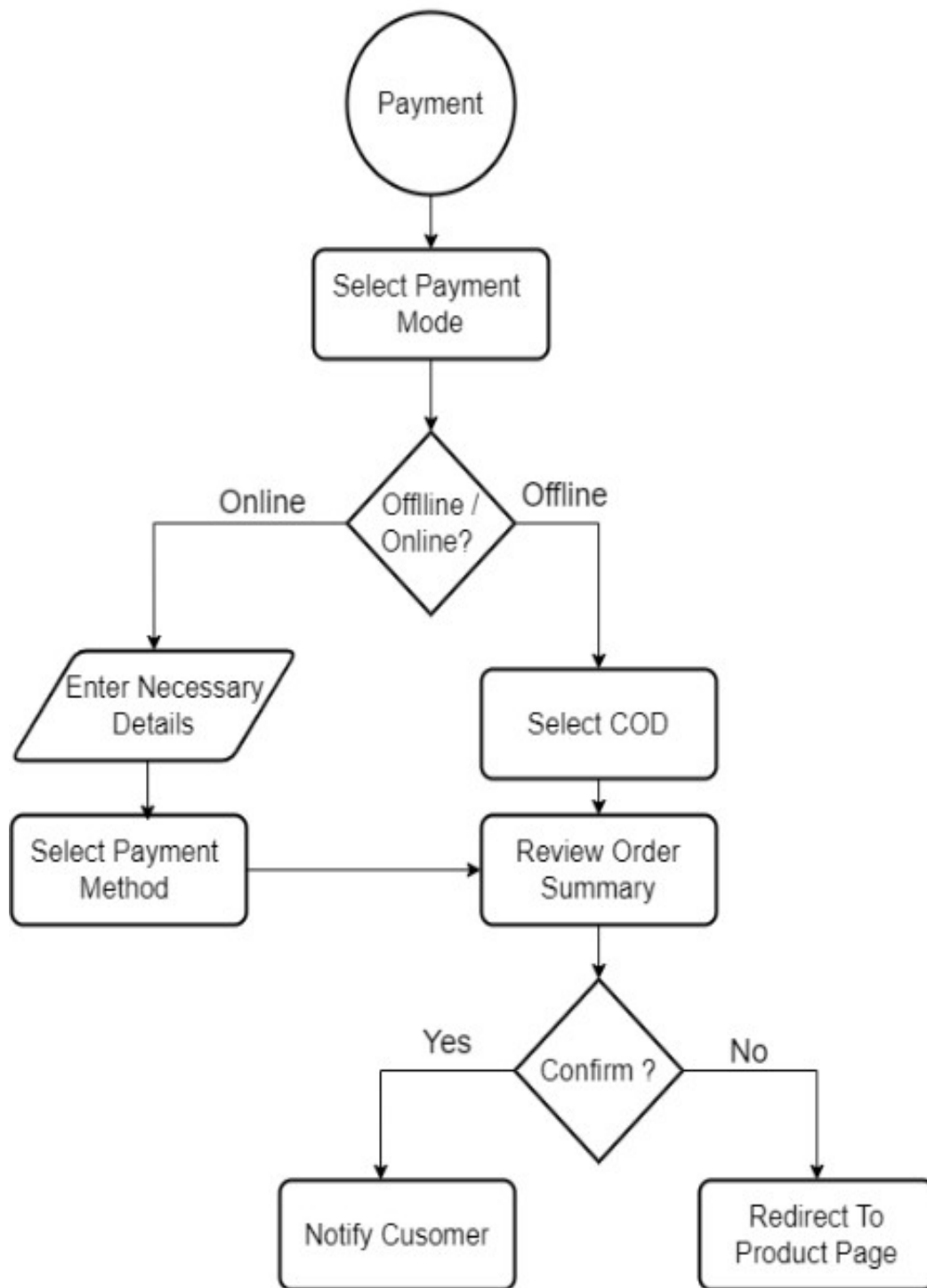


Figure 7.4 Admin DFD (Level 2)

**Customer Flowchart Diagram :-**

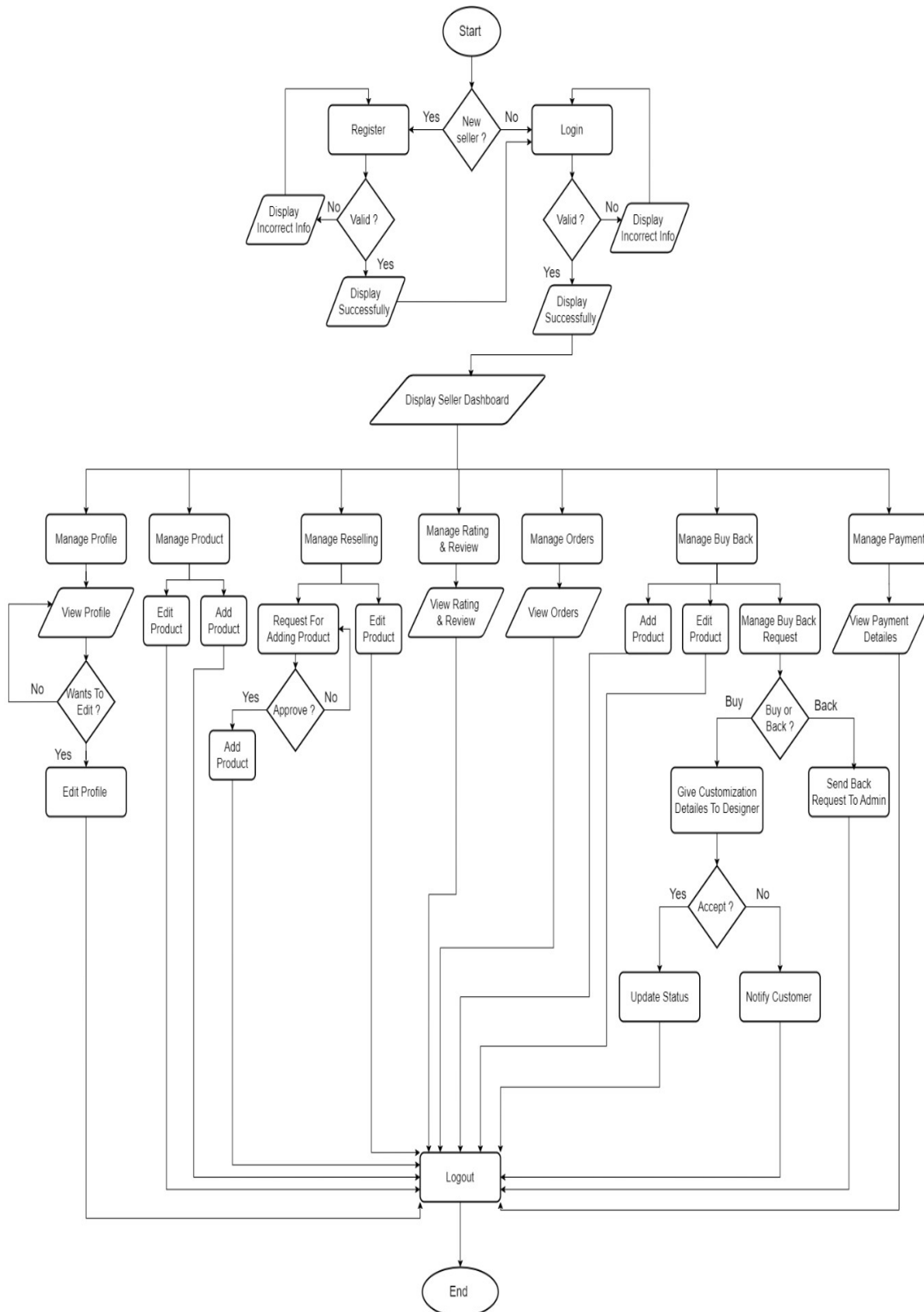






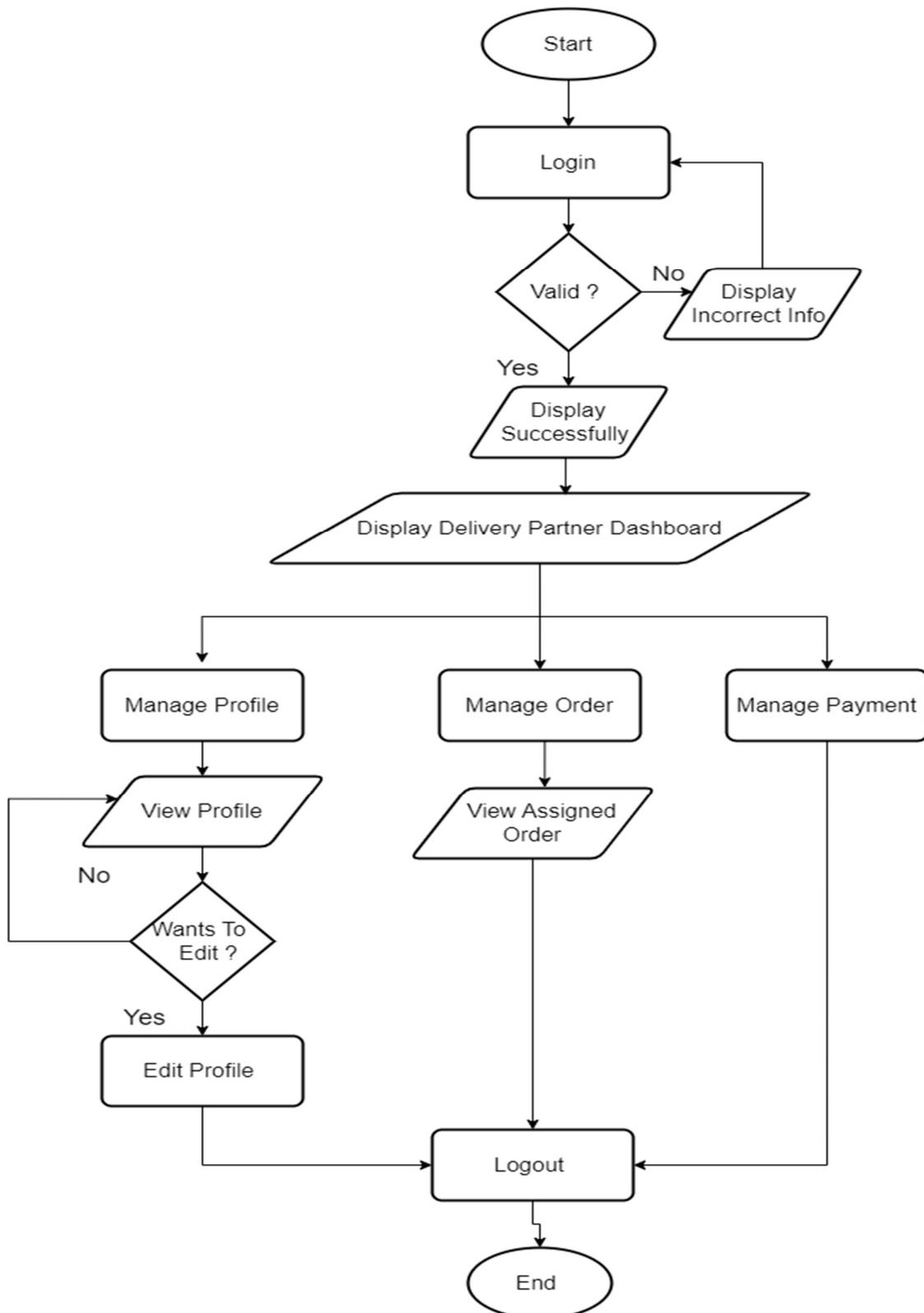
**Figure 8.1 Customer Flowchart Diagram**

## Seller Flowchart Diagram :-



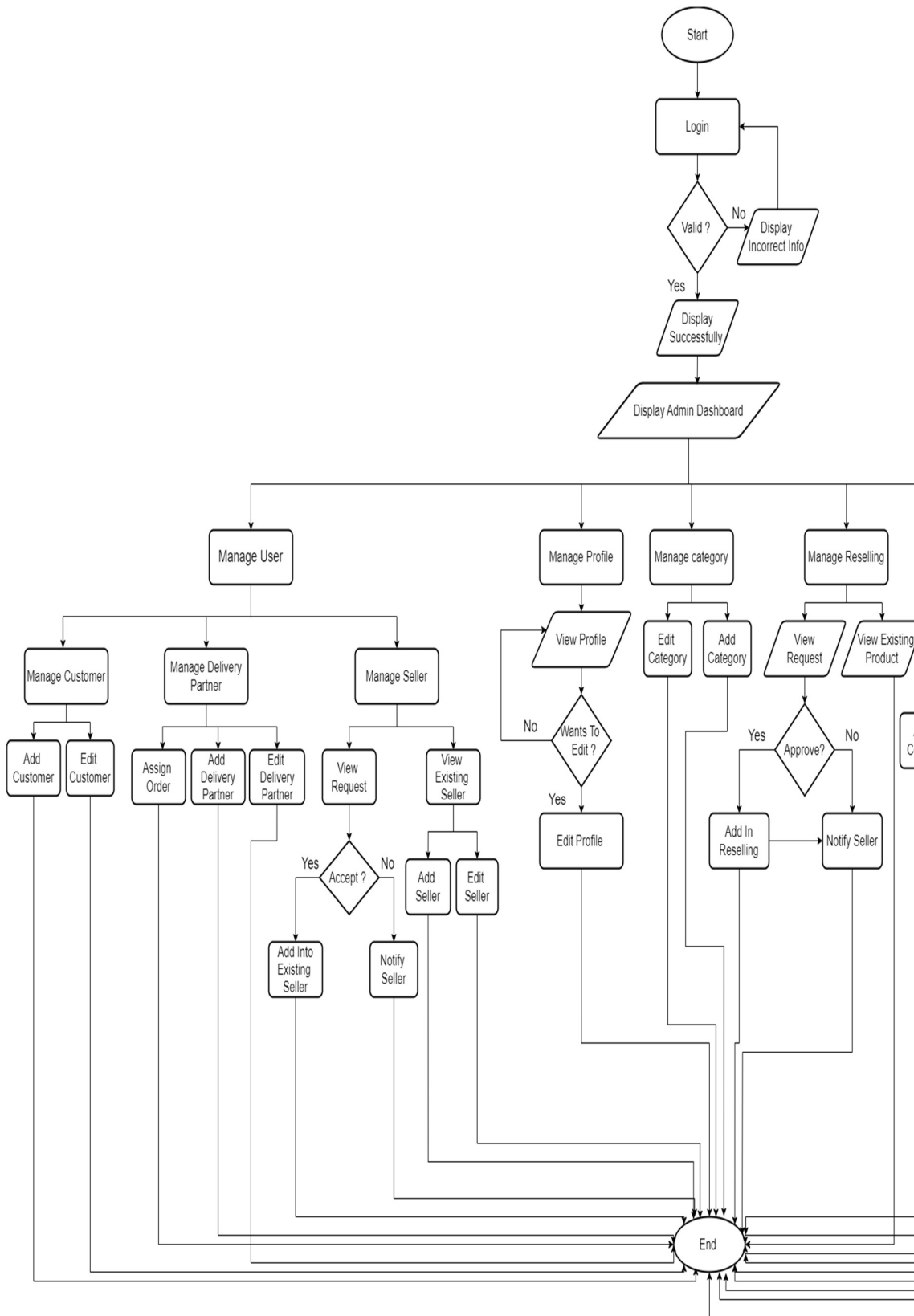
**Figure 8.2 Seller Flowchart Diagram**

**Delivery Partner Flowchart Diagram :-**

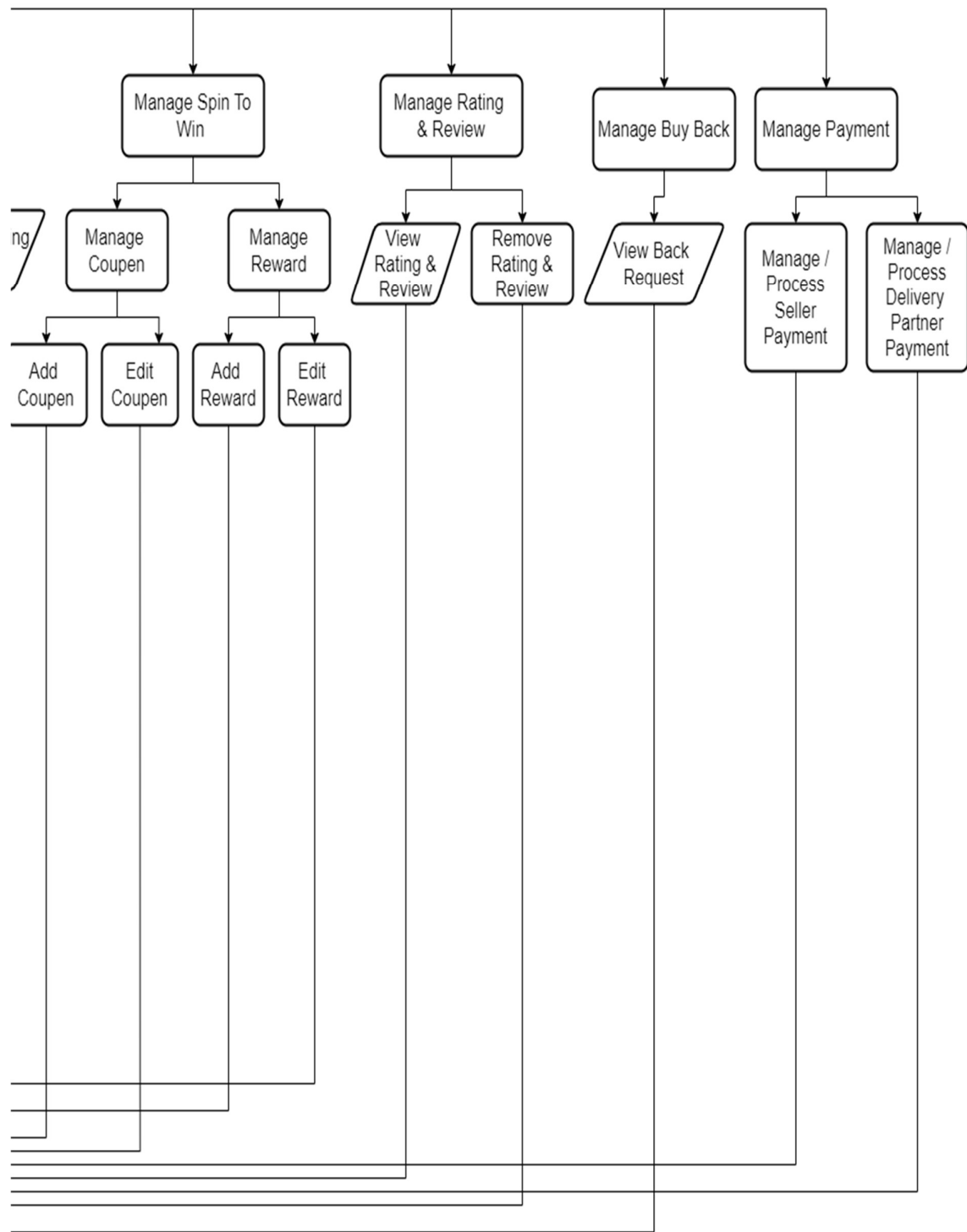


**Figure 8.3 Delivery Partner Flowchart Diagram**

# Admin Flowchart Diagram :-







**Figure 8.4 Admin Flowchart Diagram**

## Class Diagram :-

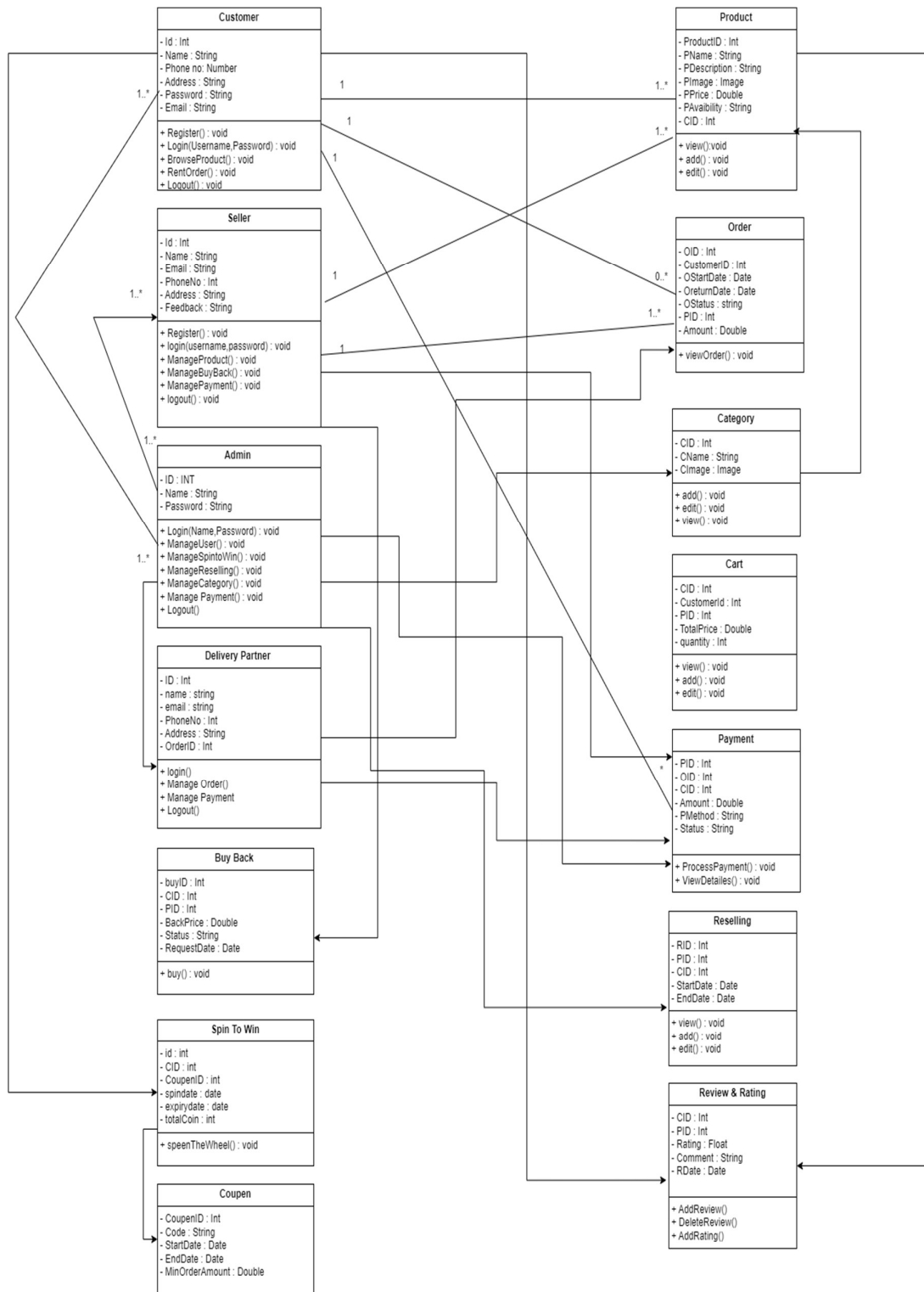


Figure 9 Class Diagram

# 1. Introduction

Our project, StyleSurfer, is a fashion rental platform that makes it simple, affordable, and eco-friendly for people to explore and rent trendy clothing. But we're not just about rentals—users can also buy and sell fashion items, helping to promote sustainable fashion. By encouraging the reuse of clothing, we aim to reduce fashion waste and create a fun, responsible shopping experience that benefits both the environment and fashion lovers.

In this section, we'll explain why this document matters and how it helps everyone from developers building the platform to customers and sellers using it. You'll get a clear breakdown of StyleSurfer's features, how it's designed to be simple and user-friendly, and the key functions that keep everything running smoothly. Whether you're a developer working behind the scenes or a business owner using the platform, this document will help you quickly find what you need.

To make things even clearer, we've included a glossary of important terms. We'll also share the references and resources that guided us in building StyleSurfer.

By the end of this section, you'll have a full understanding of what StyleSurfer is, what we're trying to achieve, and what's needed to make it successful—both technically and practically.

## 1.1 Document Purpose

This document provides all the essential details about StyleSurfer, our fashion rental platform, making it easy for everyone to understand how the system works. It covers key features, technical aspects, and the different roles users play—whether they're developers, project managers, sellers, or customers.

The goal is to give clear instructions on how to build, manage, and maintain the platform so users can easily rent, buy, or sell trendy clothes. It also explores potential future updates and improvements to make StyleSurfer even better over time. Whether you're building the platform or using it, this guide makes sure that everyone—developers and users—has what they need for a seamless and enjoyable experience.

## 1.2 Product Scope

StyleSurfer is a simple, easy-to-use fashion platform that lets people rent, buy, or sell clothes. Its main goal is to make fashion more affordable and accessible while encouraging the reuse of clothing to help the environment. Whether someone needs a stylish outfit for a special occasion or wants to earn money by selling their own clothes, StyleSurfer provides a hassle-free solution. It also makes shopping more fun with features like ‘Spin to Win’, where users can win coins and redeem the vouchers.

The platform focuses on three main groups: customers, sellers, and delivery partners. Customers can rent or buy clothes at a lower cost, giving them more flexibility in their fashion choices. Sellers can make money by renting out or selling their clothes. Delivery partners ensure that orders are delivered and returned on time. By bringing these groups together, StyleSurfer creates a smooth, enjoyable experience for fashion lovers while also promoting sustainability.

## 1.3 Intended Audience and Document Overview

Below are our intended audiences for this document, each with a description of how this guide will support them :

- Customers : People who use StyleSurfer to rent, buy, or sell fashion items. This document will help them understand how the platform works and show them how to use features that enhance their shopping experience.
- Sellers : Individuals who list their clothes for rent or sale. This document will guide them on how to manage their listings, sell items, and use the platform's tools to boost their sales or rentals.
- Delivery Partners (DP) : People responsible for delivering and returning clothing. This document explains their role and how they ensure the delivery process runs smoothly and efficiently.
- Testers : People who test the platform for bugs or issues. This document helps them understand how the system should work and highlights the features they need to test.

- **Developers** : Individuals responsible for coding and building the platform. This document provides them with technical specifications, functional and non-functional requirements, and system designs needed to develop the software.

The rest of this document is organized into several sections. It starts with an overview of the project to give a broad understanding. Then, it goes into more detailed sections about the platform's features, performance, user scenarios, design, and testing plans. Each group can focus on the sections most relevant to their role. For example, customers and sellers can read the user scenarios, while developers and testers will find the technical details useful.

## 1.4 Definitions, and Abbreviations

**Table 1.4.1 Definitions & Abbreviations**

| Abbreviation | Definition                                                   |
|--------------|--------------------------------------------------------------|
| CSS          | Cascading Style Sheets                                       |
| CRUD         | Create, Read, Update, Delete (operations for managing data)  |
| CDN          | Content Delivery Network (for optimizing website speed)      |
| DP           | Delivery Partner                                             |
| FAQ          | Frequently Asked Questions                                   |
| HTML         | Hypertext Markup Language                                    |
| JS           | JavaScript                                                   |
| MVT          | Model-View-Template (architecture used in Django)            |
| OTP          | One-Time Password (for user authentication)                  |
| SEO          | Search Engine Optimization (for enhancing online visibility) |
| SRS          | Software Requirements Specification                          |
| UI / UX      | User Interface / User Experience                             |

## 1.5 References and Acknowledgments

[1] Rent An Attire, "Rent An Attire – Premium Fashion Rental,"

Available: <https://www.rentanattire.com/>

[2] The Stylease, "The Stylease – Luxury Fashion Rental,"

Available: <https://www.thestylease.com/>

[3] Date The Ramp, “Date The Ramp – Designer Outfit Rentals,”

Available: <https://www.datetheramp.com/>

[4] Rent The Runway, “Rent The Runway – Fashion Rental Platform,”

Available: <https://www.renttherunway.com/>

[5] User Interface Style Guide,

Available: <https://example.com/ui-style-guide>

[6] Bootstrap Official Documentation,

Available: <https://getbootstrap.com/docs/>

[7] Python Official Documentation,

Available: <https://docs.python.org/3/>

[8] Django Official Documentation,

Available: <https://docs.djangoproject.com/en/stable/>

## **2. Overall Description**

### **2.1 Product Perspective**

StyleSurfer is an exciting new platform designed to transform the way people rent, buy, and sell clothes. It's not just another version of an existing system—it's built to meet the growing demand for sustainable fashion. By making it easy for users to rent stylish outfits, buy pre-loved fashion, or sell their own clothes, StyleSurfer encourages eco-friendly shopping while ensuring a smooth experience for everyone involved.

The platform is designed for three main user groups :

- Customers : People who want access to a wide variety of fashion items. They can rent outfits for special occasions or buy trendy clothes at affordable prices.
- Sellers : Individuals, boutiques, or designers who want to rent out or sell clothing. StyleSurfer provides them with powerful tools to create listings, set pricing, track orders, and manage their fashion business efficiently.
- Delivery Partners (DPs) : The people responsible for delivering and picking up clothing. They ensure that rentals and purchases are handled efficiently, making the entire process hassle-free.

### **2.2 Product Functionality**

Below are the products' functionalities on StyleSurfer :

- Add to Cart :
  - Functionality : Users can add selected products to their shopping cart for future purchases.
  - Explanation : This feature makes shopping easier by allowing users to collect multiple items in one place before checking out.

➤ Spin to Win :

- Functionality : Users can participate in a gamified experience by spinning a wheel to win rewards, such as coins or no prize.
- Explanation : The feature includes a Coupon Page, where users can redeem their earned coins to purchase various coupons, making shopping more engaging and fun.

➤ Reselling :

- Functionality : Sellers can list rental products for direct purchase by customers for a limited time.
- Explanation : This helps sellers manage their inventory while giving customers the opportunity to buy products they like at discounted prices.

➤ Sell with Us :

- Functionality : Customers who own a small shop or boutique can apply to rent out their products on the platform. Unlike regular sellers, they do not have access to the seller dashboard but can still manage their listed products and view orders from their account section.
- Explanation : This feature provides an opportunity for small business owners to reach a wider audience without needing a full seller account, making it easier for them to rent out their fashion items.

➤ Buy-Back Option :

- Functionality :
  - Buy Option : Customers can request customized fashion items and view the total cost before purchasing.
  - Back Option : Customers can return their customized products to be rented out on the platform.



- Explanation : This feature provides flexibility, allowing customers to either buy personalized items or rent them out, making fashion more sustainable and engaging.
- Payment Processing :
  - Functionality : Secure payment options ensure safe and smooth transactions.
  - Explanation : This feature builds user trust by securely handling financial information and ensuring seamless payment processing.

## 2.3 Users and Characteristics

Users Identification :

- Customers
- Sellers
- Delivery Partners
- Admins (Site Administrators)

Users Characteristics :

- Customers :
  - Frequency of Use : Regular shoppers who often rent fashion items and enjoy promotions.
  - Functions Used : Browsing clothing options, adding items to the cart, playing the Spin-to-Win game, and completing rental orders.
  - Technical Expertise : Varies; some are very comfortable with online shopping, while others may need help.

- Security Level : Basic access to manage personal information and view rental history.
- Sellers :
  - Frequency of Use : Active users who frequently list clothes for rent and manage their inventory.
  - Functions Used : Listing fashion items, tracking available stock, and checking sales data on the seller dashboard.
  - Technical Expertise : Needs moderate tech skills to manage listings and inventory effectively.
  - Security Level : Higher access than customers, allowing them to manage products and view sales information.
- Delivery Partners :
  - Frequency of Use : Regularly deliver clothing items to customers.
  - Functions Used : Manage their schedules and communicate with customers about orders.
  - Technical Expertise : Basic tech skills are sufficient for their role.
  - Security Level : Limited access, mostly related to delivery tasks.
- Admins :
  - Frequency of Use : Use the platform daily to manage everything and ensure smooth operations.
  - Functions Used : Admins can access all system parts, including managing users, sellers, delivery partners (DP), and viewing analytics.

- Technical Expertise : High technical skills are needed to troubleshoot issues and maintain website functionality.

Distinction Of Important Users :

➤ Most Important Users :

- Customers : The primary users of StyleSurfer who directly influence rental sales and overall revenue. Their satisfaction is key to the platform's success.
- Sellers : They provide the clothing for rent and are essential to offering variety. Their satisfaction and engagement are crucial for a successful marketplace.

➤ Less Important Users :

- Delivery Partners : While important for delivering items, their satisfaction is less critical compared to customers and sellers.
- Admins : While essential for managing the platform, their satisfaction doesn't directly affect the user experience. Their role is necessary for smooth operations but not as crucial as keeping customers and sellers happy.

## 2.4 Operating Environment

The StyleSurfer fashion rental website is designed to provide a smooth experience across both desktop and mobile devices. Users can access the website through modern web browsers, making it compatible with various platforms.

Here are the minimum hardware and software requirements for the website to work efficiently

➤ Hardware Requirements :

- Desktops/Laptops : A computer with at least an Intel i3 processor (or similar) and 4 GB of RAM.
- Mobile Devices : Any modern smartphone or tablet with at least 3 GB of RAM.

- Operating Systems :
  - Desktops/Laptops :
    - Windows 10 or newer
    - macOS 12 or newer
    - Any Linux distribution that supports modern browsers.
  - Mobile Devices :
    - iOS 12 or newer
- Software Stack :
  - Backend : The website will use Python with the Django framework for the server-side processing.
  - Frontend : The design will be built with HTML5, CSS, JavaScript, and Bootstrap to ensure it is responsive and user-friendly.
  - Database : SQLite3 will handle user data and product listings efficiently.

## 2.5 Design and Implementation Constraints

Constraints of StyleSurfer website are described below :

- Hardware Limitations :
  - Memory Requirements : StyleSurfer is a fashion rental website designed to work on different devices like smartphones and computers. To ensure smooth performance, the website should work well on devices with at least 3GB of RAM. This ensures that even users with less powerful devices can browse without issues.
- Database & Technology Choices :
  - Database Selection : StyleSurfer will use SQLite3 to store data about users, products, orders, and rewards. This database is simple and efficient but has some limitations—complex features that bigger databases support may not be available. The team needs to ensure that data is well-organized and retrieved efficiently.

- Framework & Tool Limitations :
  - Django Framework Constraints : Since StyleSurfer is built using Django, the development must follow Django's Model-View-Template (MVT) structure. This means all features, like user logins, rentals, and payments, must be built within Django's system. While Django is powerful for security and scalability, some features like real-time chat may be harder to implement since Django doesn't have built-in support for them.
- Security Measures :
  - User Data Protection : The website will handle sensitive information like customer addresses and payment details, so security is a top priority. To protect users, the team must use secure logins (such as two-factor authentication) and encrypt passwords and payment details. However, adding these security features will require extra development effort to ensure user safety.
- Coding & Design Standards :
  - Clean & Organized Code : Developers must follow best practices for coding, including Django's backend rules and modern front-end standards using HTML, CSS, JavaScript, and Bootstrap. This means , Using clear and meaningful names for variables and functions. Writing clean and well-documented code so future developers can easily understand and update it. Keeping the structure organized so the project remains easy to maintain over time.

## **2.6 User Documentation**

StyleSurfer will provide user-friendly documentation to ensure a smooth experience for all users. A downloadable PDF manual will be available, allowing users to access clear, step-by-step instructions on signing up, browsing products, adding items to the cart, making secure payments, and managing their accounts. Additionally, a built-in FAQ section on the website will offer quick answers to common questions, such as setting up an account, available payment options, and view orders. These resources will help users navigate the platform effortlessly, making their shopping and rental experience simple and hassle-free.

## 2.7 Assumptions and Dependencies

Major Assumptions :

- User Accounts : We expect a large number of users to sign up on StyleSurfer when we launch. This means we need to design our database to handle and manage user information efficiently.
- Product Listings : We plan to start with a wide range of products available for rent. This will influence how we organize, display, and manage these listings in our system.
- Payment Methods : We'll offer at least two payment options—online and offline. This affects how we design the checkout process and the security measures we need to implement.
- Delivery Partners : We anticipate working with five delivery partners to handle product deliveries. This will shape how we manage logistics and ensure smooth order fulfillment.
- User Engagement : We estimate that around 20% of users will participate in promotional activities like the “Spin to Win” game. This means we need to carefully design and integrate the rewards system.

Dependencies :

- Database Management System : Our project will rely on a specific database system (yet to be decided). Any limitations or issues with the database could impact how we store and retrieve data.
- Django Framework : Since we're using Django for the backend, any updates or changes to Django could require adjustments to our implementation.

- Team Skills : We assume that everyone on the team is comfortable with the technologies we're using (HTML, CSS, JavaScript, Django). If there are any gaps in knowledge, we might face some development challenges.

## 3. Specific Requirements

### 3.1 External Interface Requirements

#### 3.1.1 User Interfaces

Below are the user interfaces of StyleSurfer :

➤ Home Page :

- Description : The home page of StyleSurfer will have a clean and easy-to-use design. It will showcase the latest and most popular clothes available for rent or purchase. The homepage will include a navigation bar and other features to help users browse the website effortlessly.
- Key Elements :
  - Navigation Bar : This will have options for Login/Signup, Account, and Cart. The cart , account and logout icon will only appear if the user is logged in. If not, it will be hidden to keep the page simple.
  - Banner : A large banner will showcase special offers, seasonal trends, and popular clothing items.
  - Search Bar : A tool for users to quickly search for specific clothing by categories like style, brand, or type.
  - Quick Links : Links to important sections like the Browse Collections, reselling and other personalized pages.

➤ Product Detail Page :

- Description : Clicking on a product takes the user to a dedicated page with complete details about that item.
- Key Elements :
  - Product Descriptions : Includes information about fabric, available sizes, rental duration, and other key details.
  - Rent Now Option : A button to initiate the rental process.



- Security Deposit & Rental Duration : Displays the required security deposit and lets users select the rental period and start date.
  - Customer Reviews and Ratings : Displays feedback from previous renters to help users make informed decisions.
- Seller Dashboard :
- Description : The seller dashboard gives sellers all the necessary tools to manage their products, view orders, and handle payments.
  - Key Elements :
    - Manage Products : Allows sellers to add, edit, or remove listings and track product rentals.
    - Manage Payments : Enables sellers to view earnings .
    - Buy-Back Requests : Allows sellers to manage customer requests for customized products by approving or rejecting them.
- Delivery Partner Dashboard :
- Description : The delivery partner dashboard provides an easy-to-use interface for managing deliveries and tracking earnings. It helps delivery partners stay organized by allowing them to view and manage orders efficiently.
  - Key Elements :
    - Manage Orders : Delivery partners can view all assigned orders, update delivery status (e.g., Out for Delivery, Delivered, Returned), and manage return pickups.
    - Order Details : Displays important information such as customer details, delivery address and product return schedules.
- Admin Panel :
- Description : The admin panel is where StyleSurfer administrators oversee the platform, manage users, and ensure smooth operations.

- Key Elements :
  - Manage Users : Admins can oversee customer, seller, and delivery partner accounts.
  - Dispute Management : A section for admins to resolve user-related issues and disputes.

### **3.1.2 Hardware Interfaces**

This are Hardware Interfaces :

- Client-Side Devices :
  - Supported Devices : StyleSurfer will work on desktops, laptops, and smartphones. It supports major operating systems like Android, iOS, Windows, macOS, and Linux.
  - Interaction Type : Users can interact with StyleSurfer via touchscreens (for mobile devices) or keyboards (for desktops and laptops). The site is designed to run smoothly whether users are tapping on a phone or clicking with a mouse on a computer.

### **3.1.3 Software Interfaces**

Below are software interfaces :

- Operating System Interface :
  - StyleSurfer is platform-independent, meaning it works on any device that can run a modern web browser, like those with Windows, macOS, Linux, Android, or iOS.
  - Browser Compatibility : The platform will be compatible with popular browsers like Google Chrome, Firefox. This ensures a smooth experience for all users, regardless of their device.
- Database :
  - StyleSurfer will use SQLite3 as its database. This will store important data like user accounts, product details, order and cart details and etc...

➤ Libraries and Frameworks :

- Django (Python Framework) : Django will handle all the background processes, like user requests, database management, and serving web pages.
- Bootstrap : This will be used to make the website responsive, ensuring that it looks good and works well on all devices, including phones and computers.
- JavaScript Libraries (e.g., jQuery) : JavaScript will be used to add dynamic features like the 'Spin to Win' game and cart updates in quantity and validation (client side).

### 3.1.4 Communications Interfaces

StyleSurfer will use SMTP for email communications, including account verification, order updates, and promotions. The platform will operate over HTTP for web interactions and data exchange. The system will ensure smooth data transfer and synchronization across all user interactions.

## 3.2 Functional Requirements

Below are the functional requirements of StyleSurfer :

➤ Search and Browse :

- Functionality : Users can search for specific products or explore different categories like clothing type, brand, size, color, and price.
- Explanation : This feature helps users quickly find exactly what they're looking for. Whether they know the specific product they want or just want to browse through options, this makes the shopping process faster and easier.

➤ Registration :

- Functionality : New users can create an account by providing basic details like their name, email address, and phone number and also verify email using OTP verification.

- Explanation : Once registered, users can access their accounts to manage their orders, rentals, personal details, and more.
- Login :
- Functionality : Registered users can log in using their account details.
  - Explanation : Logging in gives users access to personalized features like their shopping cart, previous orders, and any rewards they've earned. This makes the shopping experience more customized and convenient.
- Add to Cart :
- Functionality : Users can add products to their shopping cart and review them later before completing the rental.
  - Explanation : Instead of renting items immediately, users can save products in their cart while they browse more options. This helps them manage their choices better and check out when they're ready.
- Spin to Win :
- Functionality : Users can spin a virtual wheel to win rewards, such as coins or nothing. These coins can be used to buy discount coupons on the platform.
  - Explanation : This feature adds a fun game-like experience to shopping. It encourages users to keep coming back to the platform because they can win rewards that give them discounts on future rentals. StyleSurfer provides 2 spins per month.
- Rules and Regulations with Penalties :
- Functionality : This section lists the rules that customers must follow when using the platform. If they don't follow the rules, penalties will apply.
  - Explanation : Clear rules ensure that both customers and sellers behave fairly on the platform. Penalties help maintain a smooth and trustworthy experience.

- Order for Rent :
  - Functionality : Customers can rent products by selecting details like size, color, rental duration, and choosing a delivery date.
  - Explanation : This feature allows users to personalize their rental experience. They can choose options that fit their style or needs and rent for the duration they prefer.
- Payment Processing :
  - Functionality : The platform supports both online and offline payment options. Customers can pay via Stripe (payment gateway) or choose cash on delivery.
  - Explanation : Offering both online and offline payments ensures that users can pay in the way that is most convenient for them. This makes the platform accessible to more people, including those who prefer cash payments.
- Review and Rating :
  - Functionality : After a purchase or rental, users can leave a review and rate the product with stars.
  - Explanation : Reviews help future customers decide whether they want to rent or buy a product. They also provide feedback to sellers, helping them improve their products.
- Reselling :
  - Functionality : Sellers can make their rental products available for direct purchase.
  - Explanation : This feature allows customers to buy items they want without waiting for them to be rented out. It also helps sellers manage their stock by offering exclusive products.
- Sell with Us :
  - Functionality : Customers (such as boutiques) can apply to sell their own products on the platform and access a seller dashboard to manage their listings.

- Explanation : This feature encourages more users to become sellers, which increases the variety of products available on the platform. It also builds a sense of community where customers and sellers interact directly.
- Buy-Back Options :
- Functionality :
    - Buy Option : Users can request selected customized products from sellers.
    - Back Option : Users can offer their own customized products for rent on the platform.
  - Explanation : This gives users more control over how they interact with the platform. They can either purchase unique, personalized items or earn revenue by renting out their own custom products. This creates more ways for users to engage with the service and benefit from it.

### **3.3 Behaviour Requirements**

#### **3.3.1 Use Case View**

A use case is a simple visual tool that shows how different users (called actors) interact with a system to achieve specific goals (called use cases). It helps you understand what the system is supposed to do and how people will use it.

## **4. Other Non-functional Requirements**

### **4.1 Safety and Security Requirements**

➤ Safety Requirements :

- User Support and Help Centre : StyleSurfer will have a helpful support center where users can fill out a form if they have problems or need help. This makes it easy for users to fix issues with their accounts, rentals quickly.
- FAQ : This will answer common questions about renting clothes and managing accounts. This way, users can get answers fast without waiting for a person to help them, making their experience smoother and less frustrating.
- Clear Terms and Conditions : The website will clearly show the rules and regulations for using the platform. This will include information on rental terms, payment policies, and what users are responsible for. Making this information easy to find helps users understand what they need to do.

➤ Expected Level of Security :

- User Identity Verification : We will use secure methods to verify users' identities using OTP verification through email.
- Strong Passwords : Users will need to create strong passwords that meet specific requirements (like minimum length and complexity).
- Access Control : Only authorized staff will have access to sensitive data. This means that not everyone can see or manage users' personal information, which helps protect user privacy.

### **4.2 Software Quality Attributes**

#### **4.2.1 Reliability**

- Requirement : StyleSurfer needs to be available 99.5% of the time each month, meaning it can only be down for about 36 hours in a month.

➤ How We'll Achieve This :

- Regular Updates : We will plan updates and maintenance during off-peak times to avoid downtime when users are online.
- Performance Monitoring : We will use tools to keep an eye on server performance and catch issues early before they cause downtime.

#### **4.2.2 Usability**

- Requirement : Users should be able to find their way around the StyleSurfer website. We aim for 80% of first-time users to complete their transactions without needing help.

➤ How We'll Achieve This :

- User Testing : We will hold testing sessions where real users give feedback on how the website looks and works. We'll make changes based on what they say.
- Easy Navigation : We will design a clean and simple user interface that guides users through each step of renting with easy-to-follow instructions.
- Accessibility : We will ensure that the website is user-friendly for everyone, including people with disabilities, by following accessibility guidelines.

#### **4.2.3 Maintainability**

- Requirement : The code for StyleSurfer should allow for updates to be made within 48 hours for small changes and within 96 hours for larger updates.

➤ How We'll Achieve This :

- Modular Code : We will organize the code into separate parts so it's easier to update specific features without impacting the whole system.
- Documentation : We will provide clear explanations of the code and how the system works, making it easy for future developers to understand and use it.



## Appendix A – Data Dictionary

**Table 1 Customer**

| Column Name | Data Type    | Constraint                     | Description                           |
|-------------|--------------|--------------------------------|---------------------------------------|
| Customer_id | Integer      | Primary Key ,<br>Autoincrement | Unique id for each customer           |
| Username    | Varchar(255) | Not null                       | Customers user name                   |
| Number      | Varchar(10)  | Not null                       | Customers number                      |
| Email       | Varchar(255) | Not null                       | Customers email                       |
| Address     | Text         | Null                           | Customers address                     |
| Pincode     | Varchar(6)   | Null                           | pincode                               |
| Password    | Varchar(255) | Not null                       | Customers password for account access |

**Table 2 Seller**

| Column Name | Data Type    | Constraint                     | Description                          |
|-------------|--------------|--------------------------------|--------------------------------------|
| Seller_id   | Integer      | Primary Key ,<br>Autoincrement | Unique id for each Seller            |
| Sname       | Varchar(255) | Not null                       | Seller's name                        |
| Snumber     | Varchar(10)  | Not null                       | Seller's number                      |
| Semail      | Varchar(255) | Not null                       | Seller's email                       |
| Saddress    | Text         | Not Null                       | Seller's address                     |
| Spassword   | Varchar(255) | Not null                       | Seller's password for account access |

**Table 3 DP**

| Column Name | Data Type    | Constraint                     | Description           |
|-------------|--------------|--------------------------------|-----------------------|
| DP_id       | Integer      | Primary Key ,<br>Autoincrement | Unique id for each DP |
| Dname       | Varchar(255) | Not null                       | DP's name             |
| Dnumber     | Varchar(10)  | Not null                       | DP's number           |

|           |              |          |                                  |
|-----------|--------------|----------|----------------------------------|
| Demail    | Varchar(255) | Not null | DP's email                       |
| Daddress  | Text         | Not Null | DP's address                     |
| DPassword | Varchar(255) | Not null | DP's password for account access |

**Table 4 FAQ**

| Column Name | Data Type    | Constraint                     | Description            |
|-------------|--------------|--------------------------------|------------------------|
| FAQ_id      | Integer      | Primary Key ,<br>Autoincrement | Unique id for each FAQ |
| Questions   | Varchar(200) | Not null                       | Question text          |
| Answer      | Varchar(500) | Not null                       | Answer text            |

**Table 5 Category**

| Column Name   | Data Type   | Constraint                     | Description                                   |
|---------------|-------------|--------------------------------|-----------------------------------------------|
| category_id   | Integer     | Primary Key ,<br>Autoincrement | Unique id for each category                   |
| Main category | Varchar(50) | Not null                       | Main Category ( Men , Women , Bridal , Groom) |
| Sub category  | Varchar(50) | Not null                       | Sub category ( lehnga , kurta ...)            |

**Table 6 Product**

| Column Name  | Data Type    | Constraint                     | Description                      |
|--------------|--------------|--------------------------------|----------------------------------|
| product_id   | Integer      | Primary Key ,<br>Autoincrement | Unique id for each product       |
| name         | Varchar(500) | Not null                       | Product name                     |
| description  | Text         | Not null                       | Product description              |
| price        | Decimal      | Not null                       | Product selling price            |
| Rental price | Decimal      | Not null                       | Product rental price ( per day ) |
| Size         | Varchar(20)  | Not null                       | Product size                     |
| Color        | Varchar(50)  | Not null                       | Product color                    |

|                  |                  |                                                       |                                                          |
|------------------|------------------|-------------------------------------------------------|----------------------------------------------------------|
| Brand            | Varchar(100)     | Not null                                              | Product brand name                                       |
| Quantity (stoke) | Integer unsigned | Not null check<br>(quantity >= 0)                     | Number of product<br>stoke available                     |
| Image front      | ImageField       | Required                                              | Product front view<br>image                              |
| Image back       | ImageField       | Required                                              | Product back view<br>image                               |
| Image side       | ImageField       | Required                                              | Product side view<br>image                               |
| Category         | Integer          | Foreign key<br>references Category<br>( category_id ) | Stores the reference<br>of id from the<br>Category table |
| Seller           | Integer          | Foreign key<br>references Seller ( Seller_id )        | Stores the reference<br>of id from the Seller<br>table   |
| Buy back         | Bool             | Default = false                                       | Indicates buy back<br>eligibility                        |
| Created at       | Datetime         | Auto set , not null                                   | Timestamp of<br>product creation                         |

**Table 7 RCategory**

| Column Name    | Data Type   | Constraint                     | Description                           |
|----------------|-------------|--------------------------------|---------------------------------------|
| Rcategory_id   | INTEGER     | Primary Key ,<br>Autoincrement | Unique id for each<br>Resell category |
| RMain category | Varchar(50) | Not null                       | Main Category ( Men , Women)          |
| RSub category  | Varchar(50) | Not null                       | Sub category ( lehnga , kurta ...)    |

**Table 8 Resell\_Product**

| Column Name             | Data Type        | Constraint                                                | Description                                               |
|-------------------------|------------------|-----------------------------------------------------------|-----------------------------------------------------------|
| r_id                    | Integer          | Primary Key ,<br>Autoincrement                            | Unique id for each<br>resell product                      |
| rname                   | Varchar(500)     | Not null                                                  | Resell Product name                                       |
| rdescription            | Text             | Not null                                                  | Resell Product<br>description                             |
| oprice                  | Decimal          | Not null                                                  | Resell Product<br>original price                          |
| Rdiscount<br>percentage | Integer unsigned | Not null                                                  | Discount percentage<br>at which product sell              |
| r_price                 | Decimal          | Not null                                                  | Resell Product<br>selling price ( per<br>day )            |
| rSize                   | Varchar(20)      | Not null                                                  | Resell Product size                                       |
| rColor                  | Varchar(50)      | Not null                                                  | Resell Product color                                      |
| rBrand                  | Varchar(100)     | Not null                                                  | Resell Product brand<br>name                              |
| rQuantity (stoke)       | Integer unsigned | Not null check<br>(quantity >= 0)                         | Number of resell<br>product stoke<br>available            |
| rImage front            | ImageField       | Required                                                  | Resell Product front<br>view image                        |
| rImage back             | ImageField       | Required                                                  | Resell Product back<br>view image                         |
| rImage side             | ImageField       | Required                                                  | Resell Product side<br>view image                         |
| rCategory               | Integer          | Foreign key<br>references<br>RCategory (<br>category_id ) | Stores the reference<br>of id from the<br>RCategory table |

|            |          |                                                   |                                                        |
|------------|----------|---------------------------------------------------|--------------------------------------------------------|
| rSeller    | Integer  | Foreign key<br>references Seller (<br>Seller_id ) | Stores the reference<br>of id from the Seller<br>table |
| Created at | Datetime | Auto set , not null                               | Timestamp of resell<br>product creation                |

**Table 9 Sell With Us**

| Column Name      | Data Type        | Constraint                        | Description                                      |
|------------------|------------------|-----------------------------------|--------------------------------------------------|
| cs_id            | Integer          | Primary Key ,<br>Autoincrement    | Unique id for each<br>product                    |
| csname           | Varchar(255)     | Not null                          | Customer seller<br>name                          |
| number           | Varchar(10)      | Not null                          | Customer seller<br>number                        |
| pincode          | Varchar(6)       | Not null                          | Customer Seller<br>pincode                       |
| address          | Text             | Not null                          | Customer seller<br>address of boutique /<br>shop |
| email            | Varchar(255)     | Not null                          | Customer seller<br>email                         |
| name             | Varchar(500)     | Not null                          | Product name                                     |
| description      | Text             | Not null                          | Product description                              |
| price            | Decimal          | Not null                          | Product selling price                            |
| Rental price     | Decimal          | Not null                          | Product rental price (<br>per day )              |
| Size             | Varchar(20)      | Not null                          | Product size                                     |
| Color            | Varchar(50)      | Not null                          | Product color                                    |
| Brand            | Varchar(100)     | Not null                          | Product brand name                               |
| Quantity (stoke) | Integer unsigned | Not null check<br>(quantity >= 0) | Number of product<br>stoke available             |
| Image front      | ImageField       | Required                          | Product front view<br>image                      |

|            |              |                     |                               |
|------------|--------------|---------------------|-------------------------------|
| Image back | ImageField   | Required            | Product back view image       |
| Image side | ImageField   | Required            | Product side view image       |
| Category   | Varchar(100) | Not Null            | Product Category              |
| Boutique   | Varchar(100) | Not null            | Boutique / shop name          |
| Created at | Datetime     | Auto set , not null | Timestamp of product creation |

**Table 10 Cart**

| Column Name    | Data Type | Constraint                                      | Description                                                  |
|----------------|-----------|-------------------------------------------------|--------------------------------------------------------------|
| cart_id        | Integer   | Primary Key , Autoincrement                     | Unique id for each cart                                      |
| User           | Integer   | Foreign key references Customer ( customer_id ) | Stores the reference of id from the customer table           |
| Total price    | Decimal   | Not null                                        | Total price of all the products in cart                      |
| Total security | Decimal   | Not null                                        | Total security price of all the products in cart             |
| Shipping cost  | Decimal   | Not null                                        | Shipping price                                               |
| Discount       | Decimal   | Nullable                                        | Discount price if applied voucher discount                   |
| Grand total    | Decimal   | Not null                                        | Grand total of all the prices ( security , shipping , etc..) |
| Created_at     | Datetime  | Auto now add , not null                         | Timestamp when cart is created                               |
| Updated_at     | Datetime  | Auto now , not null                             | Timestamp when cart is updated                               |

**Table 11 Cart Item**

| Column Name | Data Type | Constraint                                                         | Description                                                                            |
|-------------|-----------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| id          | Integer   | Primary Key ,<br>Autoincrement                                     | Unique id for each<br>cart item                                                        |
| Product_id  | Integer   | Foreign key<br>references Product (<br>Product_id ) ,<br>nullable  | Stores the reference<br>of id from the<br>product table if<br>product                  |
| r_id        | Integer   | Foreign key<br>references Resell<br>Product ( r_id ) ,<br>nullable | Stores the reference<br>of id from the resell<br>product table if<br>reselling product |
| Cart        | Integer   | Foreign key<br>references Cart (<br>cart_id )                      | Stores the reference<br>of id from the cart<br>table                                   |

**Table 12 Order**

| Column Name    | Data Type   | Constraint                                            | Description                                              |
|----------------|-------------|-------------------------------------------------------|----------------------------------------------------------|
| order_id       | Integer     | Primary Key ,<br>Autoincrement                        | Unique id for each<br>order                              |
| User           | Integer     | Foreign key<br>references Customer<br>( customer_id ) | Stores the reference<br>of id from the<br>customer table |
| Order_status   | Varchar(50) | Not null                                              | Status of the order                                      |
| Total price    | Decimal     | Not null                                              | Total price of items<br>in order                         |
| Total security | Decimal     | Not null                                              | Total security<br>deposite                               |
| Shipping cost  | Decimal     | Not null                                              | Shipping price                                           |
| Discount       | Decimal     | Nullable                                              | Discount price if<br>applied voucher<br>discount         |

|                |             |                         |                                         |
|----------------|-------------|-------------------------|-----------------------------------------|
| Grand total    | Decimal     | Not null                | Final order price after discount        |
| Payment method | Varchar(20) | Nullable                | Method of payment used ( COD , online ) |
| Payment date   | Datetime    | Nullable                | Date of the payment                     |
| Payment status | Varchar(20) | Nullable                | Status of the payment                   |
| Created_at     | Datetime    | Auto now add , not null | Timestamp when order is created         |
| Updated_at     | Datetime    | Auto now , not null     | Timestamp when order is last updated    |

**Table 13 OrderItem**

| Column Name          | Data Type   | Constraint                                                | Description                                                                   |
|----------------------|-------------|-----------------------------------------------------------|-------------------------------------------------------------------------------|
| id                   | Integer     | Primary Key , Autoincrement                               | Unique id for each order item                                                 |
| Order                | Integer     | Foreign key references Order ( order_id )                 | Stores the reference of id from the order table                               |
| Product_id           | Integer     | Foreign key references Product ( Product_id ) , nullable  | Stores the reference of id from the product table if product                  |
| r_id                 | Integer     | Foreign key references Resell Product ( r_id ) , nullable | Stores the reference of id from the resell product table if reselling product |
| Product order status | Varchar(20) | Not null                                                  | Order status of the product                                                   |



**Table 14 ReviewRating**

| Column Name     | Data Type | Constraint                                              | Description                                                    |
|-----------------|-----------|---------------------------------------------------------|----------------------------------------------------------------|
| ReviewRating_id | Integer   | Primary Key ,<br>Autoincrement                          | Unique id for each<br>review & rating                          |
| User            | Integer   | Foreign key<br>references Customer<br>( customer_id )   | Stores the reference<br>of id from the<br>customer table       |
| Order           | Integer   | Foreign key<br>references Order (<br>order_id )         | Stores the reference<br>of id from the order<br>table          |
| Product         | Integer   | Foreign key<br>references Product (<br>Product_id )     | Stores the reference<br>of id from the<br>Product table        |
| Sell With Us    | Integer   | Foreign key<br>references<br>SellWithUs ( cs_id )       | Stores the reference<br>of id from the<br>SellWithUs table     |
| Rproduct        | Integer   | Foreign key<br>references<br>Resell_Product (<br>r_id ) | Stores the reference<br>of id from the Resell<br>Product table |
| Rating          | Integer   | Not null                                                | Rating for each<br>product                                     |
| Review          | Text      | Not null                                                | Review for each<br>product                                     |
| Created_at      | Datetime  | Auto now add , not<br>null                              | Timestamp when<br>review rating is<br>created                  |

**Table 15 SpinToWin**

| Column Name | Data Type | Constraint                     | Description                         |
|-------------|-----------|--------------------------------|-------------------------------------|
| Spin_id     | Integer   | Primary Key ,<br>Autoincrement | Unique id for<br>spinning the wheel |

|        |          |                                                       |                                                          |
|--------|----------|-------------------------------------------------------|----------------------------------------------------------|
| User   | Integer  | Foreign key<br>references Customer<br>( customer_id ) | Stores the reference<br>of id from the<br>customer table |
| Date   | Datetime | Auto now add , not<br>null                            | Timestamp of the<br>last spin                            |
| Count  | Integer  | Not null , default = 0                                | Number of times the<br>user has spin the<br>wheel        |
| Reward | Integer  | Not null , default = 0                                | Reward (coin)<br>earned from<br>spinning                 |

**Table 16 Voucher**

| Column Name   | Data Type        | Constraint                     | Description                                         |
|---------------|------------------|--------------------------------|-----------------------------------------------------|
| voucher_id    | Integer          | Primary Key ,<br>Autoincrement | Unique id for each<br>voucher                       |
| Discount      | Integer unsigned | Not null                       | Discount percentage<br>or value                     |
| Voucher price | Integer          | Not null                       | Price of the voucher                                |
| Description   | Varchar(100)     | Not null                       | Brief description of<br>the voucher                 |
| Created_at    | Datetime         | Auto now add , not<br>null     | Timestamp when<br>voucher is created                |
| Expire_at     | Datetime         | Auto now add , not<br>null     | Timestamp when<br>voucher is going to<br>be expired |

**Table 17 RedeemVoucher**

| Column Name | Data Type   | Constraint                     | Description                     |
|-------------|-------------|--------------------------------|---------------------------------|
| id          | Integer     | Primary Key ,<br>Autoincrement | Unique id for<br>redeem voucher |
| Code        | Varchar(20) | Not null , unique              | Unique voucher<br>code          |

|             |             |                                                       |                                                             |
|-------------|-------------|-------------------------------------------------------|-------------------------------------------------------------|
| User        | Integer     | Foreign key<br>references Customer<br>( customer_id ) | Stores the reference<br>of id from the<br>customer table    |
| Voucher     | Integer     | Foreign key<br>references Voucher (<br>Voucher_id )   | Stores the reference<br>of id from the<br>voucher table     |
| Redeemed_at | Datetime    | Auto now add , not<br>null                            | Timestamp when<br>voucher is redeem                         |
| Status      | Varchar(20) | Not null                                              | Status of the<br>redeemed voucher (<br>expire , available ) |

**Table 18 DPOrder**

| Column Name   | Data Type   | Constraint                                      | Description                                           |
|---------------|-------------|-------------------------------------------------|-------------------------------------------------------|
| id            | Integer     | Primary Key ,<br>Autoincrement                  | Unique id for dp<br>order                             |
| DP            | Integer     | Foreign key<br>references DP (<br>DP_id )       | Stores the reference<br>of id from the dp<br>table    |
| order         | Integer     | Foreign key<br>references Order (<br>order_id ) | Stores the reference<br>of id from the order<br>table |
| Assigned date | Datetime    | Auto add now , not<br>null                      | Timestamp when<br>order is assigned to<br>dp          |
| Status        | Varchar(10) | Not null                                        | Status of the order (<br>accepted , rejected )        |

**Table 19 HelpRequest**

| Column Name | Data Type | Constraint                                            | Description                                        |
|-------------|-----------|-------------------------------------------------------|----------------------------------------------------|
| Help_id     | Integer   | Primary Key ,<br>Autoincrement                        | Unique id for help request                         |
| User        | Integer   | Foreign key<br>references Customer<br>( customer_id ) | Stores the reference of id from the customer table |
| issue       | Text      | Not null                                              | Description of the issue                           |
| Created_at  | Datetime  | Auto now add , not null                               | Timestamp when request is created                  |

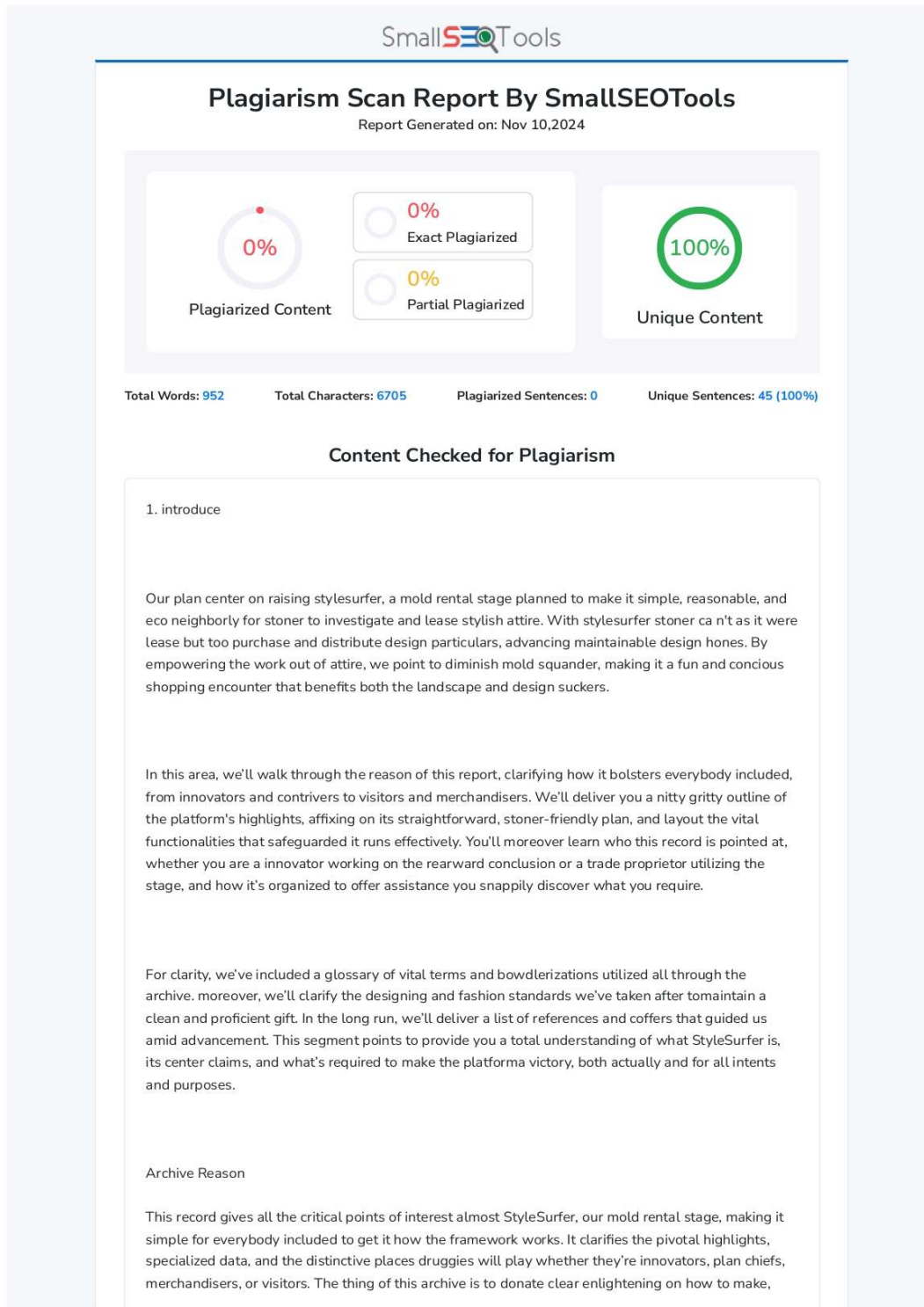
**Table 20 BuyBack**

| Column Name     | Data Type    | Constraint                                            | Description                                        |
|-----------------|--------------|-------------------------------------------------------|----------------------------------------------------|
| BuyBack_id      | Integer      | Primary Key ,<br>Autoincrement                        | Unique id for buy back                             |
| User            | Integer      | Foreign key<br>references Customer<br>( customer_id ) | Stores the reference of id from the customer table |
| Created_at      | Datetime     | Auto now add , not null                               | Timestamp when buy back is created                 |
| Updated_at      | Datetime     | Auto now add , not null                               | Timestamp when buy back is last updated            |
| Buy back status | Varchar(255) | Not null                                              | Buy back status ( buy , back , pending )           |

**Table 21 CustomDesign**

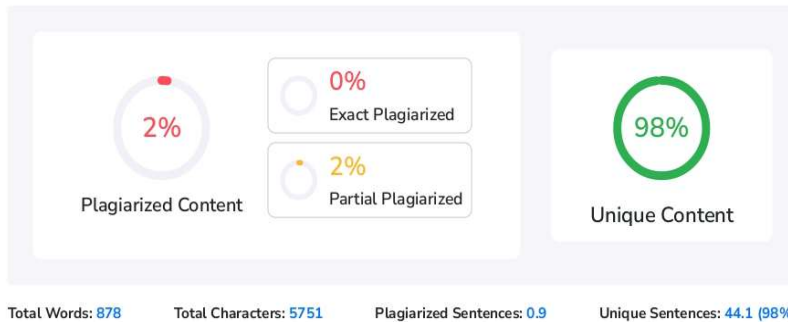
| Column Name               | Data Type | Constraint                                          | Description                                                           |
|---------------------------|-----------|-----------------------------------------------------|-----------------------------------------------------------------------|
| id                        | Integer   | Primary Key ,<br>Autoincrement                      | Unique id                                                             |
| Buy back                  | Integer   | Foreign key<br>references BuyBack<br>( BuyBack_id ) | Stores the reference<br>of id from the Buy<br>Back table              |
| Seller                    | Integer   | Foreign key<br>references Seller (<br>Seller_id )   | Stores the reference<br>of id from the Seller<br>table                |
| Product                   | Integer   | Foreign key<br>references Product (<br>Product_id ) | Stores the reference<br>of id from the<br>Product table               |
| Customized total<br>price | Decimal   | Not null                                            | Total price after all<br>the customization                            |
| Customization<br>options  | Text      | Not null                                            | Option selected for<br>customization (<br>stone , density ,<br>etc..) |

## Appendix B – Plagiarism Report



## Plagiarism Scan Report By SmallSEOTools

Report Generated on: Nov 14,2024



### Content Checked for Plagiarism

#### 2. Overall Description

##### 2.1 Product Perspective

StyleSurfer is an exciting new platform designed to change how people rent and sell clothes. It's not just an update to something that already exists, nor does it replace any old systems. Instead, StyleSurfer meets the rising need for sustainable fashion by offering a simple and easy way for users to rent, buy, or sell clothing. This platform aims to provide a smooth experience for everyone involved customers, sellers, and delivery partners while promoting a more eco-friendly approach to fashion.

The StyleSurfer system is built around three main groups of users:

- Customers: These are the individuals looking to find and enjoy a variety of fashion items. They can choose to rent or buy clothes and benefit from features like 360-degree views of products.
- Sellers: These are people who want to list their clothes for rent or sale. The platform provides them with tools to manage their listings effectively and make the most of their sales.
- Delivery Partners (DP): These individuals handle the logistics of delivering and picking up items. They play a crucial role in ensuring that the process of getting clothes to customers and returning them runs smoothly.

##### 2.2 Product Functionality

###### □ 360-Degree Product View

- Functionality: This feature allows users to view premium products from all angles in a detailed and interactive manner.
- Explanation: By providing a 360-degree view, users can better assess the product's features, quality, and appearance, enhancing their shopping experience.

###### □ Add to Cart

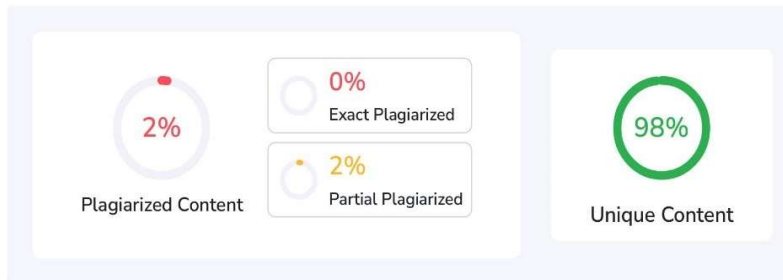
- Functionality: Users can add selected products to their shopping cart for future purchase.
- Explanation: This function simplifies the shopping process by allowing users to collect multiple items in one place before proceeding to checkout.

###### □ Spin to Win

- Functionality: Users can engage in a gamified experience by clicking on the "Spin" option and pressing the spin button to receive rewards, such as coins or nothing.
- Explanation: This feature includes a "Coupon Page," where users can redeem their earned coins to purchase various coupons, encouraging user engagement and loyalty.

## Plagiarism Scan Report By SmallSEOTools

Report Generated on: Nov 14,2024



Total Words: 840    Total Characters: 5506    Plagiarized Sentences: 0.82    Unique Sentences: 40.18 (98%)

### Content Checked for Plagiarism

Distinction of Important Users :

□ Most Important Users:

- Customers: They are the main users of StyleSurfer and directly impact rentalsales and overall revenue. Keeping them happy is crucial for the platform's success.
- Sellers: They provide the clothing items for rent and are essential for offering a good variety of options. Their engagement and satisfaction are vital for a successful marketplace.
- Designers (for Buy-Back Customization): Designers enhance the customer experience by offering customization options within the buy- back program, adding value and increasing customer satisfaction.

□ Less Important Users:

- Delivery Partners: While they play an important role in delivering items, their satisfaction is less critical compared to customers and sellers.
- Admins: They are important for managing the system, but their satisfaction doesn't directly affect user experience. Their work is essential for smooth operations but is less critical than keeping customers and sellers satisfied.

#### 2.4 Operating Environment

The StyleSurfer fashion rental website will operate in a web-based environment designed to offer a smooth experience for users on different devices. Users can access the site through modern web browsers, making it compatible with both desktop and mobile platforms.

Here are the minimum hardware and software requirements for optimal performance:

□ Hardware Requirements:

- Desktops/Laptops: An Intel i3 processor or a similar one with at least 4 GB RAM.
- Mobile Devices: Any recent smartphone or tablet with at least 3 GB of RAM.

□ Operating Systems:

- Desktops/Laptops:
  - o Windows 10 or newer
  - o macOS 12 or newer
  - o Any Linux distribution that supports modern browsers.
- Mobile Devices:

o iOS 12 or newer

□ Software Stack:

- Backend: Python using the Django framework for server-side processing.
- Frontend: HTML5, CSS, JavaScript, and Bootstrap to create a responsive design
- Database: SQLite or MySQL to handle user data , product listing.



## Plagiarism Scan Report By SmallSEOTools

Report Generated on: Nov 14,2024



### Content Checked for Plagiarism

#### 2.7 Assumptions and Dependencies

##### Major Assumptions

- User Accounts: We believe StyleSurfer will have up to many user accounts when we launch. This will affect how we design our database and manage user information.
- Product Listings: We expect to start with many products available for rent. This will impact how we organize, display, and manage these products in our database.
- Payment Methods: We plan to integrate at least 2 payment options (online, offline) for users to make payments. This will shape how we design the checkout process and the security measures we need to put in place.
- Delivery Partners: We anticipate working with 5 delivery partners to manage the delivery of products. This will affect how we design our logistics.
- Mobile Users: We believe that around 50% of users will access StyleSurfer using their mobile devices. This will guide our choices for making the site responsive and easy to use on phones.
- User Engagement: We expect that 20% of users will take part in promotional activities like the "Spin to Win" game. This will impact how we design and integrate the rewards system.

##### Dependencies

- Database Management System: The project will rely on a specific database management system (to be determined). Any limitations or issues with this database could affect how we manage and retrieve data.
- Django Framework: We are using the Django framework for the back end of the website. Any updates or changes to Django may require us to adjust our implementation.
- Team Skills: We assume that all team members have the necessary skills in the technologies we're using (HTML, CSS, JavaScript, Django). If there are any gaps in expertise, it could lead to challenges in development.

#### 3. Specific Requirements

##### 3.1 External Interface Requirements

###### 3.1.1 User Interfaces

###### Home Page:

- Description: The homepage of StyleSurfer will have a clean and easy-to-use design. It will show the latest and most popular clothes available for rent or purchase. The homepage will include a navigation bar and other features to make it easy for users to browse the website.

## Plagiarism Scan Report By SmallSEOTools

Report Generated on: Nov 14,2024



### Content Checked for Plagiarism

#### 3.1.3 Software Interfaces

##### □ Operating System Interface:

- StyleSurfer is platform-independent, meaning it works on any device that can run a modern web browser, like those with Windows, macOS, Linux, Android, or iOS.
- Browser Compatibility: The platform will be compatible with popular browsers like Google Chrome, Firefox, Safari. This ensures a smooth experience for all users, regardless of their device.

##### □ Database:

- StyleSurfer will use SQLite as its database. This will store important data like user accounts, product details.

##### □ Libraries and Frameworks:

- Django (Python Framework): Django will handle all the background processes, like user requests, database management, and serving web pages.
- Bootstrap: This will be used to make the website responsive, ensuring that it looks good and works well on all devices, including phones and computers.
- JavaScript Libraries (e.g., jQuery): JavaScript will be used to add dynamic features like the 'Spin to Win' game and cart updates.

#### 3.1.4 Communications Interfaces

Web Communication Protocols : The website will use HTTP/HTTPS protocols to communicate between users' devices and the servers. Sensitive information like passwords and payment details will be securely transferred using HTTPS

#### 3.2 Functional Requirements

##### □ Search and Browse

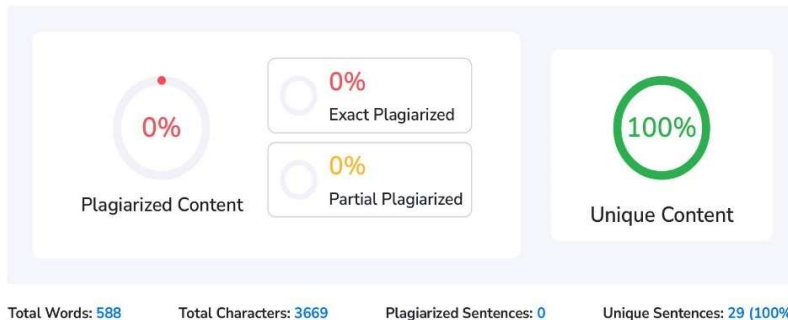
- Functionality: Users can search for specific products or explore different categories like clothing type, brand, or style.
- Explanation: This feature helps users quickly find exactly what they're looking for. Whether they know the specific product they want or just want to browse through options, this makes the shopping process faster and easier.

##### □ 360-Degree Product View

- Functionality: Premium products have an interactive feature where users can view the product from

## Plagiarism Scan Report By SmallSEOTools

Report Generated on: Nov 14,2024



### Content Checked for Plagiarism

#### 4. Other Non-functional Requirements

##### 4.1 Safety and Security Requirements

###### □ Safety Requirements:

- User Support and Help Centre: StyleSurfer will have a helpful support center where users can fill out a form if they have problems or need help. This makes it easy for users to fix issues with their accounts , rentals quickly.
- FAQ :This will answer common questions about renting clothes and managing accounts. This way, users can get answers fast without waiting for a person to help them, making their experience smoother and less frustrating.
- Clear Terms and Conditions: The website will clearly show the rules and regulations for using the platform. This will include information on rental terms, payment policies, and what users are responsible for. Making this information easy to find helps users understand what they need to do.

###### □ Expected Level of Security:

- User Identity Verification: We will use secure methods to verify users' identities, including:
- Strong Passwords: Users will need to create strong passwords that meet specific requirements (like minimum length and complexity).
- Access Control: Only authorized staff will have access to sensitive data. This means that not everyone can see or manage users' personal information, which helps protect user privacy.
- Incident Response Plan: We will create a plan to quickly respond to data breaches or security issues. This plan will outline what steps to take immediately, how to notify affected users, and how to fix the problem.
- Regular Security Training: We will provide ongoing training for the development team and support staff. This training will keep them updated on security best practices, potential threats, and how to effectively respond to security incidents.

##### 4.2 Software Quality Attributes

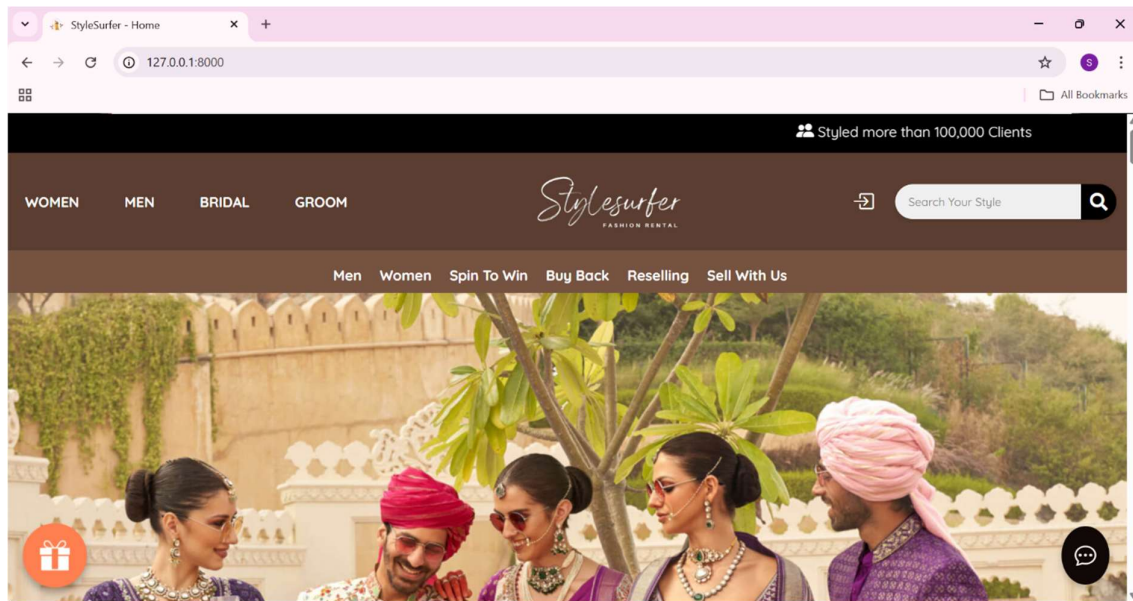
###### □ Reliability

- Requirement: StyleSurfer needs to be available 99.5% of the time each month , meaning it can only be down for about 36 hours in a month.
- How We'll Achieve This:
  - o Backup Systems: We will set up extra servers and databases so that if one fails, another can take over, keeping the website running smoothly.
  - o Regular Updates: We will plan updates and maintenance during off-peak times to avoid downtime

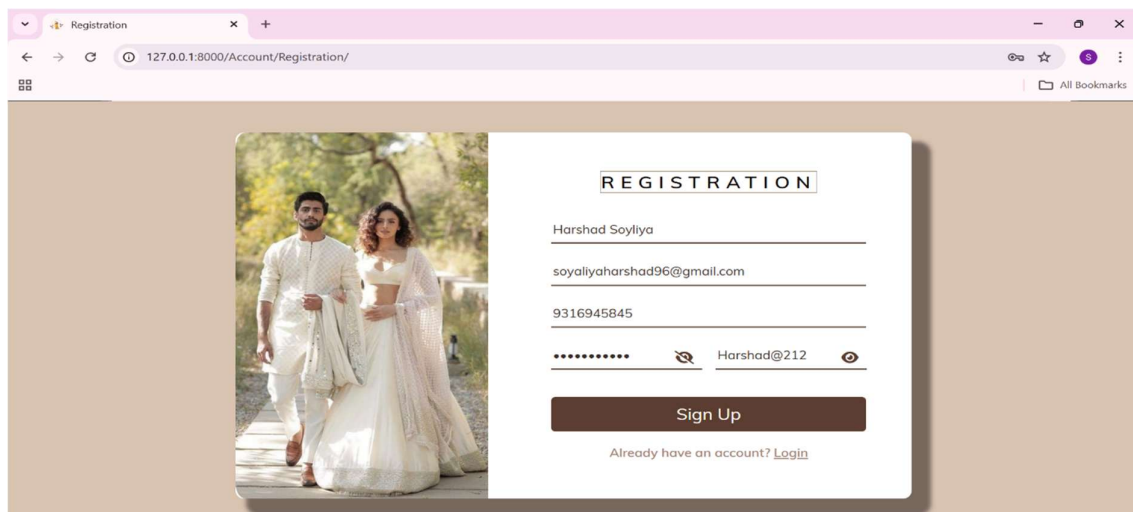
## Appendix C – User Manual

### 1) Customer :

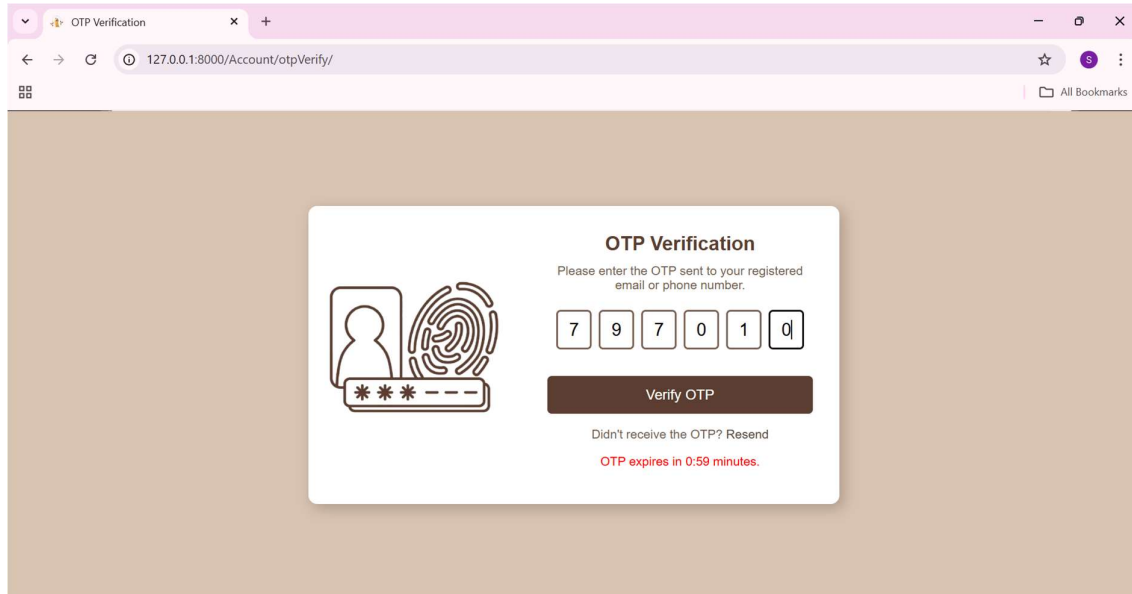
On the StyleSurfer homepage, users can search for products, browse available items, and view ongoing offers without logging in. However, to purchase or rent products and access other interactive features such as adding to cart, managing orders, or redeeming rewards, users are required to log in or sign up.



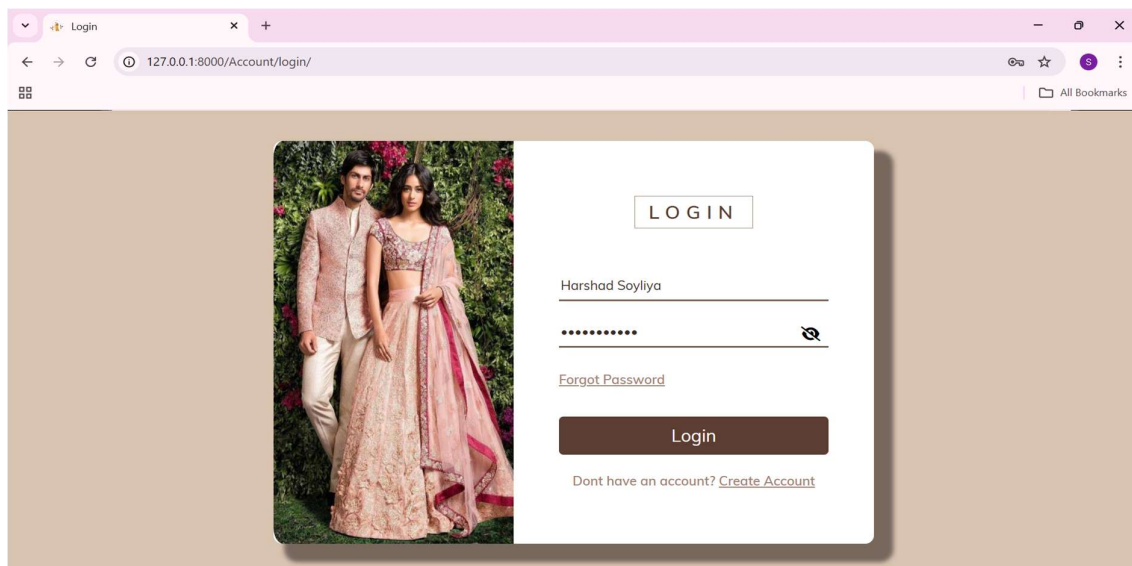
This is the registration page, where users can create a new account on StyleSurfer. To register, users need to provide their name, email, mobile number, and password. If the user already has an account, they can simply click on the login option to access their profile.



After entering their details on the registration page, users will receive a One-Time Password (OTP) on their registered email address. They must enter the correct OTP in the verification field. If the OTP is valid, the user is redirected to the login page. If the OTP is invalid, an error message is displayed, prompting the user to try again.

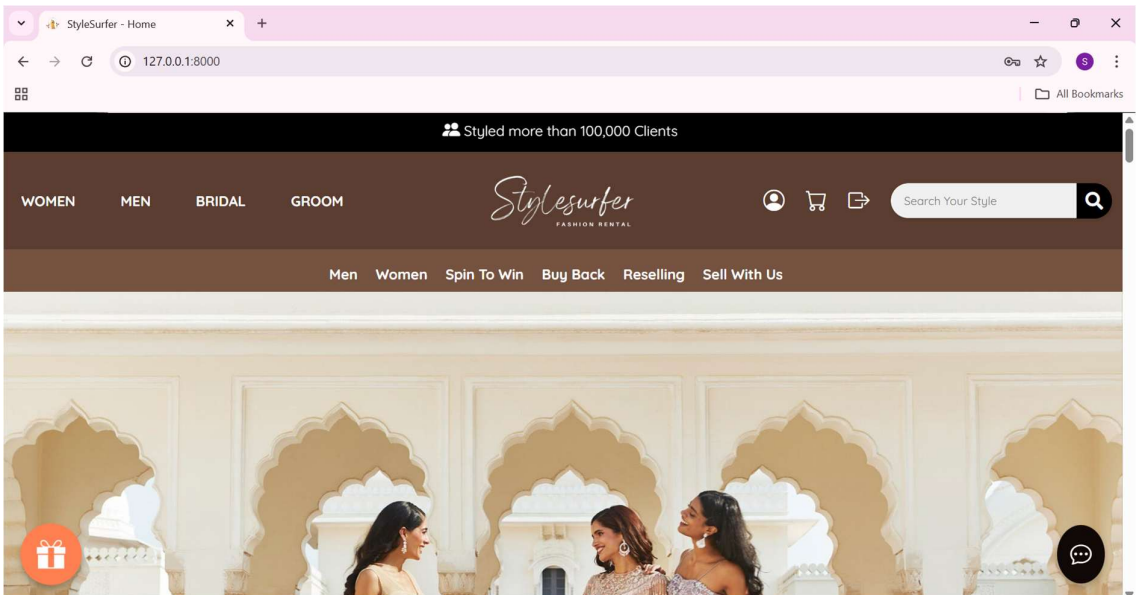


On the login page of StyleSurfer, users can access their account by entering their username and password. If a user forgets their password, they can click on the "Forgot Password" link, which redirects them to a page where they need to enter their registered email address. Upon doing so, they will receive a password reset link via email, allowing them to create a new password and regain access to their account. If the user does not have an existing account, they can click on the "Register" button, which will take them to the registration page to create a new account.

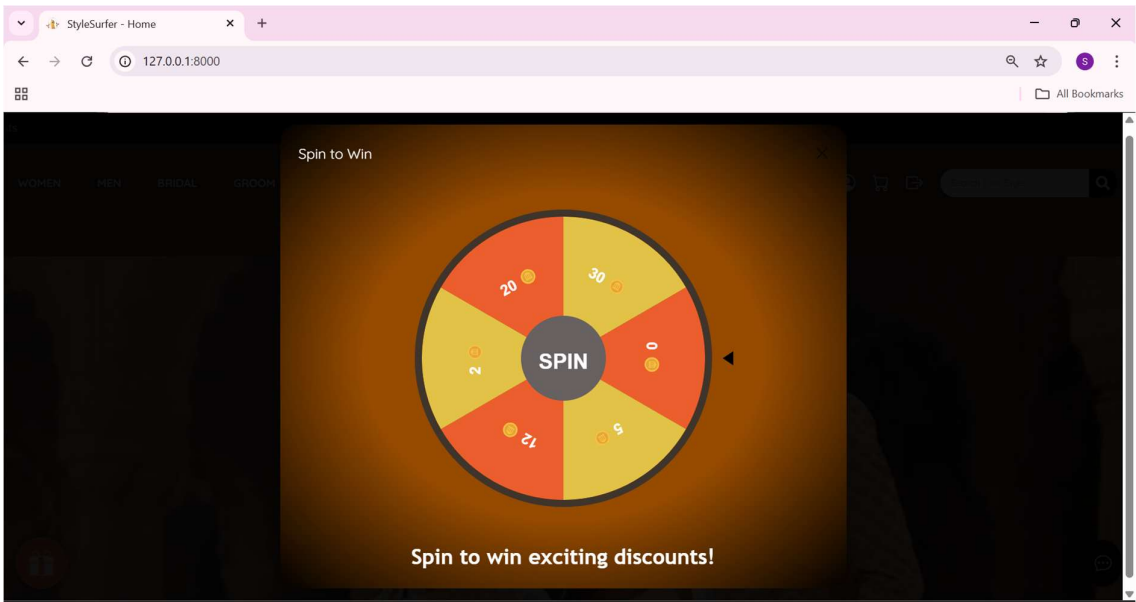


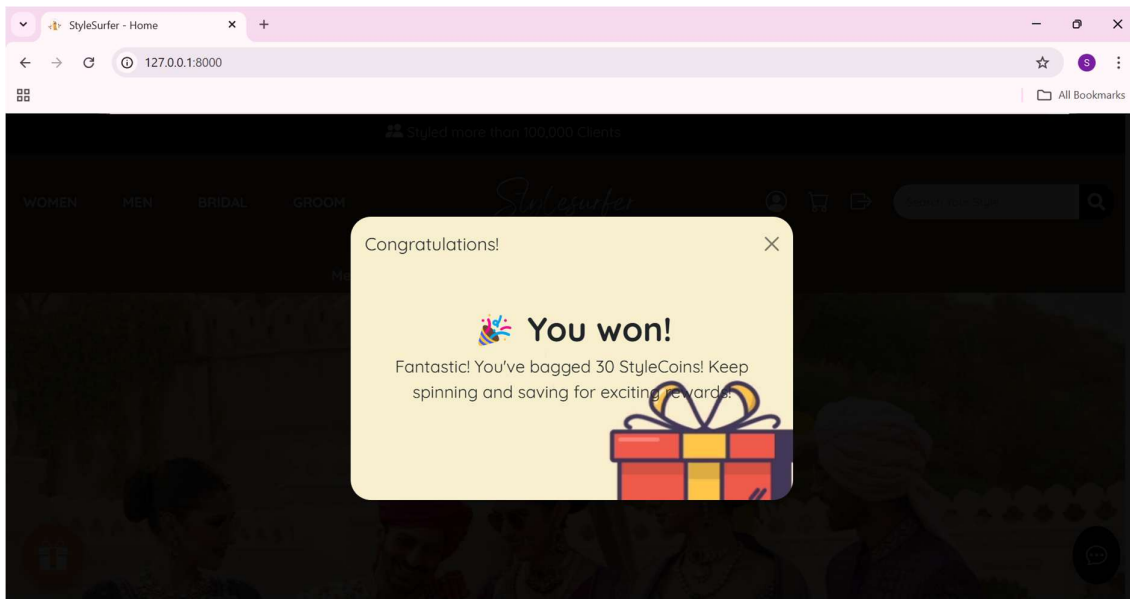


After logging in, users gain access to all interactive features on StyleSurfer. They can view and manage their cart, access their account details, add products to the cart, and proceed to purchase or rent items available on the platform.

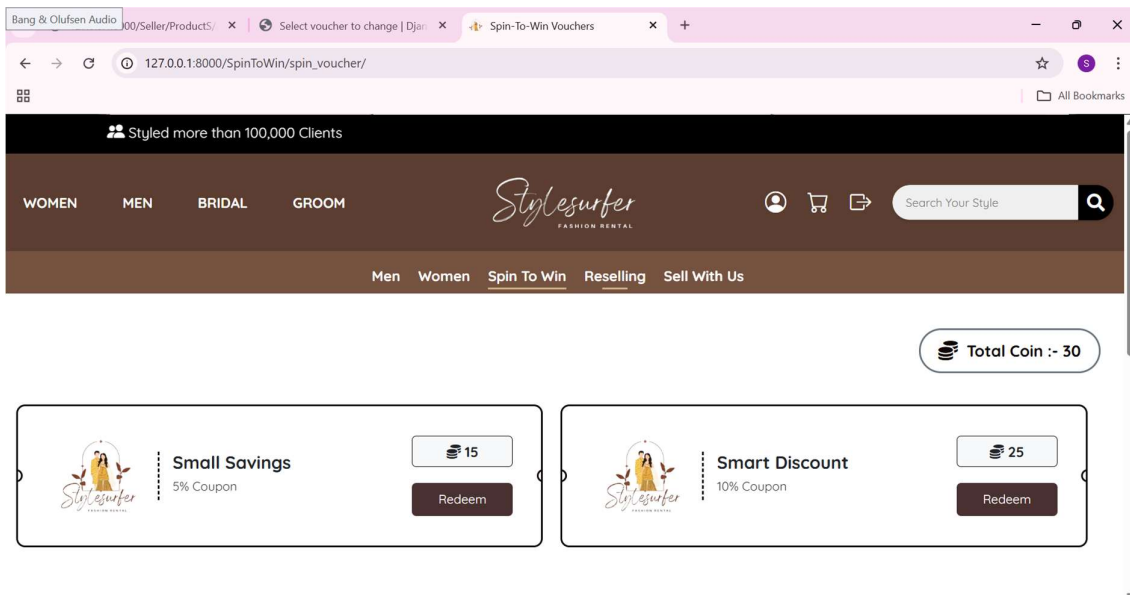


By clicking on the gift icon on the homepage, users can access the Spin to Win feature. A wheel appears on the screen, which users can spin to try their luck. Upon spinning, users have a chance to win coins. These earned coins can then be used to redeem vouchers, which can be applied for discounts or special offers on future purchases.

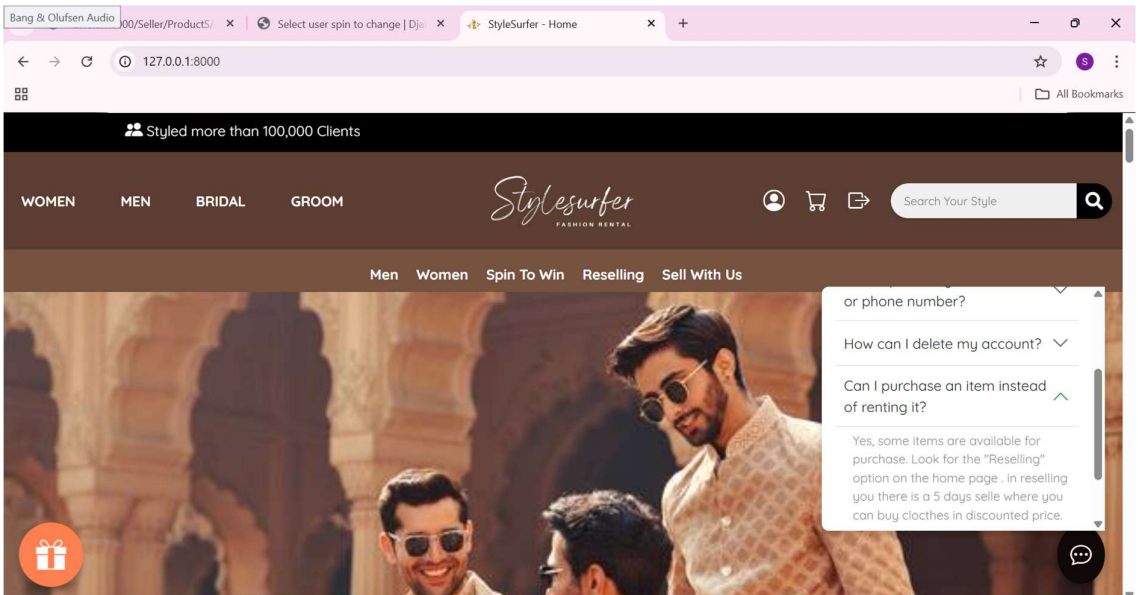




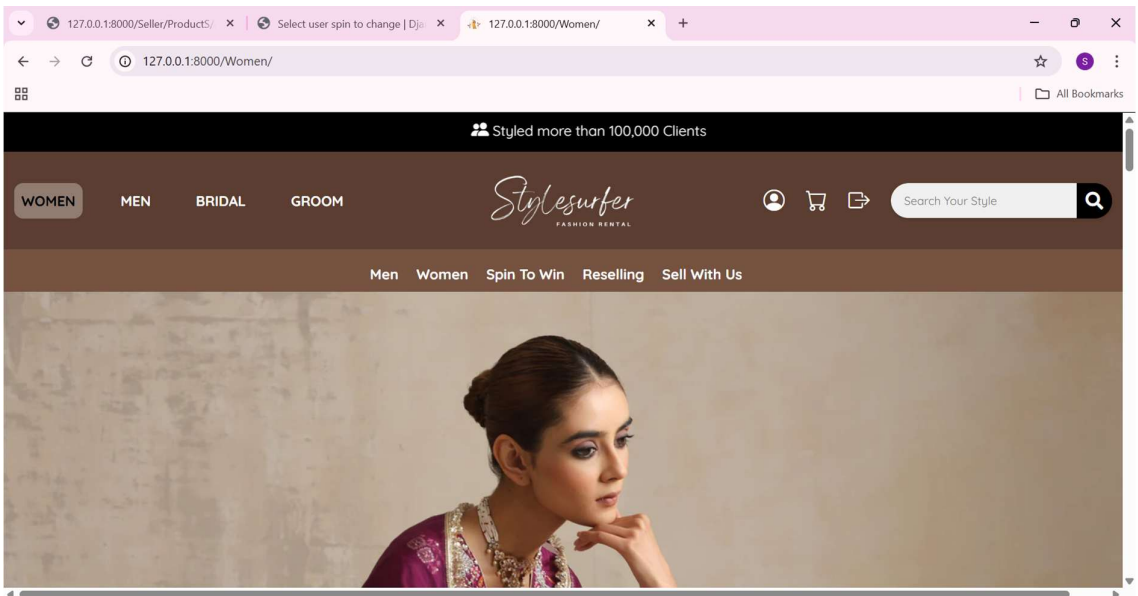
From the navigation bar, users can click on the Spin to Win option to view all the available vouchers. Based on the number of coins they have won, users can redeem these vouchers and use them for discounts or special offers while renting or purchasing products.



On the lower right side of the homepage, users will see a message icon. By clicking on it, they can access the FAQ (Frequently Asked Questions) section, where a list of common questions and their answers is displayed to help users quickly find solutions and information.

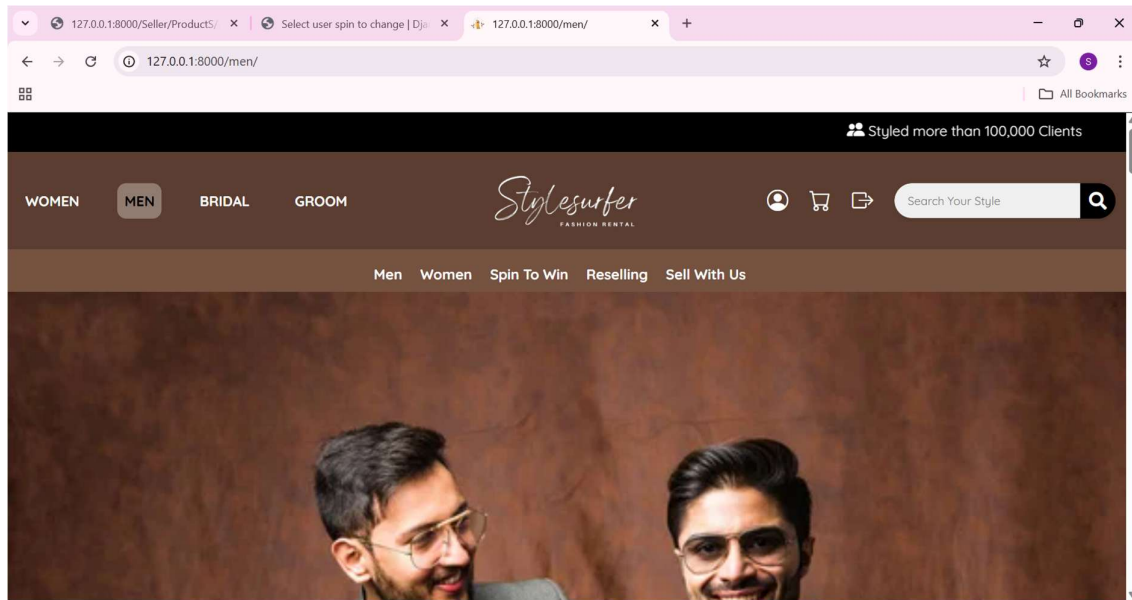


This is the Women's page, which displays various categories of women's fashion. Users can click on any specific category to view the products available under that category, making it easier to browse items based on their preferences and style.

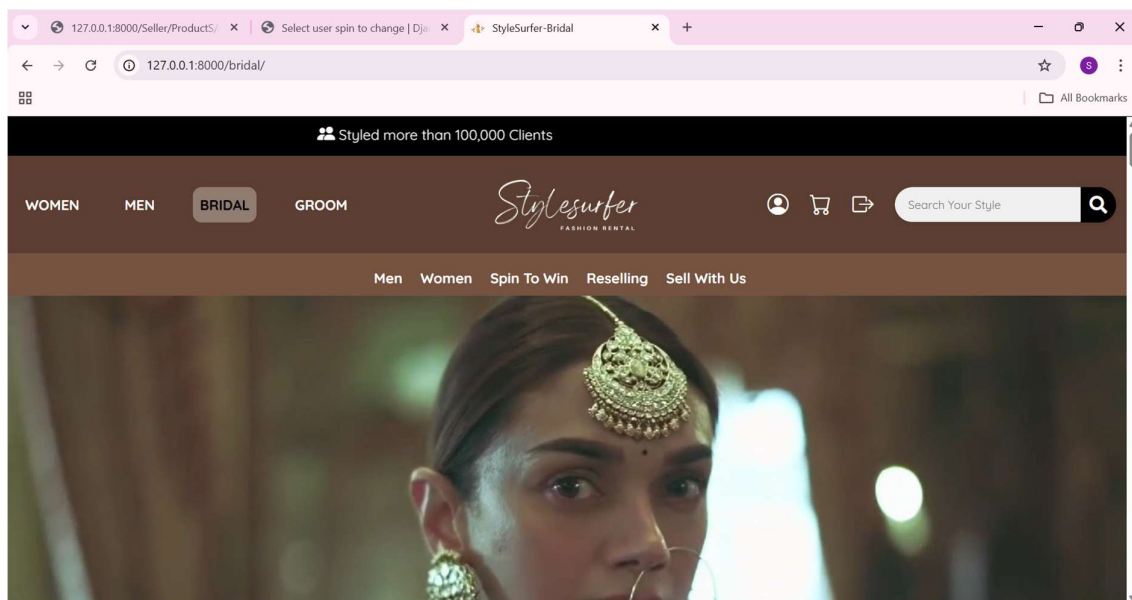




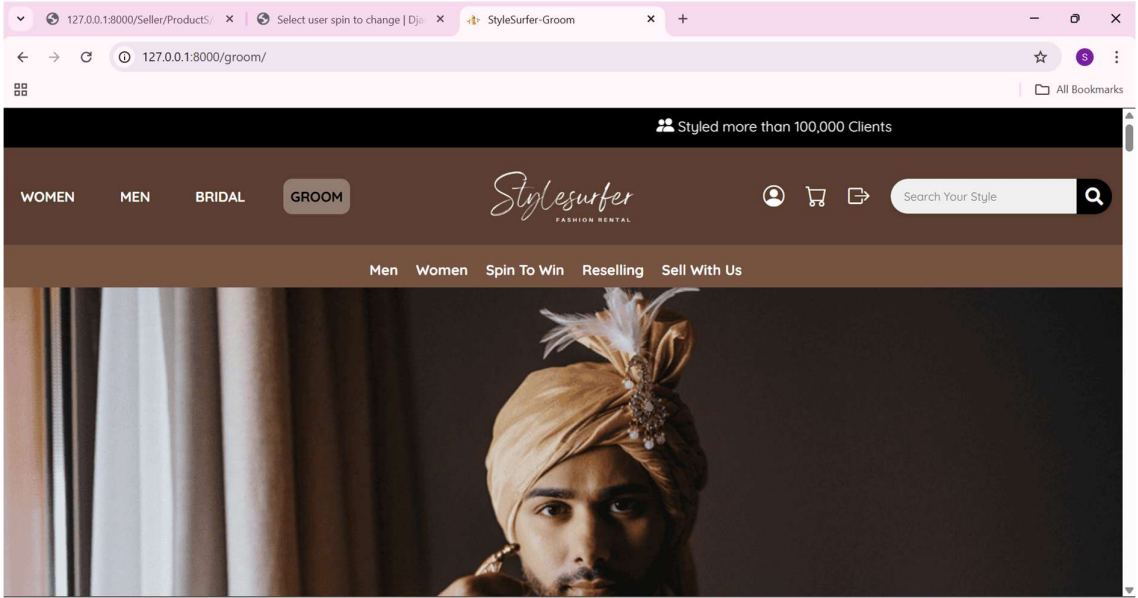
This is the Men's page, featuring different categories of men's fashion. Users can select a specific category to view all the products listed under that category, allowing for easy browsing based on their style and preferences.



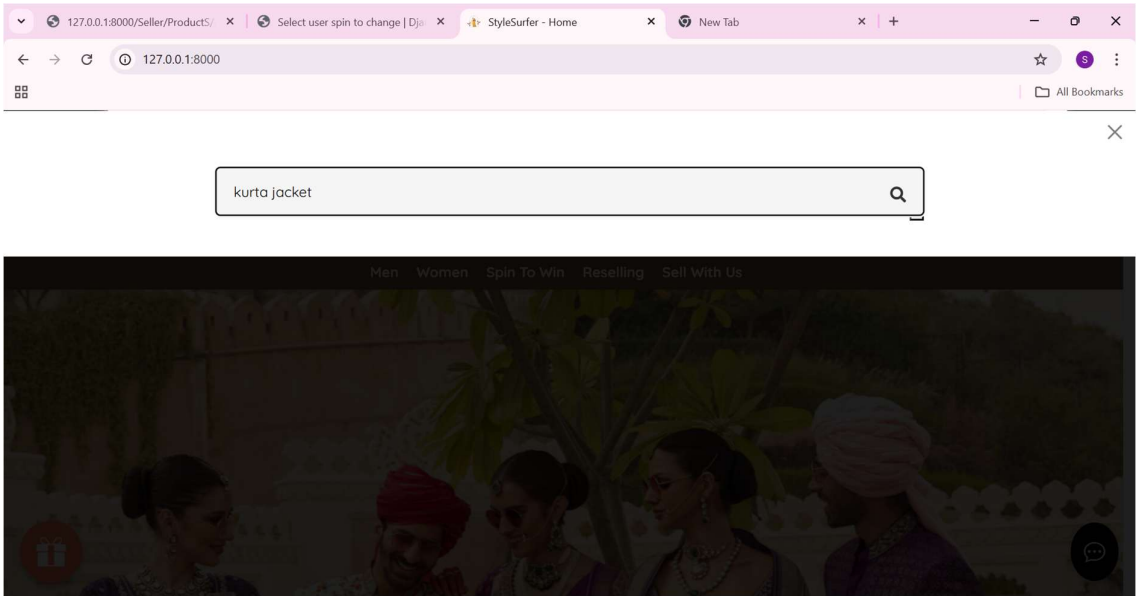
This is the Bridal page, which showcases a beautiful collection of categories tailored for wedding occasions such as Mehndi, Haldi, Sangeet, Engagement, Wedding, and more. Users can click on any category to explore outfits specially curated for each ceremony, making it easier to find the perfect look for every wedding event.



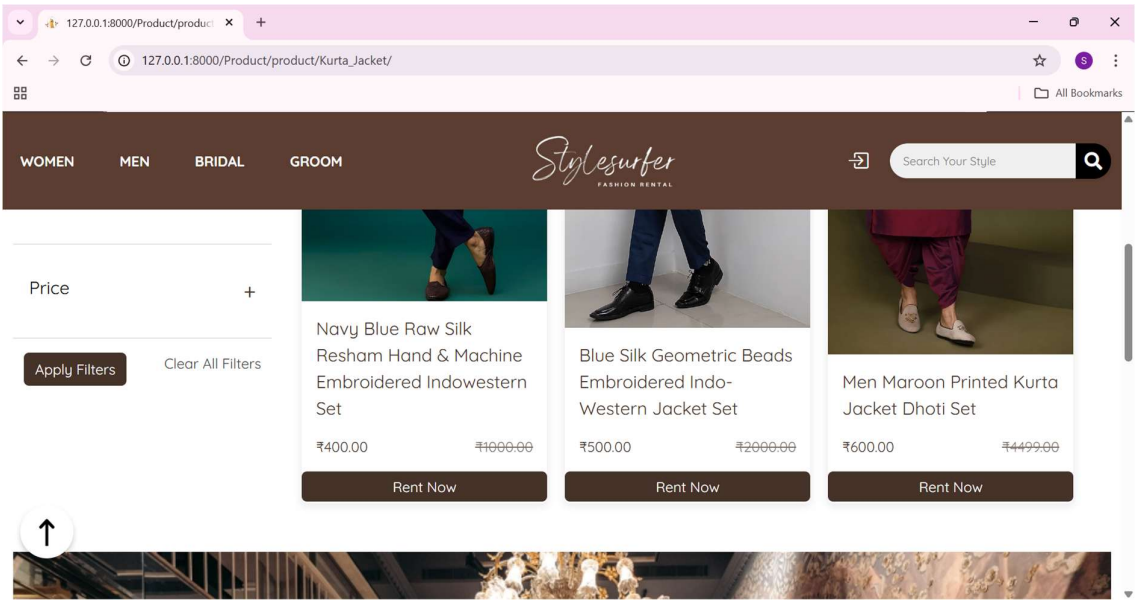
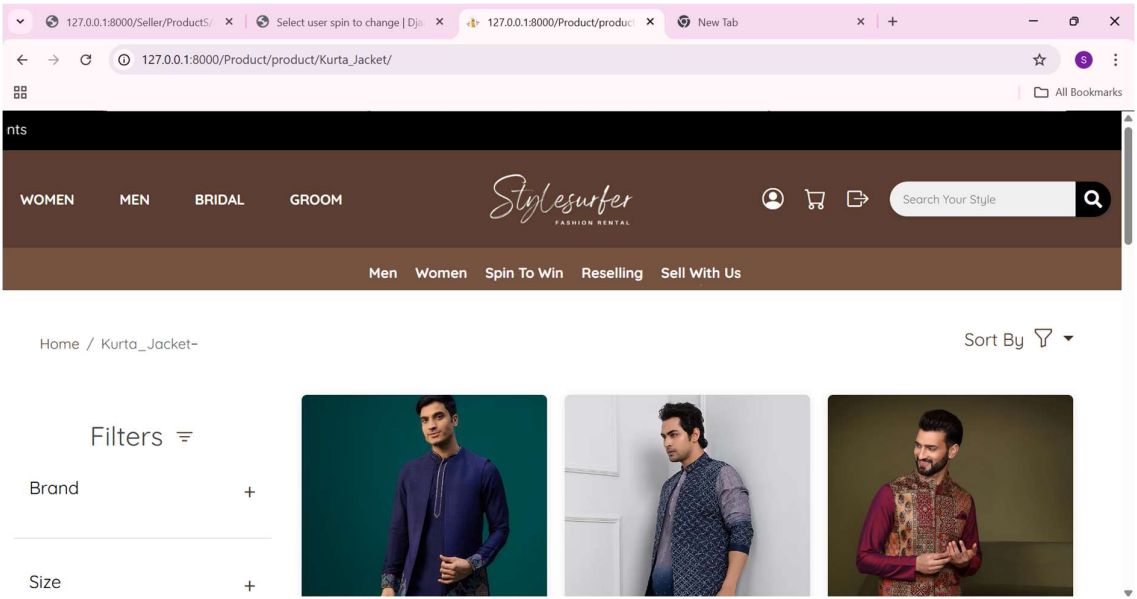
This is the Groom page, featuring a curated collection of categories for wedding occasions such as Mehndi, Haldi, Sangeet, Engagement, Wedding, and more. Each category offers a range of stylish outfits designed specifically for grooms, helping them find the perfect attire for every ceremony and celebration.



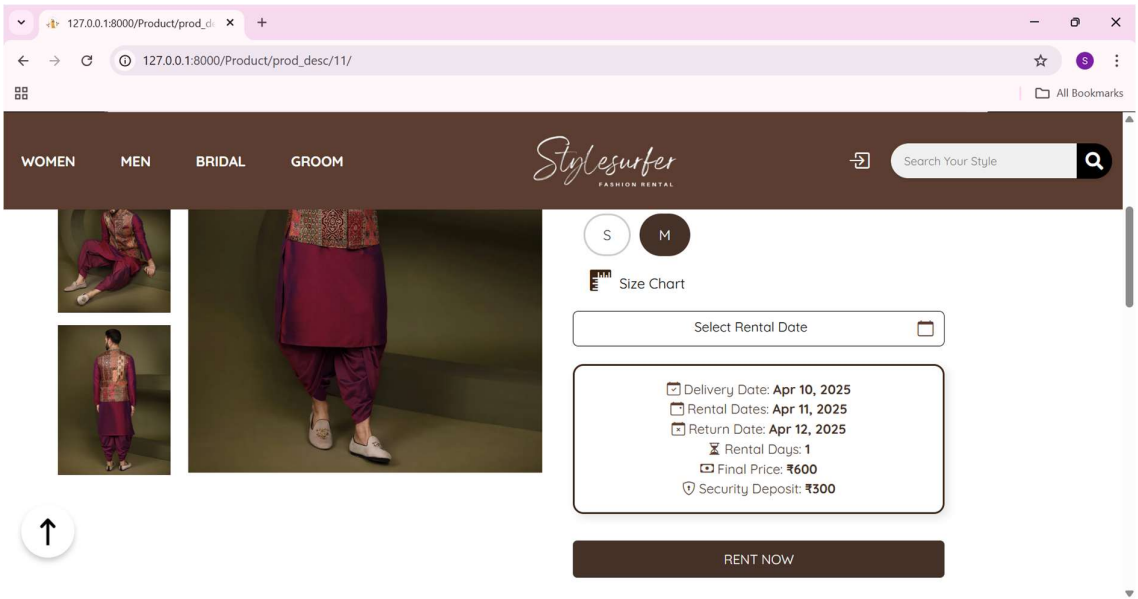
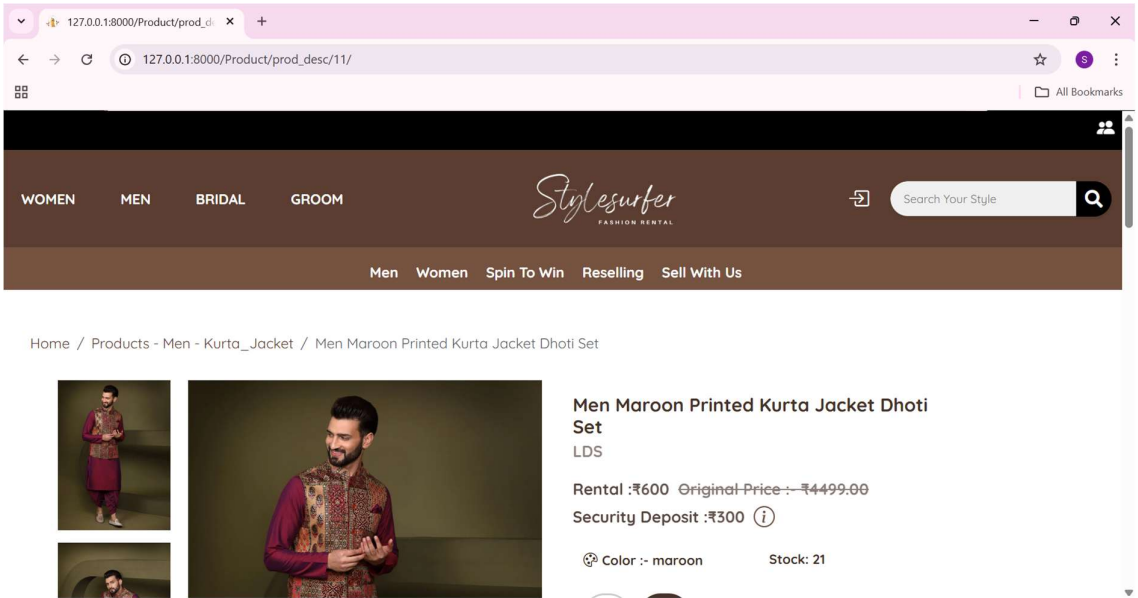
Here, users can easily search for products by entering the product name, selecting a category, or typing a brand name. This helps users quickly find items based on their preferences.

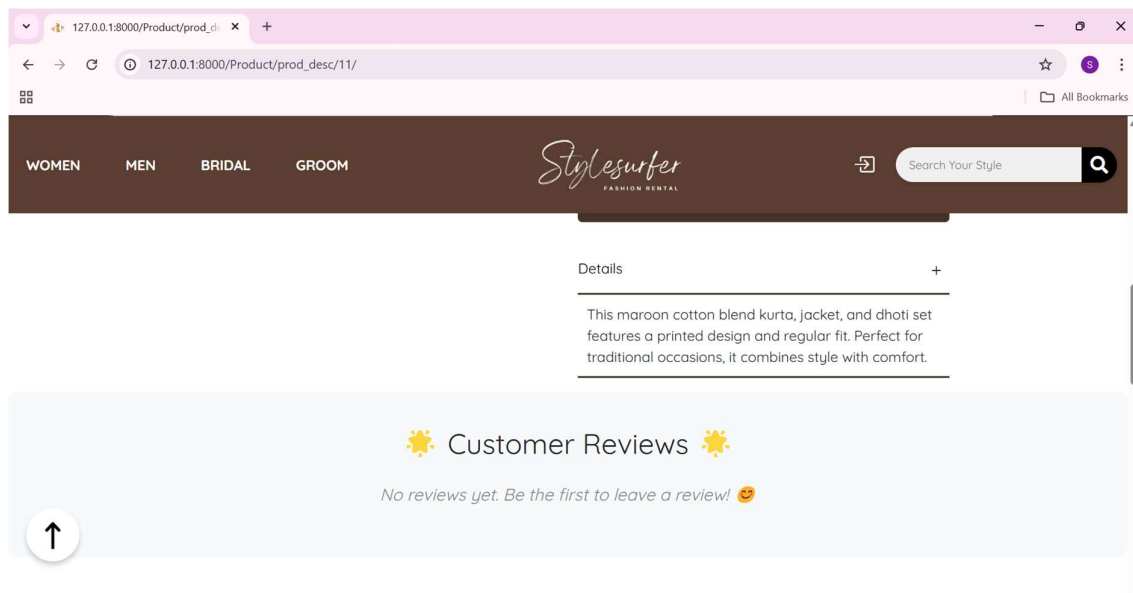


This is the product page of a selected category. Users can filter products by brand, color, size, and sort by price or new arrivals. Clicking “Rent Now” opens the product description page.

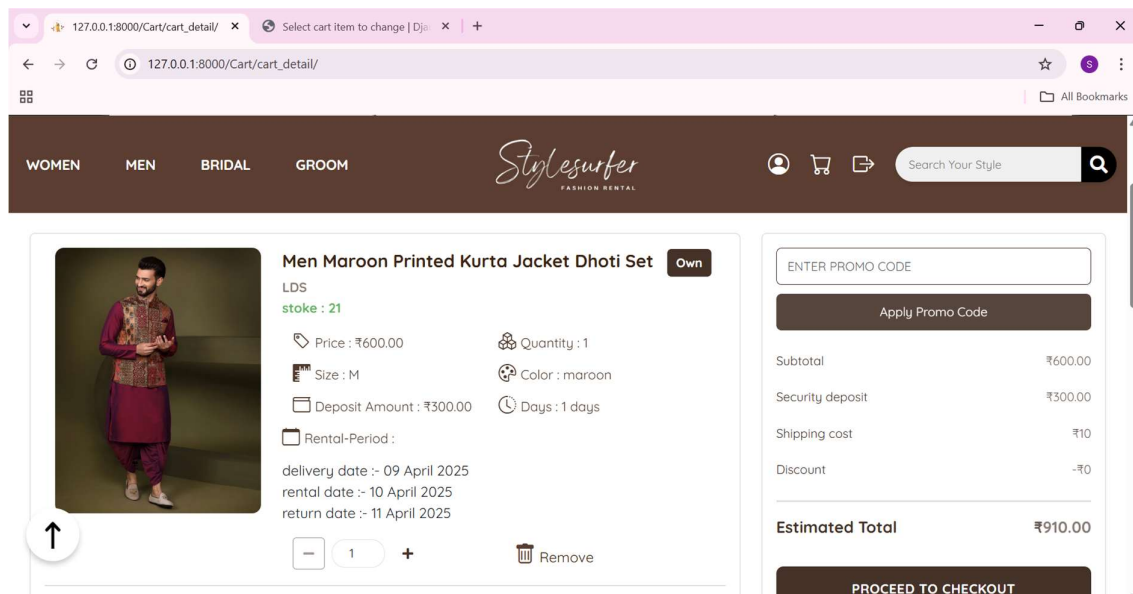


This is the description page of a product where users can view detailed information, select size and rental period, and add the product to their cart. Users can also read customer reviews for that product.





This is the cart page, where users can view all added products, apply a promo code if available, adjust size or quantity, and proceed to checkout.



This is the checkout page . & The checkout process consists of three simple steps:

➤ Step 1: Users enter their delivery address details and click Continue.

127.0.0.1:8000/order/order\_form/step=1

WOMEN MEN BRIDAL GROOM

Stylesurfer FASHION RENTAL

Search Your Style

Your Order

1 ADDRESS 2 REVIEW 3 PAYMENT

First Name harshad Last Name soyaliya

Email Mobail Number

127.0.0.1:8000/order/order\_form/step=1

WOMEN MEN BRIDAL GROOM

Stylesurfer FASHION RENTAL

Search Your Style

soyaliyaharshad96@gmail.com 9316945845

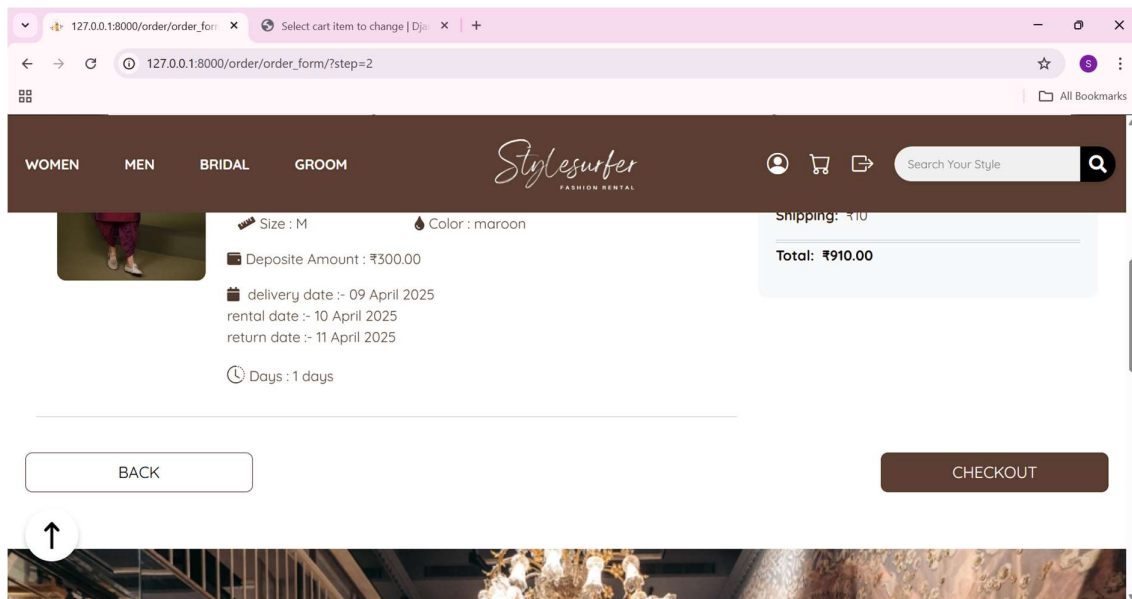
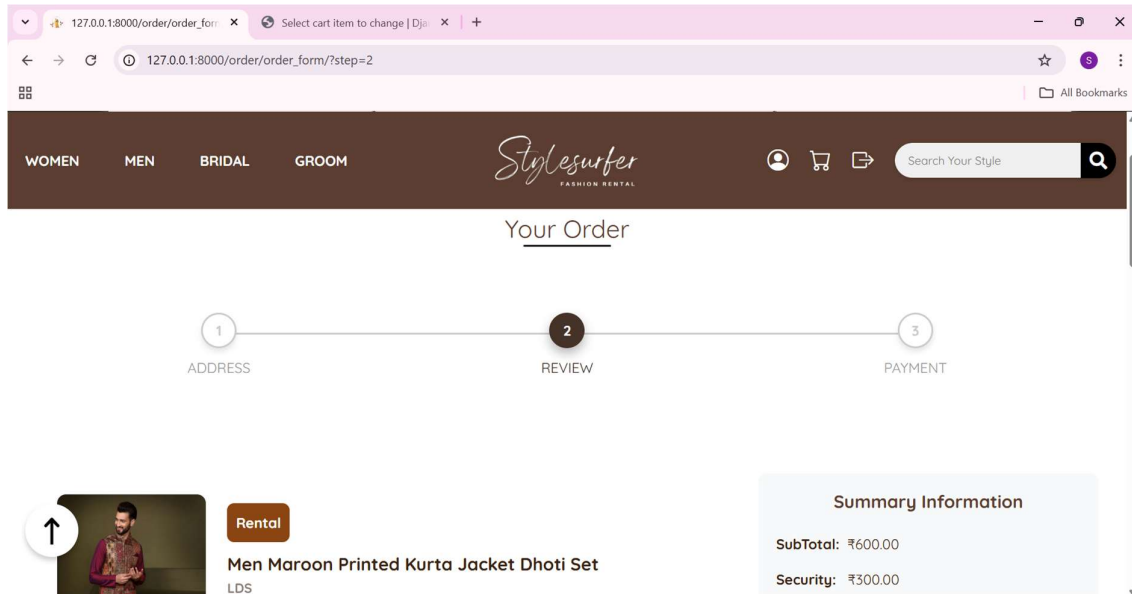
Address Line1 nana mandva Address Line2

City Rajkot State Gujarat

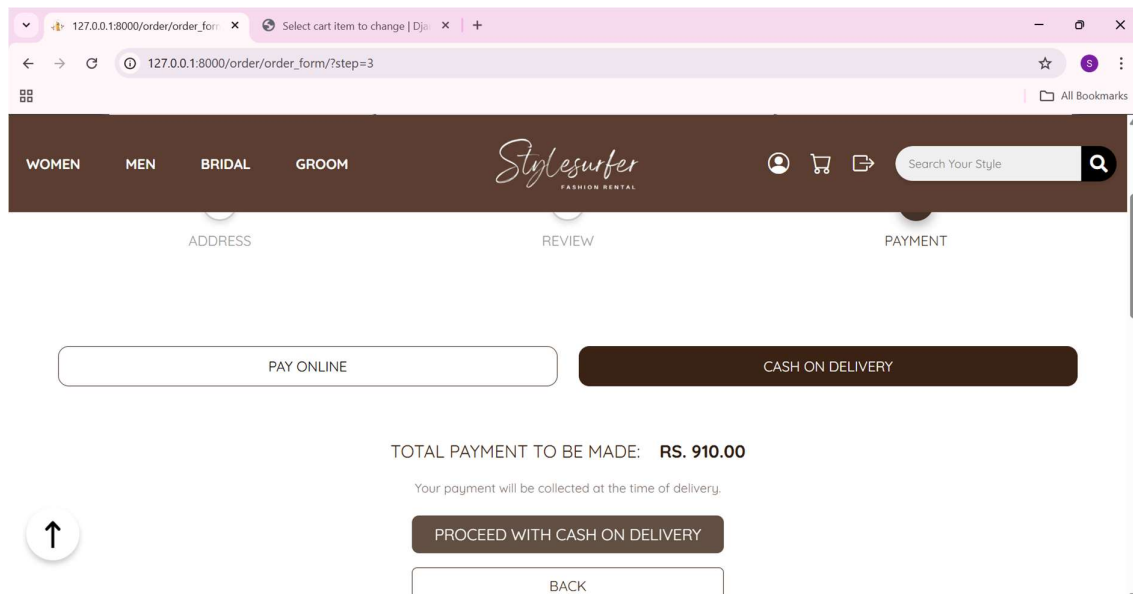
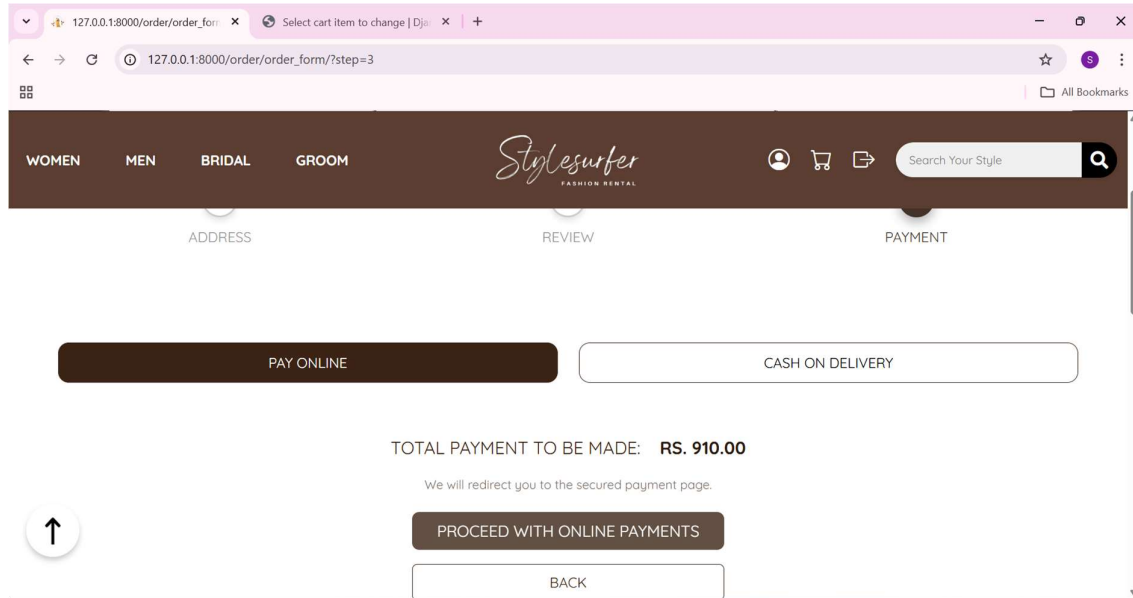
Pin Code 360311 Country India

BACK CONTINUE

- Step 2: A summary of the selected products (to rent or buy) is displayed. Users can review the items and click Checkout.

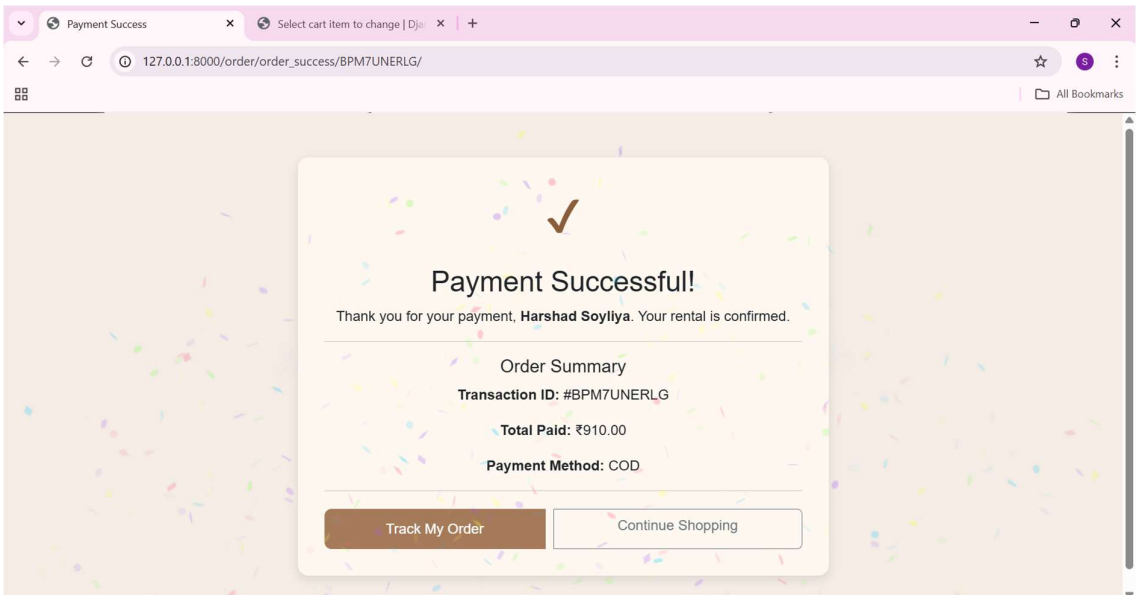


- Step 3: Users choose a payment method – either Cash on Delivery (COD) or Online Payment. For COD, users simply click Proceed to Checkout. For Online Payment, users select the option, click Proceed to Checkout, and are redirected to the payment gateway to enter their card details.

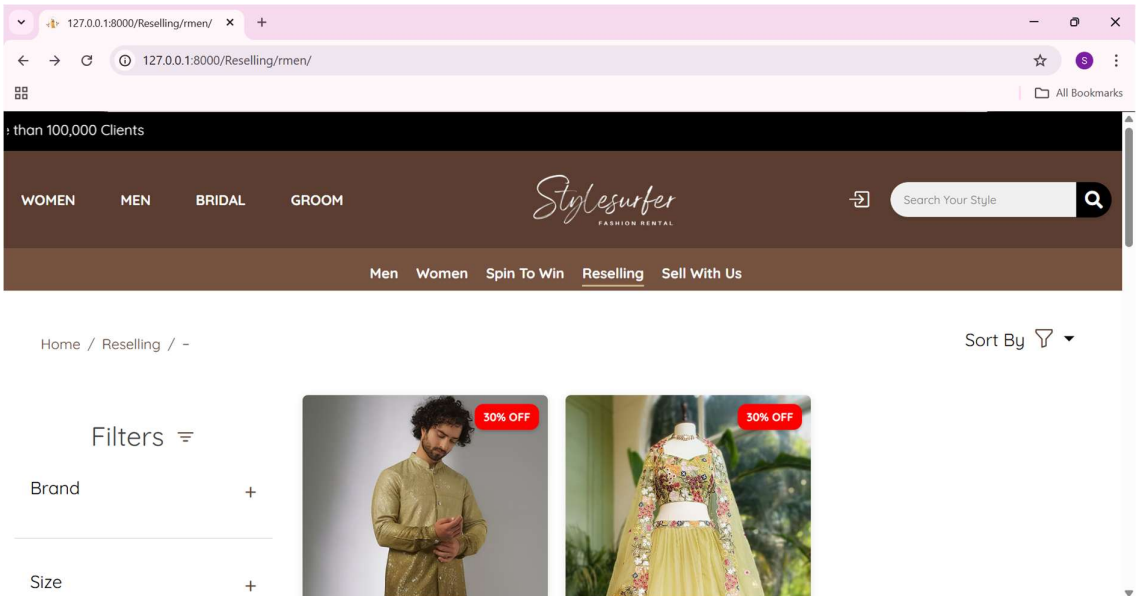




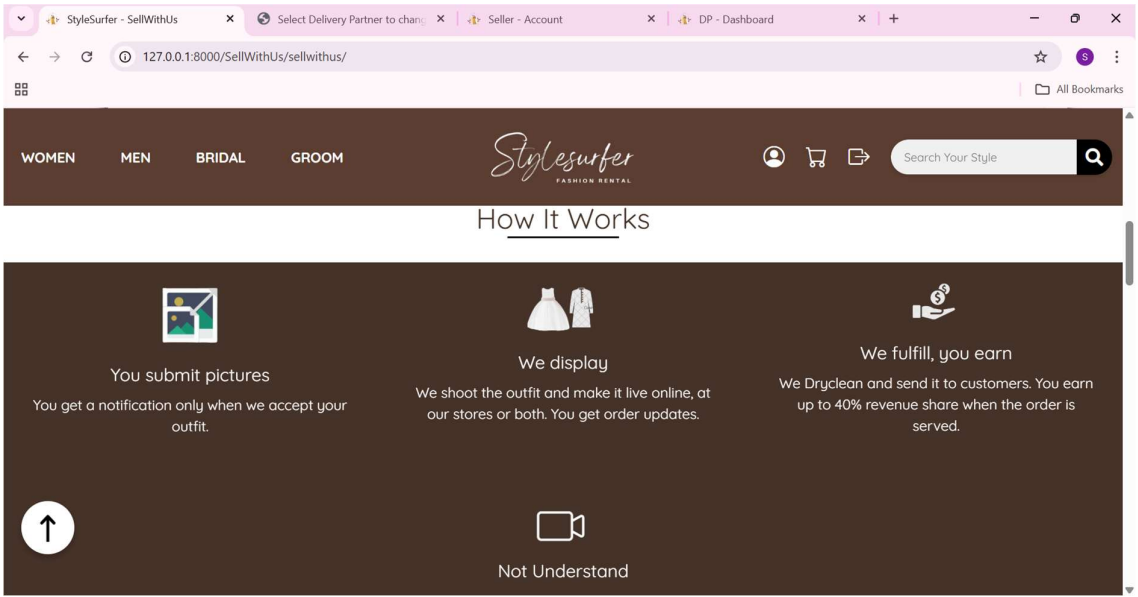
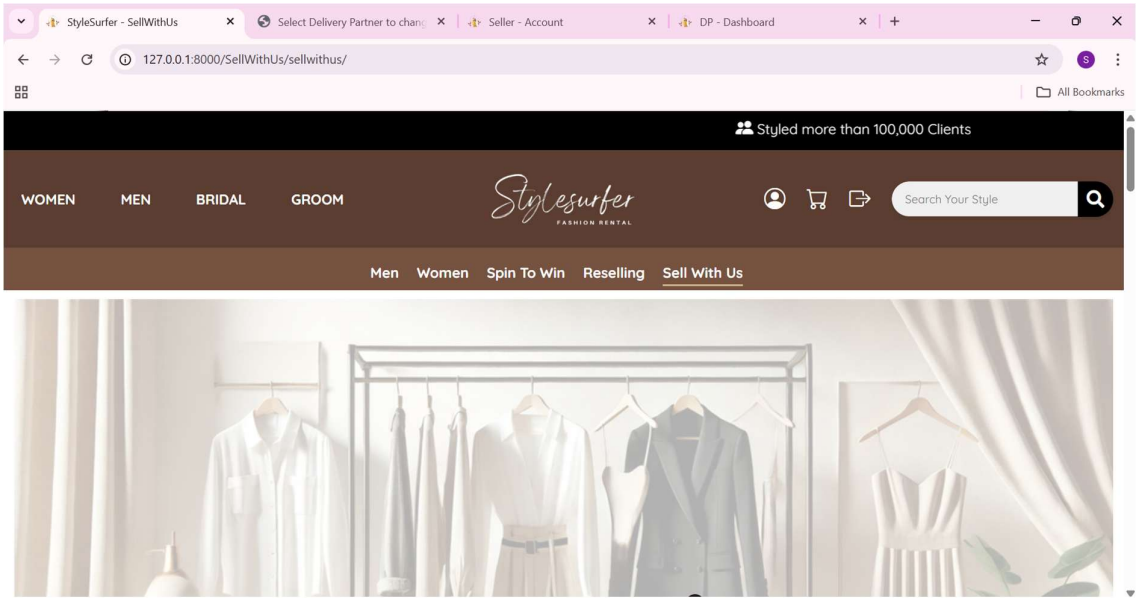
If the payment is successful, users are redirected to the Success Page, confirming their order placement. They will see an order confirmation message, along with order details and a tracking option if applicable.



This page displays resell products available for purchase at discounted prices. Users can browse and buy pre-rented fashion items.



The Sell With Us section of StyleSurfer allows users to rent out their products by filling out a registration form. The submitted details are sent to the admin for verification, and upon approval, the product goes live on the website for rental.



StyleSurfer - SellWithUs x Select Delivery Partner to chan: x Seller - Account x DP - Dashboard x + -

127.0.0.1:8000/SellWithUs/sellwithus/ ☆ All Bookmarks

WOMEN MEN BRIDAL GROOM

StyleSurfer FASHION RENTAL

Search Your Style

## Fill The Details

**Name\*** Harshad Soyaliya

**Number\*** 9316945845

**Email\*** soyaliyaharshad96@gmail.com

**Pick Up Address\*** madhapar chowk , rajkot

**City** Rajkot

**Pincode\*** 360311

StyleSurfer - SellWithUs x Select Delivery Partner to chan: x Seller - Account x DP - Dashboard x + -

127.0.0.1:8000/SellWithUs/sellwithus/ ☆ All Bookmarks

WOMEN MEN BRIDAL GROOM

StyleSurfer FASHION RENTAL

Search Your Style

**Do You Have Any Boutique Or Shop ?** ABC Boutique

**Product Name\*** blue embroidery kurta

**Product Brand/Designer Name\*** LDS

**Description\*** this blue embroidery kurta set gives festive vibes of wedding season made up with stones and embroidery

**MRP\*** 2000

**Color** blue

**Size and Quantity**

StyleSurfer - SellWithUs | Select Delivery Partner to chan... | Seller - Account | DP - Dashboard

127.0.0.1:8000/SellWithUs/sellwithus/

WOMEN MEN BRIDAL GROOM

StyleSurfer FASHION RENTAL

Search Your Style

Size and Quantity

S 10

Add Size

Category

Women - Lehnga

Front View Image Side View Image Back View Image

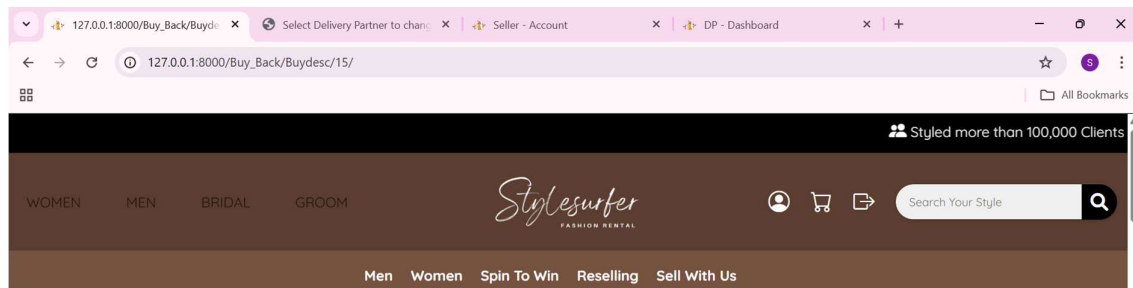
Choose File m7.jpg Choose File m8.jpg Choose File m9.jpg

For further info or feedback email us on [stylesurfer3@gmail.com](mailto:stylesurfer3@gmail.com)

☒ I agree to the [terms & conditions](#)

Submit

The Buy Back section of StyleSurfer allows users to customize their clothes with options like fabric, design, and embellishments. After use, they can return the customized outfit for a predetermined buy-back price.



### Buy\_Back - Customization Page



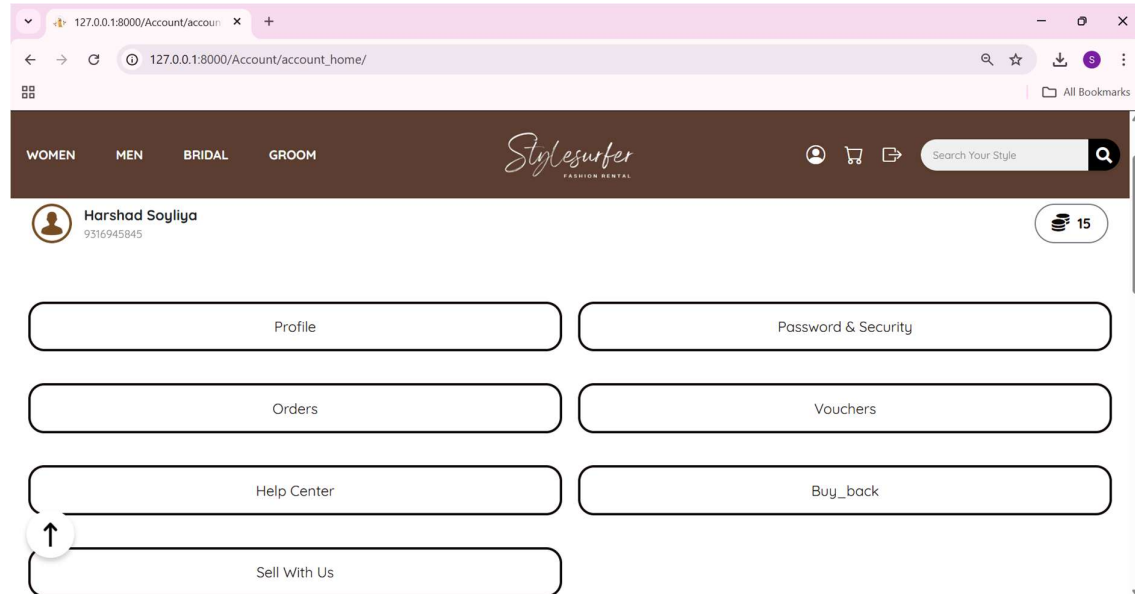

**Men Maroon Printed Kurta Jacket Dhoti Set**

LDS

~~₹4499.00~~ **Rental: ₹600.00**

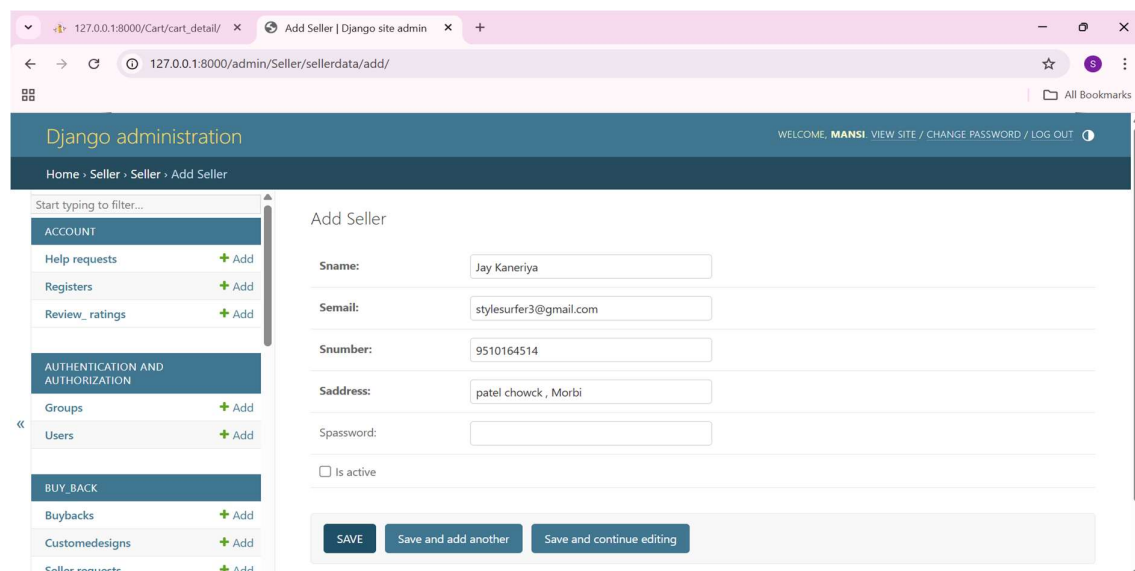
(inclusive of all taxes)

Users can access the Account Page by clicking the person icon in the navbar. Here, they can edit profile details, change password, view orders, and leave a review (if the order is delivered). They can also check redeemed vouchers, submit issue forms, view buy-back details, and manage rented products under the Sell with Us section.

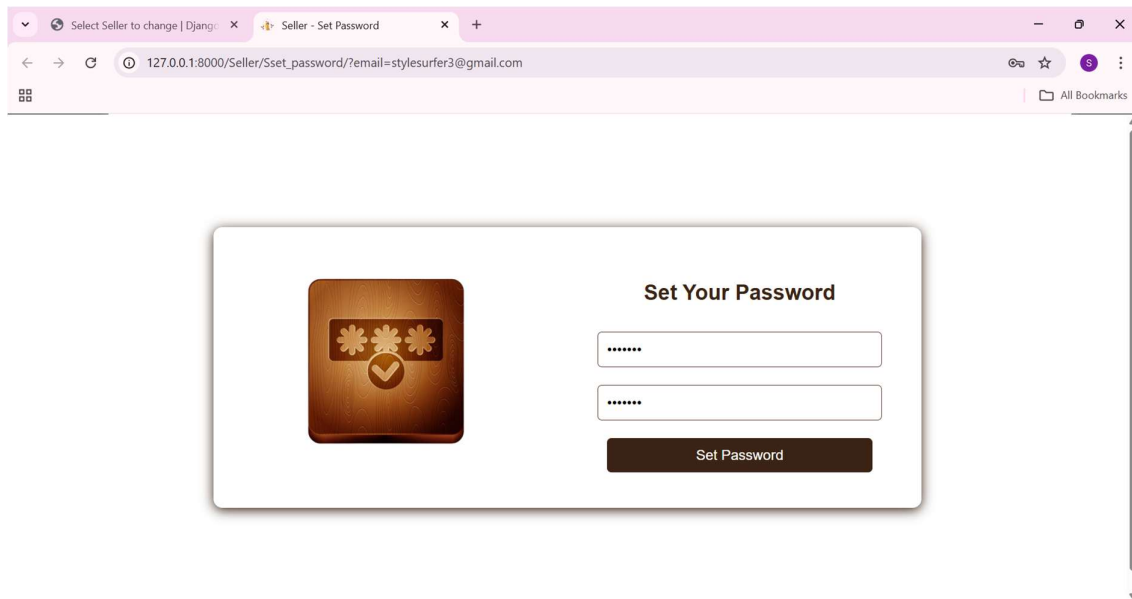


## 2) Seller :-

From the admin site , Admin adds seller details (name, email, number, address), and after saving, a set password link is sent to the seller's email.

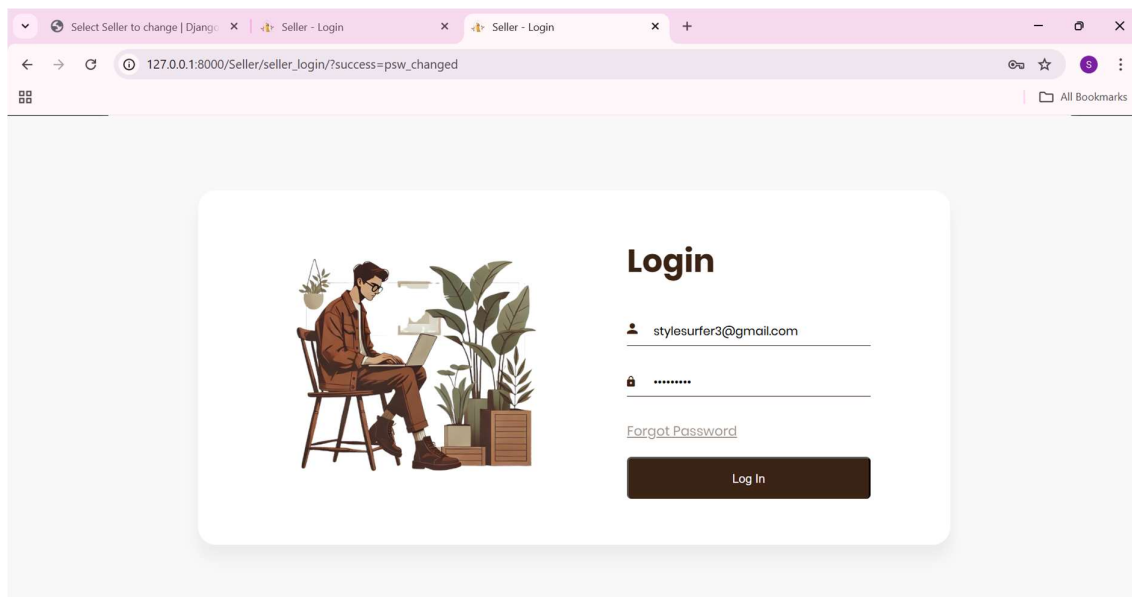


By clicking the link in the email, the seller can set a password and activate their account.



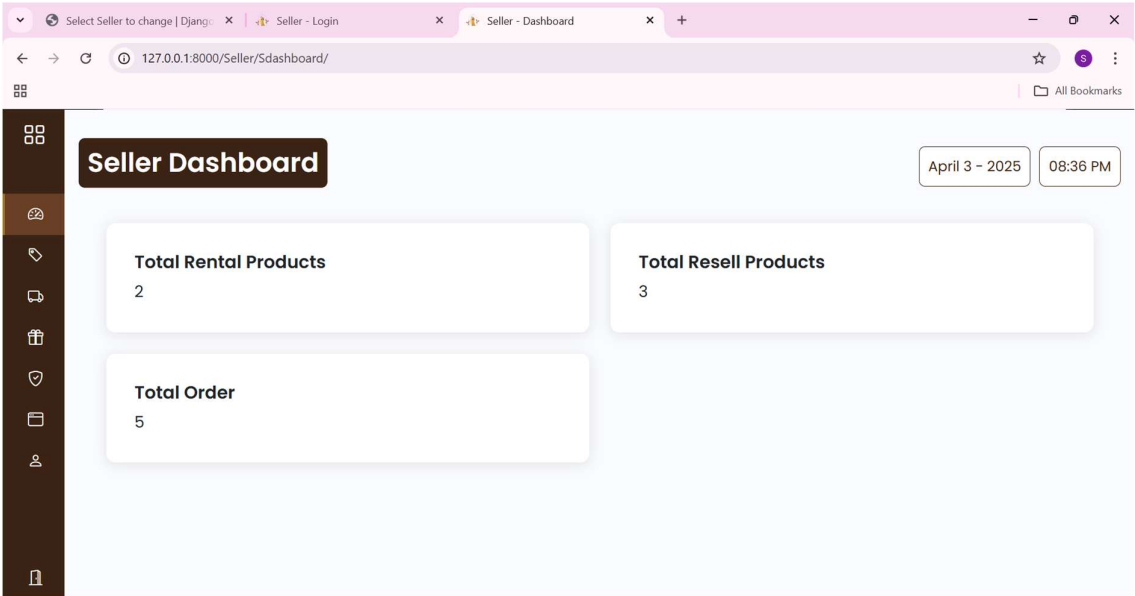
The screenshot shows a web browser window with the address bar displaying "127.0.0.1:8000/Seller/Sset\_password/?email=stylesurfer3@gmail.com". The page title is "Seller - Set Password". The main content area features a white card with a brown border. On the left side of the card is a brown square icon with three white flowers and a white checkmark. On the right side, the heading "Set Your Password" is displayed above two password input fields, each containing six asterisks. Below the input fields is a dark brown button labeled "Set Password".

After setting the password, the seller is redirected to the login page. They can log in using their registered email and password. If they click on Forgot Password, they need to enter their email, and a password reset link will be sent to change their password.

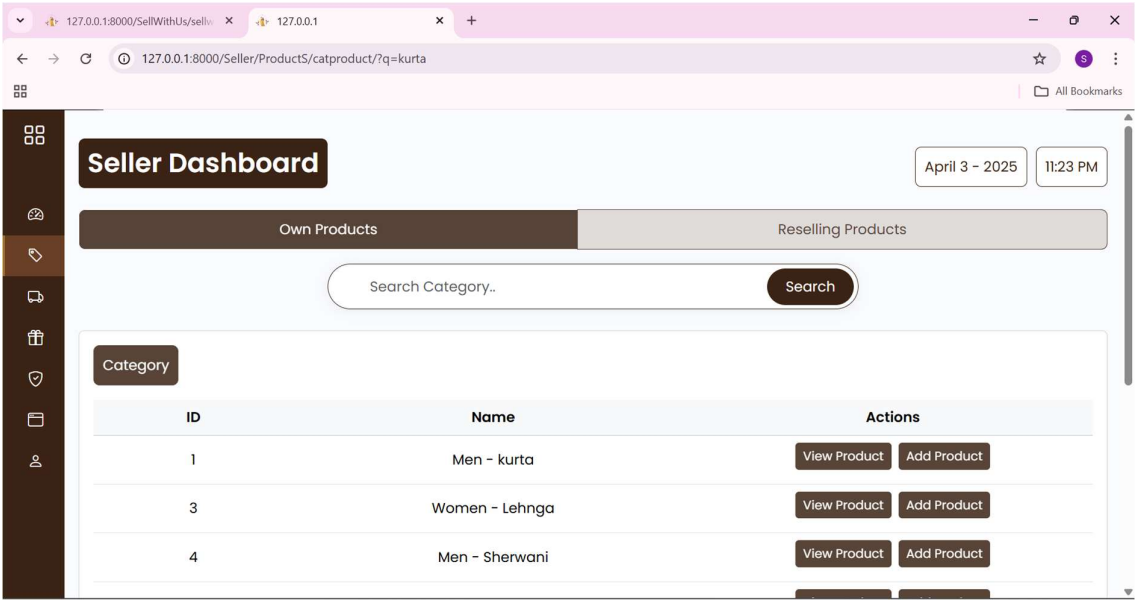


The screenshot shows a web browser window with the address bar displaying "127.0.0.1:8000/Seller/seller\_login/?success=psw\_changed". The page title is "Seller - Login". The main content area features a white card with a brown border. On the left side of the card is an illustration of a man sitting on a wooden chair, working on a laptop, with a large potted plant next to him. On the right side, the heading "Login" is displayed above two input fields: the first for the email address "stylesurfer3@gmail.com" and the second for the password, containing six asterisks. Below the input fields is a dark brown button labeled "Log In". A link labeled "Forgot Password" is positioned above the "Log In" button.

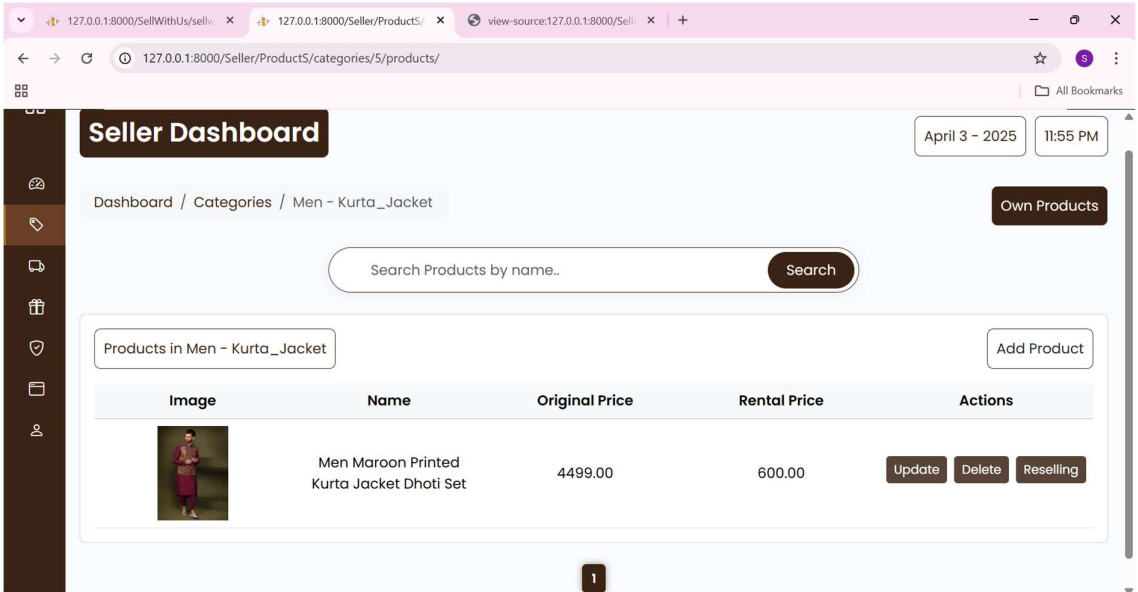
The Seller Dashboard displays an overview of their activity, including total products, total reselling products, and total orders received for their listings.



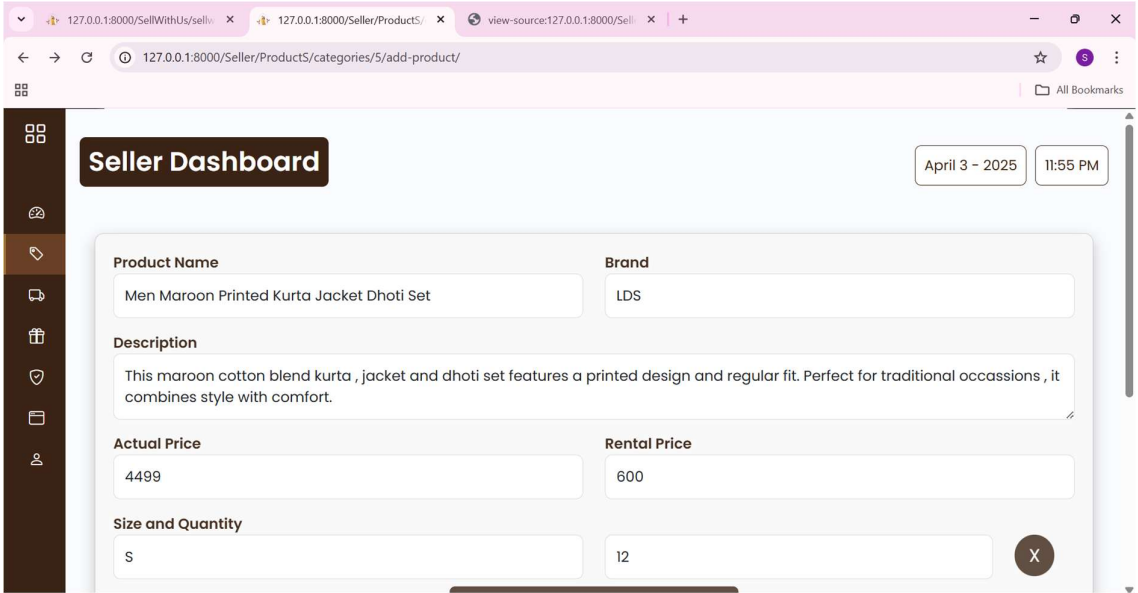
In this section, sellers can view product categories, browse products within a category, and search for specific categories easily.



In this , Sellers can view products by category, add, update, or delete products, mark items for reselling, and search for specific products.



Sellers can add a new product and optionally mark it as eligible for Buy Back.





Actual Price: 4499

Rental Price: 600

Size and Quantity: S, 12

Color: maroon

Discount Percentage: 30

Front View Image: Choose File 1.png

Side View Image: Choose File 2.png

Back View Image: Choose File 3.png

Do you want this product as a buy-back? Buy Back ☐

Add Product Cancel

In this section, sellers can view resell product categories, browse resell products within a resell category, and search for specific resell categories easily.

**Seller Dashboard**

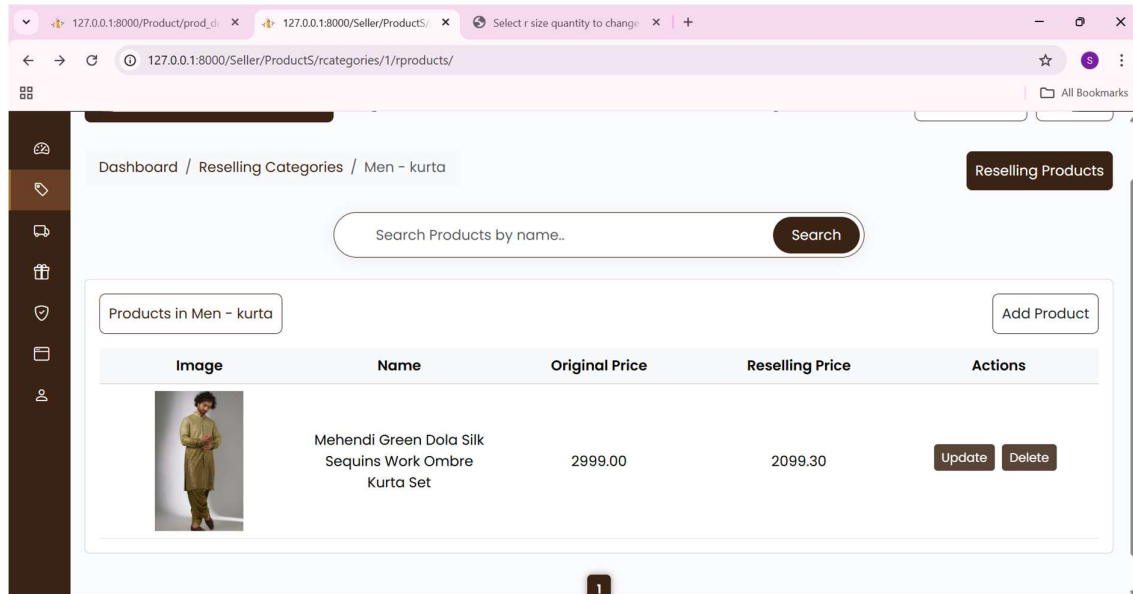
April 4 - 2025 01:46 AM

Own Products Reselling Products

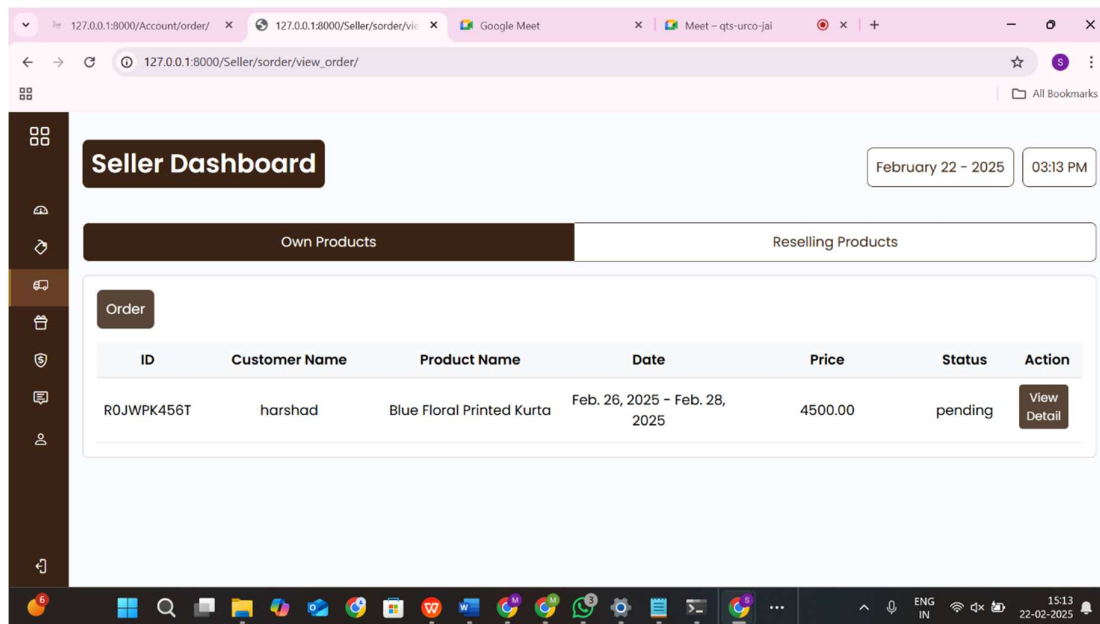
Search Category.. Search

| ID | Name           | Actions                  |
|----|----------------|--------------------------|
| 1  | Men - kurta    | View Product Add Product |
| 3  | Women - Lehnga | View Product Add Product |
| 4  | Men - Sherwani | View Product Add Product |

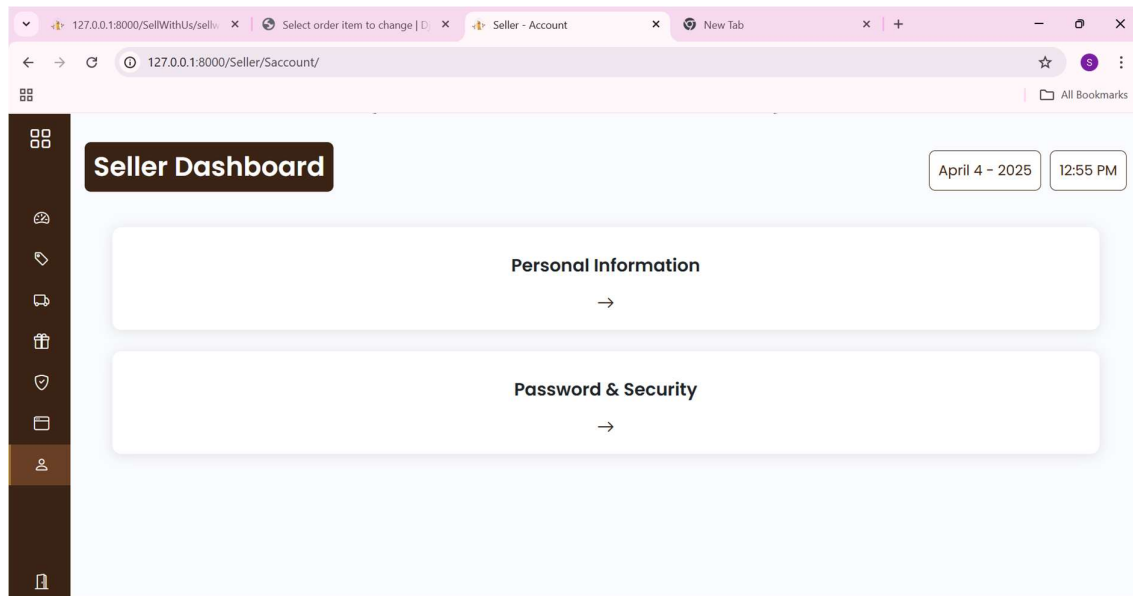
In this , Sellers can view resell products by resell category, add , update, or delete resell products, mark items for reselling, and search for specific resell products.



The Order Section in StyleSurfer allows sellers to view orders for their listed products. Sellers can track order details, manage requests, and process rentals efficiently.

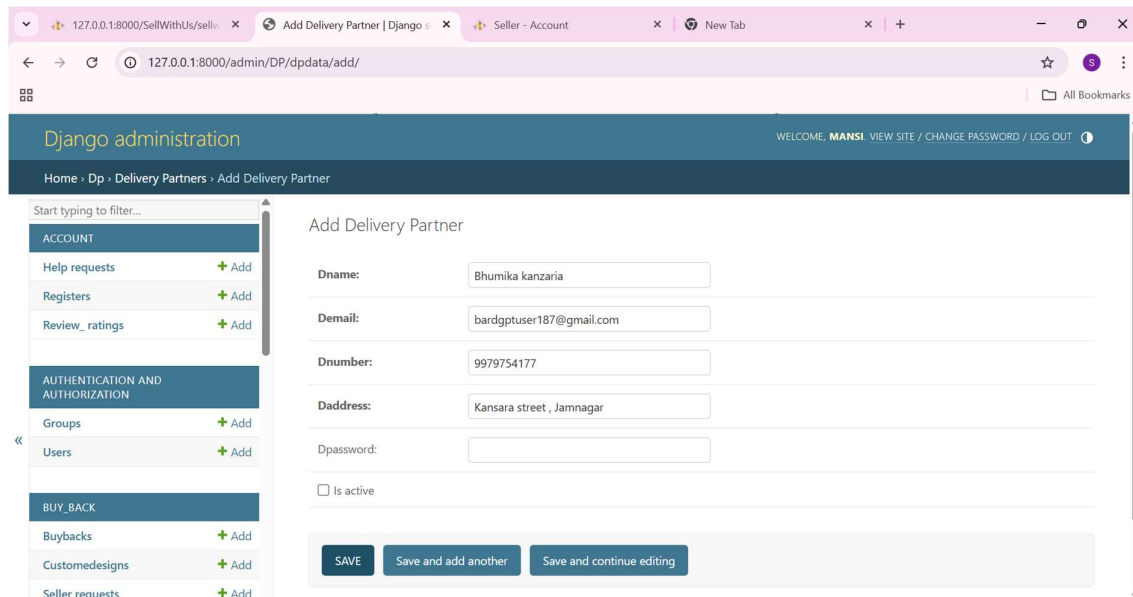


Sellers can view and edit their personal information such as name, email, etc., and also have the option to change their password.

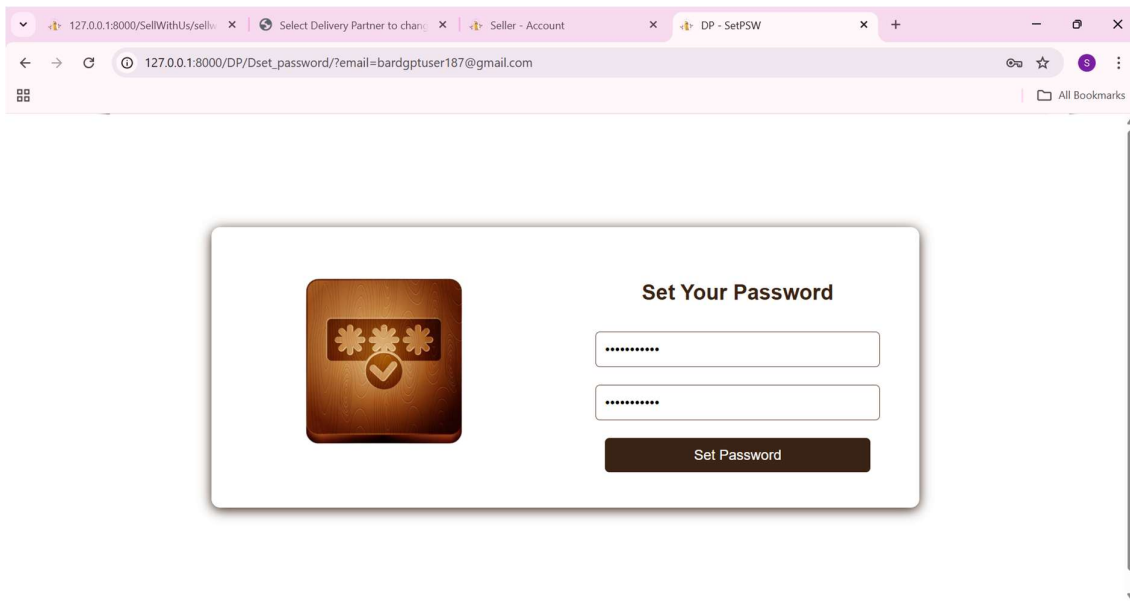


### 3) DP :-

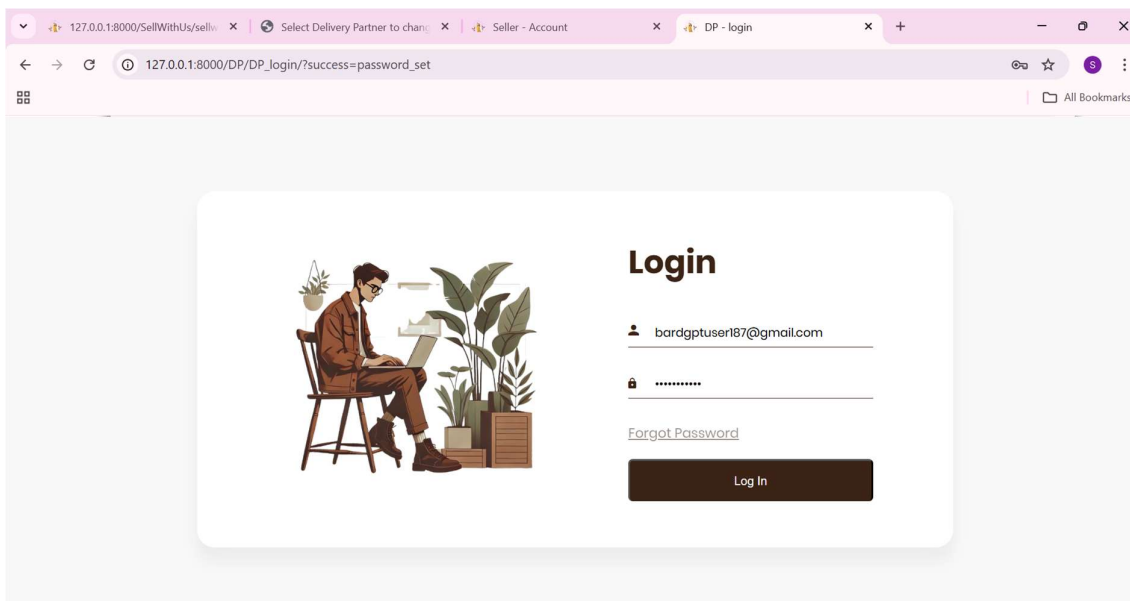
From the admin site , Admin adds dp details (name, email, number, address), and after saving, a set password link is sent to the dp's email.



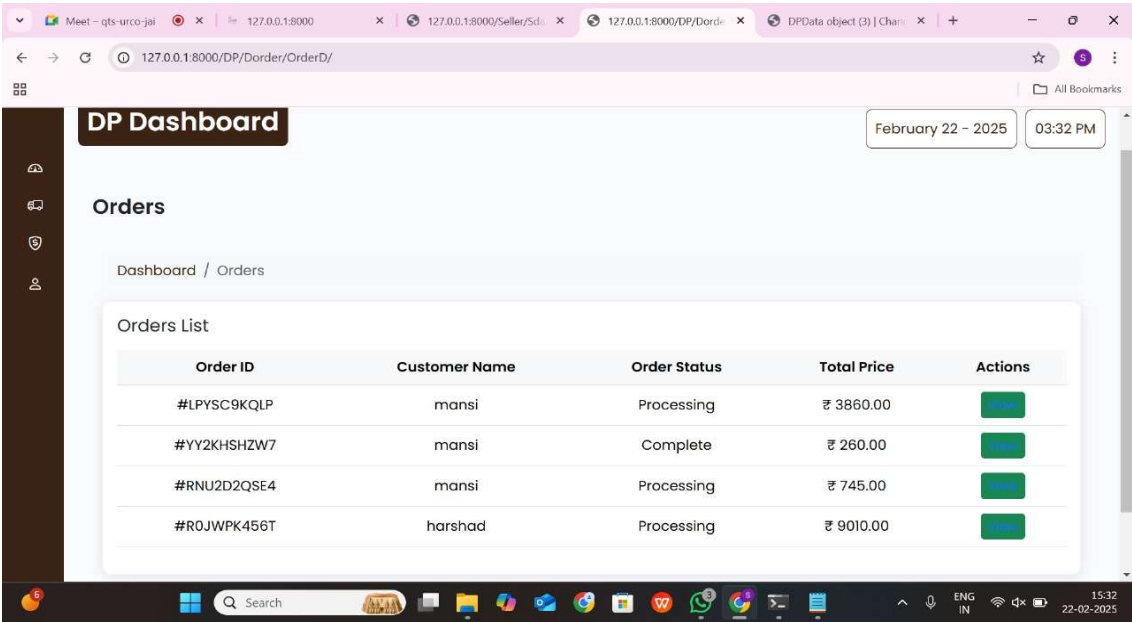
By clicking the link in the email, the dp can set a password and activate their account.



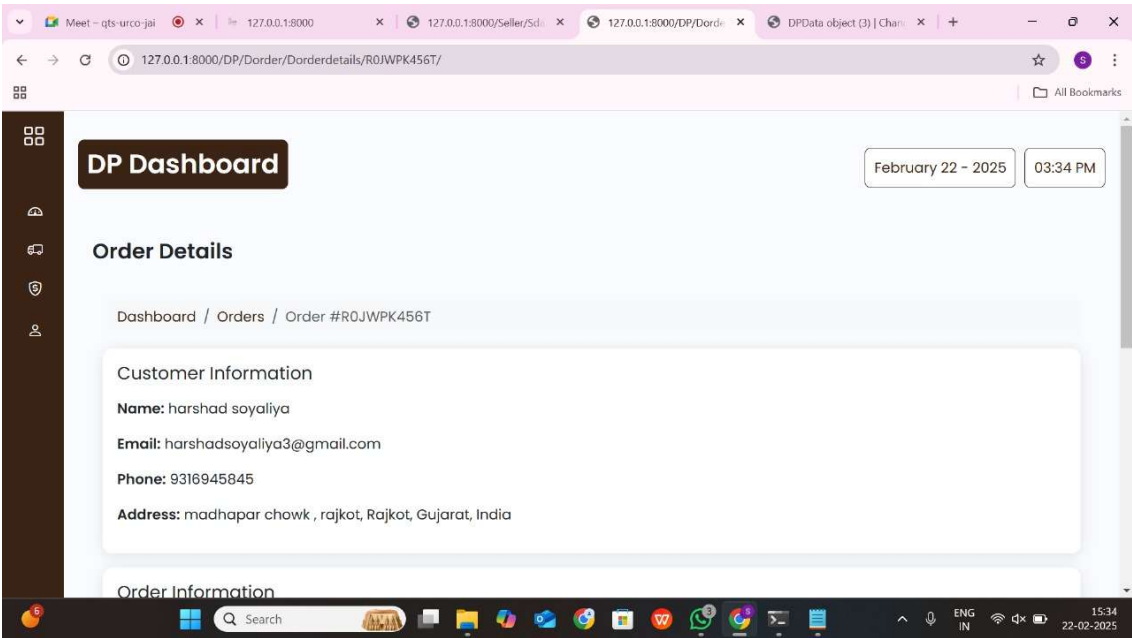
After setting the password, the dp is redirected to the login page. They can log in using their registered email and password. If they click on Forgot Password, they need to enter their email, and a password reset link will be sent to change their password.



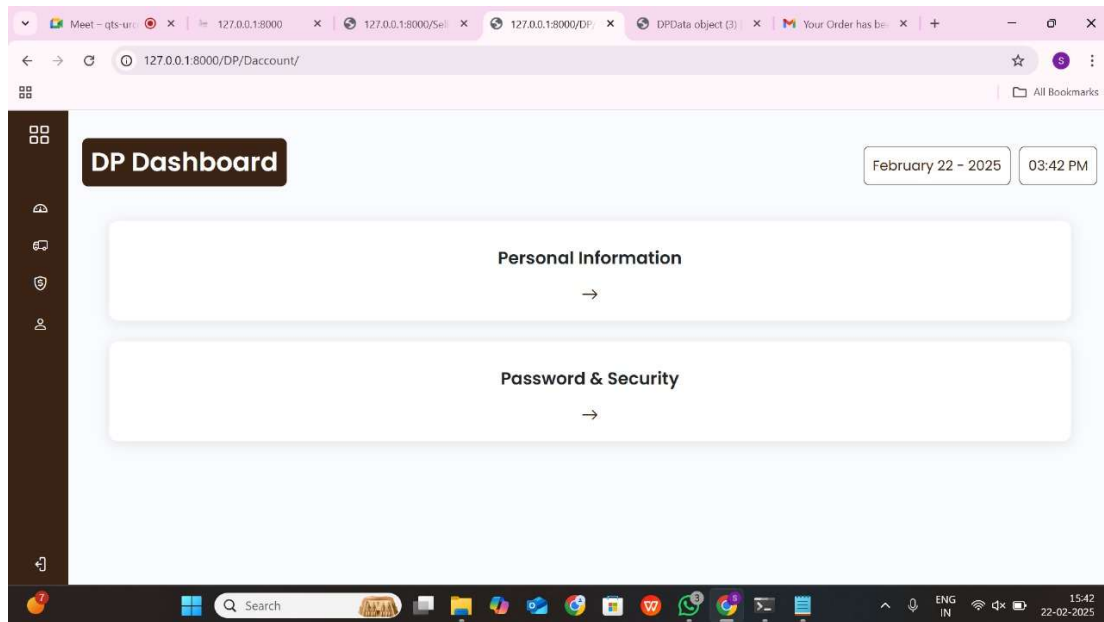
The Order Page for Delivery Partners in StyleSurfer allows them to view assigned orders, track deliveries, and manage the order status efficiently.



The Detailed Order View page for Delivery Partners in StyleSurfer displays full order information. If the order is accepted, a notification is sent to the customer, and the "Complete Order" button becomes active. If rejected, the order is reassigned to another delivery partner.

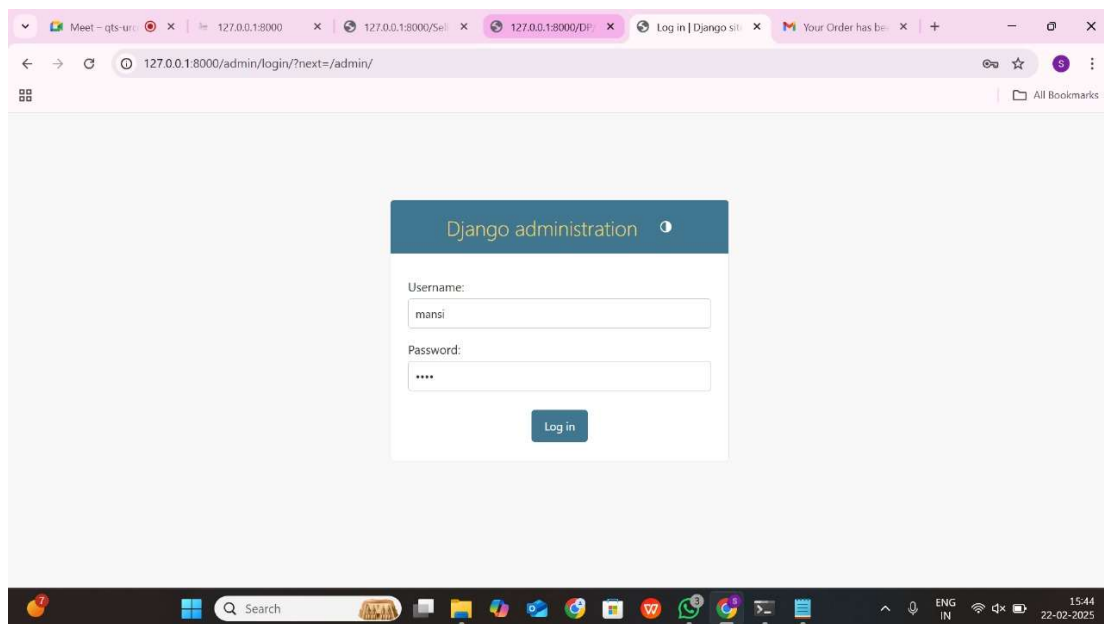


DP can view and edit their personal information such as name, email, etc., and also have the option to change their password.



#### 4) Admin :-

This is the login page of admin .



This is the dashboard of admin . admin can manipulate data .

